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Abstract

ABSTRACT.

Chapter 1

Introduction

Background and Motivation

1.1 Research Question

Research problem: identifying how global lng maritime network nodes' criticality evolves in time, distinguishing the probability that a node is active from its criticality conditioned on being active.

1. How will the structure of the global LNG maritime network evolve between 2020 and 2024?

2. What characteristics determine the probability that a terminal or chokepoint will be operational in a given month?

3. Which latent dimensions collectively describe the various measures of criticality?

4. How do structural, infrastructural and dynamic characteristics influence the criticality of nodes?

5. How do the mechanisms of criticality differ between terminals and chokepoints?

1.2 Contributions

Empirical (new monthly lng maritime dataset), methodology (separation: activity vs conditional criticality), theoretical (multidimensionality of criticality).

1.3 Thesis structure

This thesis is structured as follows...

Chapter 2

Literature review

2.1 Section 1

Global LNG Maritime trade networks: LNG organization, description of nodes' roles and routes.

2.2 Section 2

Potential subsections:

Criticality and Vulnerability in network analysis: comparison of importance, centrality, criticality, vulnerability, network science measures' roles.

Dynamic and temporal networks: temporal dependence, nodes inactivity and persistence.

Statistical models for sparse network processes: hurdle models

Bayesian reduced-rank models: multivariate bayesian models, shrinkage...

Research gap: combine dynamicity, multidimensional criticality, terminals and chokepoints analysis together

2.2.1 Subsection 2.1

Chapter 3

Data and network construction

This chapter describes the data sources and processing steps used to construct the dynamic LNG maritime network analysed in this thesis. The final dataset combines observed LNG tanker voyages, import and export terminals, reconstructed maritime routes, and the strategic chokepoints traversed by those routes. Together, these components represent the global LNG transportation system as a directed and weighted network evolving at a monthly frequency from January 2020 to December 2024.

The construction process involved harmonising the source data, matching observed voyages to their departure and arrival terminals, reconstructing the corresponding maritime routes, identifying the chokepoints crossed by each route, and aggregating trade flows and network relationships at the monthly level. The resulting dataset contains the nodes, connections, flow volumes, route characteristics, and infrastructure attributes required for the descriptive and Bayesian analyses developed in the following chapters.

3.1 Data sources

Despite the growing importance of LNG for international energy trade and energy security, publicly available data with high spatial and temporal resolution remain limited. Official and industry sources generally report LNG trade at the country level and at monthly or annual frequencies, whereas more granular information is often available only through commercial databases. Moreover, to the best of my knowledge, no single

open-access dataset jointly represents individual LNG voyages, terminal infrastructure, reconstructed maritime routes, and the strategic chokepoints traversed by those routes.

The dataset used in this thesis was therefore constructed by integrating voyage-level LNG trade records and terminal infrastructure data with reconstructed maritime routes and additional geographic information on maritime chokepoints. The following subsections describe the four main components of this integrated dataset.

3.1.1 LNG trade and infrastructure data

The main data source is the Liquefied Natural Gas from Tanker, Terminal, and Trade (LNG-T3) dataset developed by Zhou et al. (2026). LNG-T3 is an open-access dataset providing global LNG infrastructure and trade information at daily resolution for the period from January 2020 to December 2024. It was constructed by combining Automatic Identification System (AIS) vessel-position and draught reports, historical port-call records, LNG tanker specifications, and terminal metadata. The complete dataset contains 17,592 individual LNG tanker voyage events and provides information on tanker movements, terminal activity, and country-to-country trade flows.

Two components of LNG-T3 were used directly in this study: the voyage-level records and the terminal inventory. The voyage data form the empirical basis for identifying directed LNG trade connections and measuring the volumes transported between terminals. The terminal inventory defines the infrastructure nodes of the network and provides the geographic and structural attributes used to characterise them. These two components were linked through the terminal identifiers included in the original dataset.

LNG-T3 was preferred to aggregate country-level trade statistics because its voyage-level resolution makes it possible to identify the specific origin and destination terminals associated with each LNG shipment. Its 2020–2024 coverage also provides a consistent five-year observation window that captures both established trade patterns and major changes in the global LNG market, including those following the disruption

of European gas supplies in 2022.

LNG voyages

The LNG-T3 voyage data contain individual movements of LNG tankers between departure and arrival terminals. For each voyage, the available information includes the tanker identifier, the origin and destination terminals, the departure and arrival dates, and the estimated volume of LNG transported. These records were used to identify the direction, timing, and magnitude of LNG flows within the network.

Each voyage was represented as a directed movement from its departure terminal to its arrival terminal. The voyage date was then used to assign the observation to a monthly period between January 2020 and December 2024. When multiple voyages occurred between the same origin–destination pair during the same month, their transported volumes were aggregated. The resulting monthly observations define the directed and weighted trade connections between terminal nodes.

Voyage identifiers were also used to retain information on the number of shipments associated with each connection. This makes it possible to distinguish between connections characterised by a small number of high-volume voyages and those supported by more frequent tanker movements. Once the corresponding maritime routes had been reconstructed, route distance and chokepoint crossings were linked back to the individual voyage records and subsequently aggregated at the monthly level.

LNG terminals

The LNG-T3 terminal inventory provides the identifiers, names, geographic coordinates, countries, and infrastructure characteristics of LNG facilities worldwide. Terminals constitute the origin and destination nodes of the trade network and were linked to the voyage records using their unique identifiers.

The infrastructure attributes retained for the analysis include the terminal's country, infrastructure type, processing capacity, number of infrastructure units, and start year. Infrastructure type distinguishes facilities according to their import or export

function, while capacity provides an indicator of their potential scale of operation. The number of infrastructure units supplies an additional measure of facility size and complexity. The start year was used to construct terminal age as a time-varying variable, calculated for each monthly observation as the difference between the observation year and the year in which the terminal began operating.

Terminal coordinates were used both to position terminal nodes geographically and to reconstruct the maritime route associated with each observed origin–destination pair. Before route reconstruction, terminal identifiers and coordinates were checked for consistency across the voyage and infrastructure data. This harmonisation ensured that every retained voyage could be linked to a valid departure terminal and arrival terminal.

3.1.2 Maritime routes and chokepoint data

While the LNG-T3 voyage records identify the origin and destination of each shipment, they do not provide the actual maritime path connecting the two terminals. For this reason, maritime routes were reconstructed from the geographic coordinates of the departure and arrival terminals using the `SeaRoute` routing software (Eurostat, 2021).

SeaRoute software

SeaRoute applies Dijkstra's algorithm (Dijkstra, 1959) to identify the shortest path between two locations on a predefined network representing frequently used shipping lanes, which accounts for coastlines, navigable waterways, and supported maritime passages. Specifically, the 5-km resolution of the maritime network produced through the `marnet` module was used: edges shorter than this scale are progressively contracted and geometrically similar connections are removed, reducing network complexity while preserving the principal maritime routes. Additionally, in line with the thesis' objectives, transit through both the Suez and Panama canals was enabled, allowing the routing algorithm to include these passages when they formed part of the shortest available maritime path.

A route was generated for each unique origin–destination terminal pair and linked to all voyages associated with that pair. Afterwards, the reconstructed routes were merged with the chokepoints locations to identify the strategic maritime passages traversed by each trade connection.

Chokepoints geometries

The initial set of 28 maritime chokepoints, together with their names, identifiers, and reference coordinates, was obtained from the IMF PortWatch chokepoint database (International Monetary Fund, 2026), but because the PortWatch locations are represented primarily as point features, they could not be used directly to determine whether a reconstructed maritime route effectively crossed a passage. A geometry registry was therefore compiled to associate each chokepoint with an appropriate spatial representation. The registry, including the source and construction method for each geometry, is provided in Appendix A.

The final geometries were assembled from several sources. Published polygons for the Suez Canal, Bab el-Mandeb Strait, and Cape of Good Hope were directly obtained from the World Bank Development Data Partnership's Red Sea Monitoring project (World Bank Data Lab, 2025). Detailed IHO or SeaVoX geometries for the Malacca, Hormuz, Gibraltar, Dover, and Makassar straits were obtained through the Marine Regions Gazetteer and its geospatial services (Flanders Marine Institute, nd). For the Lombok, Sunda, Torres, and Bering straits, transit geometries were reconstructed from the WGS84 coordinates reported in official International Maritime Organization routeing documents (International Maritime Organization, 2019, 2014; Russian Federation and United States, 2017).

Where no suitable published polygon was available, operational geometries were constructed specifically using either a buffered centreline, as in the Panama Canal and Bosporus Strait, or an oriented crossing gate positioned across the relevant navigable passage. The placement of these geometries was informed by the reference coordinates provided by PortWatch (International Monetary Fund, 2026), geographic information

from Marine Regions (Flanders Marine Institute, nd), and navigational references from relevant IMO documents, the NGA GeoNames database, and NOAA/NCEI hydrographic survey records (International Maritime Organization, 1998; National Geospatial-Intelligence Agency, nd; National Centers for Environmental Information, nd).

Figure 1 shows the locations of the 28 candidate chokepoints and representative examples of the four types of geometry used to identify route crossings: a published polygon, a polygon reconstructed from IMO routeing coordinates, an operational crossing gate, and a buffered centreline.

All geometries were transformed into a common geographic coordinate system and stored in a harmonised geospatial format. To accommodate small discrepancies between the reconstructed `SeaRoute` paths and the chokepoint geometries, a uniform spatial-tolerance buffer of 25 km was applied around each geometry. A reconstructed origin–destination route was classified as traversing a chokepoint when it intersected the corresponding buffered geometry.

Of the 28 candidate chokepoints, 24 were traversed by at least one reconstructed route under the final 25-km specification and were therefore retained as nodes in the monthly network. The robustness of the route–chokepoint assignments was assessed by comparing this specification with exact geometric intersection and with alternative buffers of 5 and 10 km. The complete sensitivity analysis is reported in Appendix B.

Table 1 summarises the principal data sources, their units of observation, the main variables retained, and their role in the construction of the network.

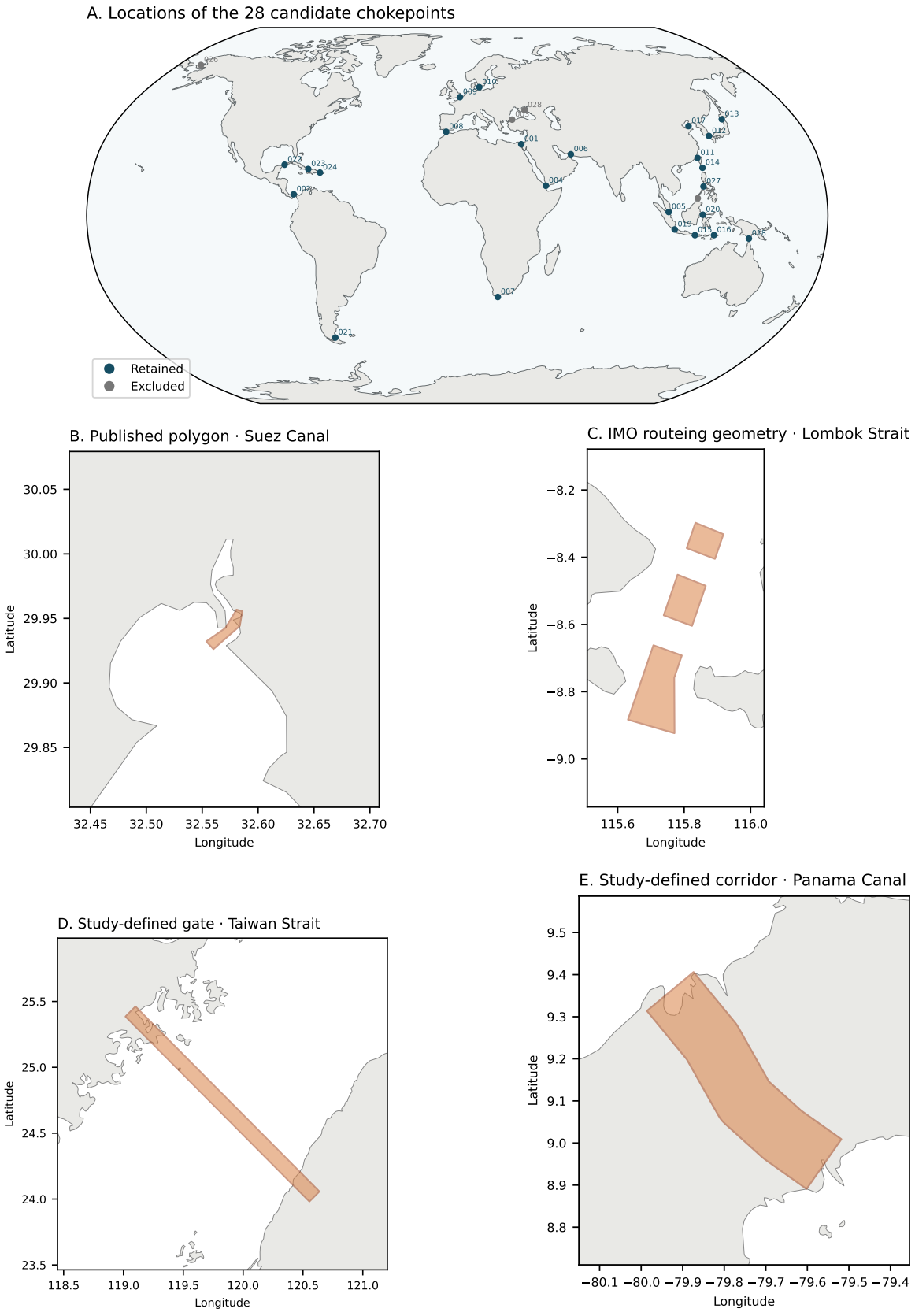


Figure 1: PortWatch reference locations of the 28 candidate chokepoints and representative route-crossing geometries. The enlarged panels show the Suez Canal (published polygon), Lombok Strait (geometry reconstructed from IMO routeing coordinates), Taiwan Strait (operational crossing gate), and Panama Canal (buffered operational centreline).

Table 1: Summary of data sources

Source	**Unit of observation**	**Main variables**	**Role in the analysis**
LNG-T3 tanker voyages	Individual voyage	Departure and arrival terminals, voyage dates, tanker identifier, estimated LNG volume	Identification of directed LNG trade connections and monthly flow volumes from January 2020 to December 2024
LNG-T3 terminal inventory	LNG terminal	Terminal identifier, name, coordinates, country, infrastructure type, capacity, unit count, start year	Definition and characterisation of terminal nodes
`searoute`	Unique origin–destination pair	Reconstructed maritime path, route distance	Reconstruction of the geographic route associated with each trade connection
Chokepoint geographic data	Maritime chokepoint	Name, type, and geographic location	Definition of chokepoint nodes, identification of route crossings

3.2 Data preparation

Before constructing the network, the voyage and terminal datasets were harmonised to ensure that each observed shipment could be associated with a unique origin and destination node.

As an initial validation step, voyage dates, LNG volumes, terminal capacities, and unit counts were parsed into consistent formats. No invalid or missing values were found in these fields.

The LNG-T3 terminal inventory contains 545 records corresponding to 471 unique terminal names. Sixty-five names occur more than once, generally because the same facility has multiple infrastructure-status records. Whereas most repeated records have identical coordinates, three terminal names have two coordinate variants each.

Records were first distinguished by name and coordinates. One record was then

selected to represent each terminal name in the final terminal map, based on the infrastructure status reported for that record in the LNG-T3 inventory. The selection prioritised records classified as *operating*, followed by those classified as *under construction*, *idle*, *mothballed*, and *proposed*. Each of the 471 terminals in the final map was then assigned a unique node identifier.

Voyage origins and destinations were matched to the consolidated terminal map using the terminal names reported in the LNG-T3 voyage data. The corresponding node identifiers and coordinates were attached to each voyage record, and the matching status of the two endpoints was recorded separately. All origin and destination values in the 17,592 voyage records were successfully matched to the terminal map.

The original voyage table contains 8,645 laden export voyages and 8,947 return movements. Since the latter take place after delivery and therefore report zero transported LNG volume, only the laden export voyages were retained to construct the LNG trade network. The retained voyages were also required to have a valid departure date, matched origin and destination identifiers, and a strictly positive LNG volume. All export records satisfied these conditions, meaning that these checks did not produce any further exclusions.

Potential duplicate export observations were identified using the departure and arrival dates, tanker IMO number, voyage type, origin and destination terminals, and transported volume. After removing three duplicate entries, the final dataset used to construct the network contains 8,642 laden LNG voyages.

3.2.1 Terminal infrastructure characteristics

Terminal characteristics were obtained from the LNG-T3 infrastructure inventory and linked to the consolidated terminal identifiers. The retained attributes include country, region, geographic coordinates, infrastructure type, operational status, annual processing capacity, number of infrastructure units, and start year. No missing values were found for these attributes among the terminals retained in the final network.

Additionally, terminal age was constructed as a time-varying characteristic by subtracting the terminal's reported start year from the year of each monthly observation:

$$\mathrm{Age}_{i,t} = \mathrm{Year}_t - \mathrm{StartYear}_i \,. \tag{1}$$

3.2.2 Temporal aggregation

Each retained voyage was assigned to a monthly period according to its departure date. The resulting observation window extends from January 2020 to December 2024, corresponding to the complete five-year coverage of the LNG-T3 dataset and yielding 60 monthly network observations.

To preserve months in which a node had no observed activity, complete node–month panels were subsequently constructed by combining every retained node with all 60 monthly periods. Flow and voyage measures for node–month observations with no activity were assigned a value of zero to distinguish these structural zeros from missing or non-applicable node characteristics, which remained coded as unavailable.

3.3 Dynamic network construction

The global LNG maritime network was represented as a sequence of monthly directed and weighted graphs,

$$G_t = (V, E_t, W_t), \qquad t = 1, \ldots, 60, \tag{2}$$

where V is the time-invariant set of terminal and chokepoint nodes, E_t is the set of directed connections observed in month t, and $W_t = [w_{ij,t}]$ is the corresponding weighted adjacency matrix. An element $w_{ij,t} > 0$ represents the total LNG volume transported from node i to node j during month t, while $w_{ij,t} = 0$ indicates that no such connection was observed.

3.3.1 Nodes and edges

The final node set contains 183 nodes and can be written as

$$V = V^T \cup V^C, \qquad |V^T| = 159, \qquad |V^C| = 24, \tag{3}$$

where V^T denotes the set of LNG terminals and V^C the set of maritime chokepoints.

The 159 terminal nodes are the facilities that appear as an origin or destination of at least one retained laden voyage. Of the 28 candidate chokepoints included in the geographic registry, 24 were traversed by at least one reconstructed route and were therefore retained in the observed network.

Each commercial voyage defines an underlying directed movement from an export terminal to an import terminal. To incorporate the maritime infrastructure traversed during transportation, the reconstructed route of voyage r was represented as an ordered path

$$P_r = (v_{r,0}, v_{r,1}, \ldots, v_{r,K_r}), \tag{4}$$

where $v_{r,0}$ is the departure terminal, v_{r,K_r} is the arrival terminal, and any intermediate nodes are the chokepoints crossed by the route. The voyage was then expanded into the consecutive directed connections

$$E(P_r) = \{(v_{r,0}, v_{r,1}), (v_{r,1}, v_{r,2}), \ldots, (v_{r,K_r-1}, v_{r,K_r})\}. \tag{5}$$

A route that crossed no retained chokepoint generated a direct terminal-to-terminal edge. Routes crossing one or more chokepoints generated terminal-to-chokepoint, chokepoint-to-chokepoint, and chokepoint-to-terminal edges according to the order in which those nodes were encountered.

Let $\mathcal{R}_t$ denote the set of voyages during month t, and q_r the LNG volume transported by voyage r. The edge weight from node i to node j at month t is defined as

$$w_{ij,t} = \sum_{r \in \mathcal{R}_t} q_r \, \mathbb{I}\left\{(i,j) \in \mathcal{E}(P_r)\right\}, \tag{6}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. Thus, the full volume of a shipment was assigned to every segment traversed by its route. Because shipments crossing multiple segments would consequently be counted more than once in the sum of all edge weights, aggregate monthly trade volume was calculated directly from the original voyage observations.

Voyages sharing the same directed pair of nodes within a month were aggregated into a single edge. In addition to LNG volume, each monthly edge records the number of voyage traversals, the number of distinct origin–destination routes using the connection, and their route identifiers.

3.3.2 Node activity

Node activity was defined from the presence of incident flow in each monthly graph. For node i in month t, the binary activity indicator was defined as

$$a_{i,t} = \mathbb{I}\left\{\sum_{j \in V} w_{ij,t} + \sum_{j \in V} w_{ji,t} > 0\right\}. \tag{7}$$

A terminal was therefore classified as active when it recorded positive incoming or outgoing LNG flow during the month. Since terminal direction was fully consistent with the infrastructure classification reported in LNG-T3, active export terminals recorded outgoing flow and active import terminals recorded incoming flow.

A chokepoint was classified as active when at least one voyage traversed it during the month, generating positive incident flow on one or more adjacent route segments. Chokepoints not traversed by any voyage in a given month were retained in the balanced node–month panel and assigned an activity value of zero.

3.4 Construction of node-level measures

A set of monthly node-level measures was constructed to represent different dimensions of criticality in the LNG transportation network. Since terminals and chokepoints perform different functions, their criticality was evaluated using separate but complementary measures and network representations.

On the one hand, terminals represent the origins and destinations of commercial LNG flows. Their criticality was therefore defined in terms of their importance as suppliers or recipients, the concentration of their trading relationships, and the dependence associated with those relationships. On the other hand, chokepoints represent intermediate passages along maritime routes, and their criticality was consequently defined in terms of their position within the physical transportation network, the volume of LNG passing through them, and their role in connecting otherwise separated parts of the network or in mediating important transportation paths.

Terminal criticality measures were calculated from the original terminal-to-terminal voyage flows, thereby preserving the direct commercial relationship between origins and destinations and avoiding the repeated counting of shipments across multiple route segments. Chokepoint criticality measures were instead calculated on the complete segment-level network, including both terminals and chokepoints, so that the position of each passage within the full sequence of maritime movements could be represented. Additional structural indicators were calculated for both node types.

3.4.1 Terminal criticality measures

Terminal criticality was represented through measures of directional network importance, counterparty concentration, trade dependence, and voyage distance. These measures were constructed monthly and, where relevant, adapted to the terminal's observed role as an LNG exporter or importer.

Terminal role

Let $q_{ij,t}$ denote the total LNG volume transported from terminal i to terminal j during month t. Monthly outgoing flow, incoming flow, and throughput were defined as

$$\begin{aligned} O_{i,t} &= \sum_{j} q_{ij,t}, \\ I_{i,t} &= \sum_{j} q_{ji,t}, \\ T_{i,t} &= O_{i,t} + I_{i,t}. \end{aligned} \tag{8}$$

A terminal was classified as active in month t when $T_{i,t} > 0$. Among active terminals, the observed monthly role was classified as exporter when $O_{i,t} > 0$ and $I_{i,t} = 0$, importer when $I_{i,t} > 0$ and $O_{i,t} = 0$, and bidirectional when both flows were positive.

For active terminal–months, export share was defined as

$$s_{i,t}^{\text{out}} = \frac{O_{i,t}}{T_{i,t}}, \tag{9}$$

so that exporters have $s_{i,t}^{\text{out}} = 1$, importers have $s_{i,t}^{\text{out}} = 0$, and bidirectional terminals take values between zero and one.

Counterparty concentration

The concentration of a terminal's trade across its counterparties was measured using the Herfindahl–Hirschman Index (HHI), a standard measure of concentration based on the squared shares of individual counterparties (Hirschman, 1945).

Let $\mathcal{D}_{i,t} = \{j : q_{ij,t} > 0\}$ be the set of destination terminals receiving LNG from terminal i in month t, and let $\mathcal{S}_{i,t} = \{j : q_{ji,t} > 0\}$ be the set of origin terminals supplying it. The outgoing and incoming terminal-level concentration indices were calculated, respectively, as

$$\begin{aligned} H_{i,t}^{\text{out}} &= \sum_{j \in \mathcal{D}_{i,t}} \left(\frac{q_{ij,t}}{O_{i,t}} \right)^2, \\ H_{i,t}^{\text{in}} &= \sum_{j \in \mathcal{S}_{i,t}} \left(\frac{q_{ji,t}}{I_{i,t}} \right)^2. \end{aligned} \tag{10}$$

The same procedure was applied after aggregating counterparties by country, producing corresponding country-level concentration indices. Values approaching one indicate that flows are concentrated in a small number of counterparties, whereas lower values indicate greater diversification.

A role-specific concentration measure was then assigned according to the terminal's observed direction of trade:

$$H_{i,t}^{\text{role}} = \begin{cases} H_{i,t}^{\text{out}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} = 0, \\ H_{i,t}^{\text{in}}, & \text{if } I_{i,t} > 0 \text{ and } O_{i,t} = 0, \\ s_{i,t}^{\text{out}} H_{i,t}^{\text{out}} + \left(1 - s_{i,t}^{\text{out}}\right) H_{i,t}^{\text{in}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} > 0. \end{cases} \tag{11}$$

Directional PageRank

Terminal network importance was measured using PageRank, which considers both the magnitude of a terminal's connections and the importance of the terminals to which it is connected (Page et al., 1999). The measure was calculated separately for importing and exporting roles on the monthly directed and flow-weighted terminal network: import PageRank was calculated using the observed direction of LNG flows, whereas export PageRank was calculated after reversing all edges. Intuitively, import PageRank assigns greater importance to terminals receiving flows from other important terminals, while export PageRank captures the corresponding importance of LNG suppliers.

For the observed network, weighted import PageRank satisfies

$$PR_{i,t}^{\text{imp}} = \frac{1-\alpha}{N_t} + \alpha \sum_{\substack{j \\ (j,i) \in E_t}} PR_{j,t}^{\text{imp}} \frac{q_{ji,t}}{\sum_k q_{jk,t}}, \tag{12}$$

where N_t is the number of active terminals in month t, and the damping factor was set to $\alpha = 0.85$, following the conventional value used by Brin and Page (1998). In this formulation, α controls the contribution propagated through the network's directed flows, while $1 - \alpha$ determines the uniform redistribution component. Export PageRank, $PR_{i,t}^{\text{exp}}$, was obtained by applying the same calculation to the reversed

monthly network.

The role-specific PageRank measure was defined as

$$PR_{i,t}^{\text{role}} = \begin{cases} PR_{i,t}^{\text{exp}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} = 0, \\ PR_{i,t}^{\text{imp}}, & \text{if } I_{i,t} > 0 \text{ and } O_{i,t} = 0, \\ s_{i,t}^{\text{out}} PR_{i,t}^{\text{exp}} + \left(1 - s_{i,t}^{\text{out}}\right) PR_{i,t}^{\text{imp}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} > 0. \end{cases} \tag{13}$$

Trade dependence

For each directed terminal pair, the dependence of an importing terminal j on an exporter i was measured as the share of j's total imports supplied by i:

$$d_{ij,t} = \frac{q_{ij,t}}{I_{j,t}}. \tag{14}$$

To summarise the dependence generated by exporter i, this bilateral measure was weighted by the share of the exporter's flow directed towards each destination:

$$D_{i,t}^{\text{exp}} = \sum_{j} \left(\frac{q_{ij,t}}{O_{i,t}} \right) d_{ij,t}. \tag{15}$$

The resulting measure is high when an exporter directs a substantial share of its LNG towards terminals that depend strongly on its supply.

For an importing terminal i located in country $c(i)$, dependence was measured as its share of the total LNG volume received by all terminals in that country:

$$D_{i,t}^{\text{imp}} = \frac{I_{i,t}}{\sum_{k:\, c(k)=c(i)} I_{k,t}}. \tag{16}$$

The role-specific dependence measure was subsequently defined as

$$D_{i,t}^{\text{role}} = \begin{cases} D_{i,t}^{\text{exp}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} = 0, \\ D_{i,t}^{\text{imp}}, & \text{if } I_{i,t} > 0 \text{ and } O_{i,t} = 0, \\ s_{i,t}^{\text{out}} D_{i,t}^{\text{exp}} + \left(1 - s_{i,t}^{\text{out}}\right) D_{i,t}^{\text{imp}}, & \text{if } O_{i,t} > 0 \text{ and } I_{i,t} > 0. \end{cases} \tag{17}$$

Mean voyage distance

Mean voyage distance was calculated as the unweighted arithmetic mean of the reconstructed distances of voyages associated with each terminal. For an exporting terminal, this corresponds to

$$\overline{L}_{i,t}^{\text{out}} = \frac{1}{n_{i,t}^{\text{out}}} \sum_{r \in \mathcal{R}_{i,t}^{\text{out}}} L_r, \tag{18}$$

where $\mathcal{R}_{i,t}^{\text{out}}$ is the set of voyages departing from terminal i in month t, $n_{i,t}^{\text{out}}$ is the cardinality of this set, and L_r is the reconstructed distance of voyage r.

For an importing terminal, the corresponding measure was defined as

$$\overline{L}_{i,t}^{\text{in}} = \frac{1}{n_{i,t}^{\text{in}}} \sum_{r \in \mathcal{R}_{i,t}^{\text{in}}} L_r, \tag{19}$$

where $\mathcal{R}_{i,t}^{\text{in}}$ is the set of voyages arriving at terminal i in month t with cardinality $n_{i,t}^{in}$. The role-relevant mean voyage distance was subsequently lagged by one month before being included among the model covariates.

3.4.2 Chokepoint criticality measures

Chokepoint criticality was represented through weighted betweenness, PageRank, and the share of monthly network flow. Together, these measures capture complementary dimensions of a passage's importance, including its position along high-flow paths, its centrality in the direction of LNG movement, and the volume of LNG transported through it.

Weighted betweenness centrality

Betweenness centrality measures the extent to which a node lies on shortest paths between other nodes (Brandes, 2001). Since edge weights in the LNG network represent transported volumes rather than geographic or economic distances, they could not be used directly as path lengths. A distance measure was therefore assigned to

each directed edge as the inverse of its positive LNG flow:

$$\ell_{ij,t} = \frac{1}{w_{ij,t}}, \qquad w_{ij,t} > 0. \tag{20}$$

where $w_{ij,t}$ is the LNG volume transported from node i to node j during month t. Under this transformation, edges carrying larger LNG volumes are treated as stronger and therefore shorter connections.

Normalised weighted betweenness for node i was then calculated as

$$B_{i,t} = \frac{1}{(N_t - 1)(N_t - 2)} \sum_{\substack{j \neq i \neq k \\ j \neq k}} \frac{\sigma_{jk,t}(i)}{\sigma_{jk,t}}, \tag{21}$$

where $N_t = |V_t|$ is the number of active nodes in month t and $\sigma_{jk,t}$ in the denominator is the number of shortest directed paths from node j to node k, calculated using the edge distances defined in Equation 20. The term $\sigma_{jk,t}(i)$ denotes the number of these paths that pass through node i. A high value of $B_{i,t}$ indicates that a chokepoint lies on a large share of the shortest paths defined by the distribution of LNG flows and therefore occupies an important intermediary position in the monthly transportation network.

PageRank

PageRank provides a complementary measure of chokepoint importance by accounting for both the magnitude and direction of LNG flows and the importance of the nodes from which those flows originate. For node i, it is defined as

$$PR_{i,t} = \frac{1 - \alpha}{N_t} + \alpha \sum_{\substack{j \\ (j,i) \in E_t}} PR_{j,t} \frac{w_{ji,t}}{\sum_k w_{jk,t}}, \tag{22}$$

where $\alpha = 0.85$ is the damping factor.

Share of monthly network flow

For a terminal, monthly node throughput equals the sum of its incoming and outgoing flows. For a chokepoint, the same shipment appears on both an incoming and an outgoing segment as it passes through the node. Let $V^{\mathrm{CP}} \subseteq V$ denote the set of chokepoint nodes. For each $i \in V^{\mathrm{CP}}$, throughput was defined as half of its total incident flow:

$$F_{i,t} = \frac{1}{2}\left(\sum_{j} w_{ji,t} + \sum_{k} w_{ik,t}\right), \qquad i \in V^{\mathrm{CP}}. \tag{23}$$

The share of monthly network flow associated with node i was then calculated as

$$S_{i,t} = \frac{F_{i,t}}{\sum_{n \in V} F_{n,t}}, \qquad i \in V^{\mathrm{CP}}. \tag{24}$$

This measure represents the share of total node throughput accounted for by a chokepoint in a given month.

3.4.3 Common structural indicators

In addition to the node-type-specific criticality measures, a common set of structural indicators was calculated for both terminals and chokepoints. These measures describe whether a node or one of its incident connections contributes to maintaining the connectivity of the monthly transportation network.

The indicators were calculated separately for each month using the undirected projection U_t of the active directed network G_t. Edge direction was disregarded because articulation points and bridges concern the preservation of connectivity rather than the direction of LNG movement. Self-loops were excluded because they do not contribute to connectivity.

Articulation points

A node was classified as an articulation point when its removal increased the number of connected components in the monthly undirected network. The corresponding binary indicator was defined as

$$A_{i,t} = \mathbb{I}\left\{\kappa\left(U_t^{(-i)}\right) > \kappa\left(U_t\right)\right\}, \tag{25}$$

where $\kappa(U_t)$ denotes the number of connected components in U_t, and $U_t^{(-i)}$ is the graph obtained by removing node i and its incident edges.

A separate indicator identifies nodes directly connected to an articulation point:

$$CA_{i,t} = \mathbb{I}\left\{\exists\, j \in \mathcal{N}_{i,t} \text{ such that } A_{j,t} = 1\right\}, \tag{26}$$

where $\mathcal{N}_{i,t}$ is the set of immediate neighbours of node i in U_t. This distinction separates nodes whose own removal could disrupt network connectivity from nodes directly exposed to a structurally critical neighbour.

Global and local bridges

An edge was classified as a global bridge when its removal increased the number of connected components in the undirected network. Denoting the set of global bridges in month t by $\mathcal{B}_t^G$, the number of global bridges incident to node i was calculated as

$$GBC_{i,t} = \sum_{e \in \delta_t(i)} \mathbb{I}\left\{e \in \mathcal{B}_t^G\right\}, \tag{27}$$

where $\delta_t(i)$ is the set of edges incident to node i in month t. The corresponding binary indicator was defined as

$$GB_{i,t} = \mathbb{I}\left\{GBC_{i,t} > 0\right\}. \tag{28}$$

Local bridges were defined as edges whose endpoints had no common neighbour. Because every global bridge also satisfies this condition, global bridges were removed

from the local-bridge set so that the two measures represented distinct structural mechanisms. Denoting the resulting set of local-only bridges by $\mathcal{B}_t^L$, the corresponding node-level count was calculated as

$$LBC_{i,t} = \sum_{e \in \delta_t(i)} \mathbb{I}\left\{e \in \mathcal{B}_t^L\right\}. \tag{29}$$

To account for differences in node degree, the shares of incident edges classified as global and local-only bridges were also calculated:

$$\begin{aligned} GBS_{i,t} &= \frac{GBC_{i,t}}{d_{i,t}}, \\ LBS_{i,t} &= \frac{LBC_{i,t}}{d_{i,t}}, \end{aligned} \qquad d_{i,t} > 0, \tag{30}$$

where $d_{i,t}$ is the degree of node i in the monthly undirected network. Together, these measures distinguish connections whose removal would disconnect part of the network from connections that locally bridge otherwise weakly connected neighbourhoods.

3.5 Construction of the model-ready panels

Three model-ready panels were constructed from the monthly network data described above: a node activity panel, a terminal criticality panel, and a chokepoint criticality panel. Each panel has a balanced structure in which every unit is represented in every month from January 2020 to December 2024, including months with no observed activity. This preserves transitions between active and inactive states and provides a consistent temporal basis for the subsequent models.

For each node, one-month lags of activity, throughput, and voyage count were calculated from the corresponding values in the preceding month. Variables with strongly right-skewed distributions were also included in logarithmic form using the transformation $\log(1 + x)$, which retains zero-valued observations. Depending on the panel, these variables comprise lagged throughput, lagged voyage count, terminal processing capacity, and lagged mean voyage distance.

Given the temporal coverage of the data, lagged variables are unavailable for January 2020. The binary indicator `lag_available` identifies observations for which the required preceding-month information can be calculated.

Lagged neighbourhood activity was constructed using an undirected historical neighbourhood: for each month t, the neighbours of a node are those connected to it by at least one directed edge observed up to and including month $t-1$. The resulting variables record the number of historical neighbours, the number that were active in $t-1$, and the corresponding active share. Historical neighbourhoods were used because defining a node's neighbourhood exclusively from edges observed in $t-1$ would make every identified neighbour active by construction, causing the active-neighbour share to be mechanically equal to one. The cumulative definition instead retains previously observed neighbours regardless of whether they were active in $t-1$, thereby capturing variation in the recent activity of a node's established connections. Connections first observed after $t-1$ are excluded to prevent the use of future information: neighbourhood-activity measures remain unavailable until a node has acquired at least one historical neighbour.

Finally, the common set of model covariates included seasonal terms, represented through sine and cosine transformations of the calendar month. This cyclical representation allows December and January to be treated as adjacent rather than as the endpoints of a linear sequence.

A complete description of the variables included in each model-ready panel, together with their transformations and availability, is provided in Appendix C.

3.5.1 Node activity panel

The node activity panel contains all 183 network nodes, comprising 159 LNG terminals and 24 maritime chokepoints, observed over 60 months, resulting in $183 \times 60 = 10{,}980$ node–month observations.

The response variable is the binary indicator of current node activity. The model

covariates comprise preceding-month activity, throughput, voyage count, historical-neighbour activity, and the seasonal terms defined above. Time-invariant or slowly varying node characteristics were obtained from the consolidated node inventory and include node type and, where applicable, country, infrastructure type, start year, and annual processing capacity. Months without observed participation were retained with `active` equal to zero and the corresponding flow and voyage measures set to zero.

3.5.2 Terminal criticality panel

The terminal criticality panel is restricted to the 159 LNG terminals, producing $159 \times 60 = 9{,}540$ terminal–month observations.

Its outcomes comprise the terminal criticality measures defined in Section 3.4.1, namely role-specific PageRank, role-specific counterparty concentration at the terminal and country levels, and role-specific dependence. They also include the common structural indicators defined in Section 3.4.3, which identify whether a terminal is an articulation point, is incident to a global bridge, or is directly connected to an articulation point.

Criticality measures based on realised commercial activity are defined only for active terminal–months and remain unavailable for inactive observations. Structural indicators, by contrast, are retained for every terminal–month and are equal to zero when the terminal does not belong to the active monthly network.

The terminal panel additionally includes the preceding-month terminal role and role-relevant mean voyage distance, together with country, infrastructure type, annual processing capacity, number of infrastructure units, start year, and terminal age. Continuous covariates are retained in both their original and, where appropriate, log-transformed forms.

3.5.3 Chokepoint criticality panel

The chokepoint criticality panel contains the 24 chokepoints included in the final network over the same 60-month period, yielding $24 \times 60 = 1{,}440$ chokepoint–month

observations.

The outcomes comprise weighted betweenness centrality, PageRank, and the share of monthly network flow defined in Section 3.4.2, together with the common structural indicators introduced in Section 3.4.3. Additional measures of degree, route exposure, and voyage traversals were retained as supplementary network characteristics.

Finally, the panel includes the common lagged and seasonal covariates described above, together with lagged mean voyage distance for voyages traversing each chokepoint.

3.6 Data and methodological limitations

The construction of the LNG maritime network is subject to limitations arising both from the underlying LNG-T3 data and from the procedures used to reconstruct routes and derive network measures.

As discussed by Zhou et al. (2026), LNG-T3 relies extensively on Automatic Identification System messages, which are self-reported by vessels and may contain errors, delays, or incomplete information. AIS coverage also varies geographically according to the availability and density of receiver stations. Coverage is generally more extensive in Europe and North America than in regions such as Southeast Asia, where archipelagic geography and a lower density of receivers may generate spatial blind spots. These differences can introduce geographic variation in data quality.

Incomplete AIS trajectories and missing port-call records may result in partially reconstructed or unobserved voyages, while inaccurate draught reporting may affect the classification of whether a tanker is loaded. To mitigate these sources of uncertainty, LNG-T3 adopts a conservative validation procedure and retains only voyages considered suitable for estimating trade flows. Although this improves the reliability of the individual records, it may result in a systematic underestimation of aggregate LNG trade relative to industry and government statistics (Zhou et al., 2026).

Additional uncertainty concerns the estimation of cargo volumes. LNG-T3 assumes that tankers depart fully loaded, which provides a consistent basis for global

aggregation but does not capture possible partial loadings associated with operational constraints, contractual arrangements, or market conditions. Estimated volumes may therefore overstate the cargo transported on some individual voyages. At the aggregate level, however, this potential upward bias coexists with the potential downward bias arising from incomplete voyage coverage, and their combined effect cannot be determined unambiguously.

A further limitation arises from the reconstruction of maritime routes, as the available voyage records identify departure and arrival terminals but do not contain the complete trajectories followed by individual tankers. The routes used in this study, generated using `SeaRoute`, are modelled approximations rather than documented voyage trajectories and may not reproduce routing decisions influenced by weather, congestion, regulatory restrictions, security conditions, canal availability, or commercial considerations.

Small spatial discrepancies between the reconstructed paths and the operational chokepoint geometries were addressed through a uniform 25-km tolerance. However, the sensitivity analysis reported in Appendix B shows that the assignments for Korea Strait and Mindoro Strait are particularly dependent on the selected tolerance. Chokepoint exposure should therefore be interpreted as an approximation derived from reconstructed routes rather than direct evidence that individual vessels crossed each passage.

As a final clarification of scope, the resulting network represents physical LNG flows observed during 2020–2024 and the transportation paths reconstructed for those flows. Accordingly, the resulting measures characterise realised network activity and estimated route exposure within the study period.

Chapter 4

Exploratory network analysis

This chapter examines the structure and evolution of the reconstructed LNG maritime network from January 2020 to December 2024, by describing changes in monthly trade flows and connectivity, patterns of node activity, and the network positions of terminals and chokepoints using the measures introduced in Chapter 3. The analysis provides a descriptive overview of the main temporal and cross-sectional patterns in the data and examines the relationships among the candidate variables, to identify relevant patterns, potential redundancy, and shared variation that may inform the subsequent modelling analysis.

To address these objectives, the chapter is organised as follows. Section 4.1 examines the temporal evolution of the monthly LNG network; Section 4.2 then describes node activity and temporal sparsity over the observation period; Section 4.3 focuses on terminals, considering their observed trade roles and the measures of centrality, concentration, and dependence associated with their network positions; Section 4.4 examines chokepoints through their flow exposure, PageRank, and weighted betweenness; Section 4.5 compares the structural positions of terminals and chokepoints using articulation-point and bridge-based measures; finally, Section 4.6 examines associations among the candidate variables and discusses their implications for the subsequent modelling analysis.

As described in Section 3.2.2, the node–month and terminal–month datasets retain

both active and inactive observations. In this chapter, summary statistics for network-position measures are generally calculated over active observations, while analyses of node activity and temporal sparsity retain inactive months. The relevant sample is specified in the text or table notes whenever necessary.

4.1 Network evolution

The observed LNG maritime network changes in size over the study period. The numbers of active nodes and directed edges generally increase from 2021 to late 2023, reaching their respective maxima in December 2023, before declining during the second half of 2024. The ranges and averages of these network characteristics, together with monthly trade activity and directed density, are summarised in Table 2.

Despite this evolution in network size, directed density remains within a relatively narrow range, ranging from 0.0122 to 0.0180. The monthly LNG network is therefore consistently sparse: only a small fraction of all possible directed connections among active nodes is observed in any given month.

Directed density is calculated among active nodes, so changes in its denominator must be considered when interpreting the results: a month with fewer edges can still have higher density if fewer nodes are active. Network expansion or contraction should therefore not be interpreted as a corresponding change in the degree of connectivity among active nodes.

Table 2: Selected monthly network and node-activity summaries, 2020–2024.

Measure	Minimum	Mean	Maximum
Active nodes	72	96.5	126
Active directed edges	88	136.5	192
Export voyages	70	144.0	237
Exported volume (MCM)	11.48	24.01	38.77
Directed density among active nodes	0.0122	0.0148	0.0180

Note: Exported volume is expressed in million cubic metres (MCM).

While Table 2 summarises the monthly networks quantitatively, Figure 2 illustrates the geographic structure of a representative month. June 2021 was selected as an

illustrative snapshot because its 98 active nodes and 138 directed edges are close to the corresponding monthly averages reported in Table 2.

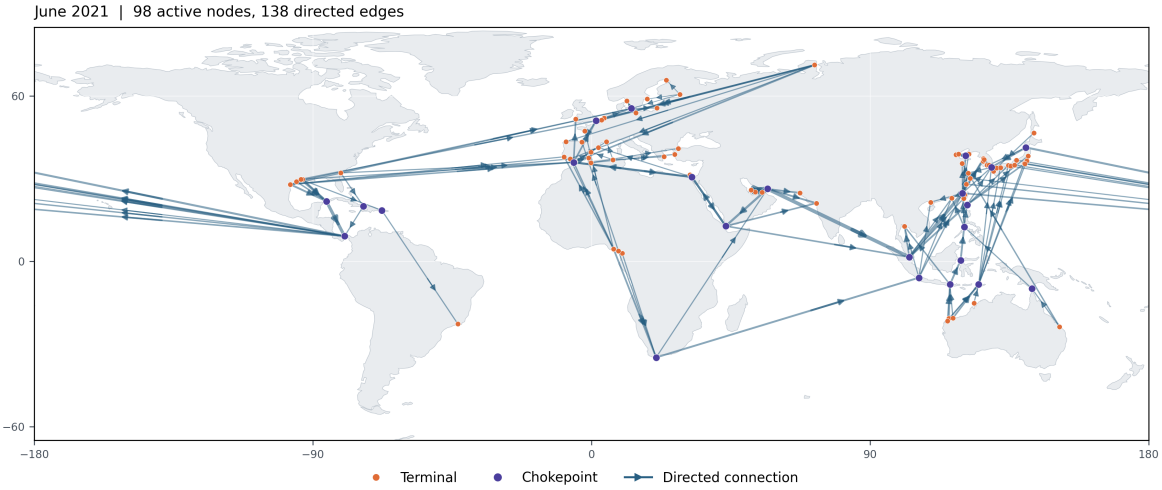


Figure 2: Geographic snapshot of the observed LNG network in June 2021.

Note: Points show active terminals and chokepoints; arrows indicate the direction of observed network connections. Line width represents monthly LNG volume on each directed edge; connections are schematic rather than sailed routes.

To examine how the network's size and trade volume evolve over time, Figure 3 compares the monthly evolution of active nodes, directed edges, and observed LNG export volume, with each series indexed to its mean over the study period.

The three measures show broadly similar changes over time, but export volume exhibits substantially greater month-to-month variation, showing that changes in network size are not necessarily proportional to changes in the trade volume carried by the network.

4.2 Node activity and temporal sparsity

Using the definition of node activity introduced in Section 3.3.2, of 10,980 potential node-month observations, 5,792 (52.75%) are active and 5,188 (47.25%) inactive.

As shown in Figure 3, the number of active nodes generally increases through 2023 and declines towards the end of 2024. Correspondingly, the monthly share of inactive nodes ranges from 31.15% to 60.66% (Figure 4).

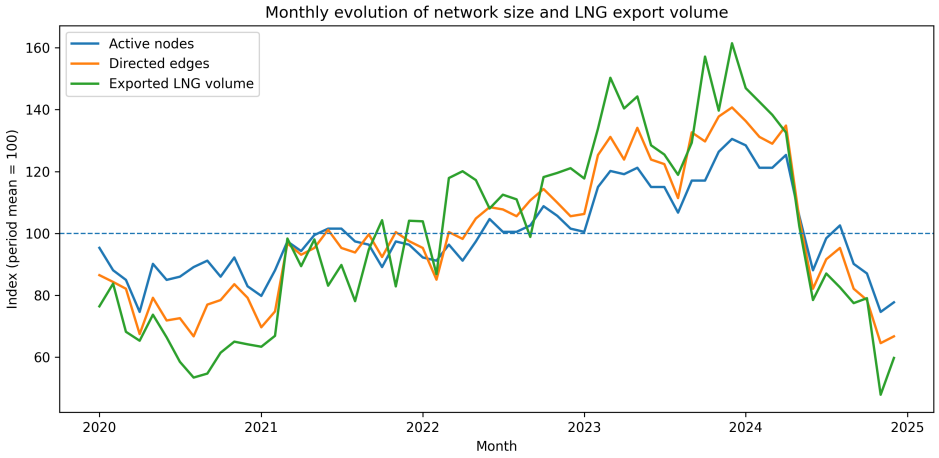


Figure 3: Monthly evolution of network size and observed LNG export volume, 2020–2024.

Note: Each series is indexed to its monthly mean over the 2020–2024 period, which is set equal to 100. Export volume is calculated from the observed export voyages.

Terminals account for 9,540 node–month observations, of which 4,552 are active and 4,988 inactive (52.29%). The 24 chokepoints account for 1,440 observations, of which 1,240 are active and 200 inactive (13.89%). This difference is consistent with the nodes' distinct roles in the network: a terminal participates as an origin or destination of a voyage, whereas a chokepoint can be traversed by voyages linking many different terminal pairs. Aggregate inactivity is therefore primarily attributable to terminals, although chokepoint activity is not constant. These different base rates suggest that activity may need to be examined separately by node type.

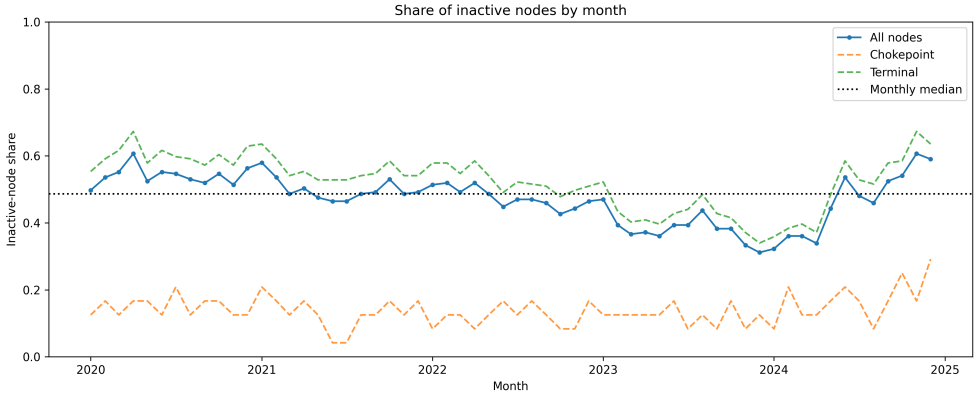


Figure 4: Monthly inactive-node shares for all nodes and by node type.

Inactivity is substantively relevant to the interpretation of centrality: an inactive node may receive a zero in the balanced node-month dataset even though the criticality measure is chiefly meaningful when it participates in trade. For this reason, the subsequent analysis of node criticality focuses on active node-month observations, avoiding the conflation of inactivity with low criticality.

4.3 Terminal network position

This section examines how terminals participate in observed LNG trade and how their positions differ across the monthly networks. It first describes terminal roles and trade activity, then considers centrality, counterparty concentration, and dependence.

4.3.1 Terminal roles and trade activity

Terminal participation varies considerably across the 60 monthly networks: of the 4,552 active terminal–month observations, 3,234 are classified as importers and 1,318 as exporters.

Country-level roles were assessed over the full 2020–2024 period using each country's net trade position. Qatar, the United States, and Russia recorded the largest cumulative export volumes, while China, France, and Japan recorded the largest cumulative import volumes. A net importer or exporter role does not imply that a country trades exclusively in one direction. For instance, Malaysia was a net importer despite recording exports, and the United States was a net exporter despite recording imports.

Table 3 compares throughput, voyage counts, and the numbers of terminal counterparties for active terminal–month observations, distinguished by importer and exporter role.

Exporter observations are characterised by higher levels of trade activity across all three measures. Median monthly throughput is 0.505 million cubic metres for exporters, compared with 0.328 million cubic metres for importers, while the corresponding median numbers of voyages are four and two. Exporters also interact with

Table 3: Trade activity of active terminal–month observations by terminal role, 2020–2024.

	Importer ($n = 3{,}234$)		Exporter ($n = 1{,}318$)	
Measure	Median	95th percentile	Median	95th percentile
Throughput (million cubic metres)	0.328	1.226	0.505	3.792
Voyages	2	7	4	22
Terminal counterparties	1	4	3	14

The analysis is restricted to active terminal–month observations. Throughput values are rounded to three decimal places.

a larger number of terminal counterparties, with a median of three compared with one for importers. These differences are even more pronounced at the 95th percentile, particularly for voyage counts and numbers of counterparties.

More generally, throughput is strongly right-skewed across active terminal–month observations. Its overall median is 346,000 cubic metres, and the 95th percentile is approximately 1.90 million cubic metres, while Rasgas LNG Terminal 3 reaches the observed maximum of 10.91 million cubic metres in October 2023.

4.3.2 Centrality, concentration and dependence

The terminal measures discussed below were defined in Section 3.4.1. Here, their distributions are examined across active terminal–month observations to explore different aspects of terminals' positions and trade relationships in the observed LNG network. These pooled distributions are complemented by terminal-level summaries of the monthly measures over the entire period, which distinguish the positions of individual terminals and help assess whether high scores are sustained over time or occur only occasionally.

First, role-specific PageRank has a median of 0.0142 and a 95th percentile of 0.0363. The 95th percentile is approximately 2.6 times the median, indicating substantial variation in the scores assigned to active terminals. The highly right-skewed distribution suggests that a relatively small number of terminal–month observations occupy particularly prominent positions within the monthly trade network, while most receive

lower scores.

These scores should nevertheless be interpreted as relative measures within each monthly network. Because PageRank is normalised separately each month, its values are influenced by both the distribution of trade relationships and the number of active terminals. Comparisons across months therefore describe changes in relative network prominence rather than changes in an absolute measure of importance.

To examine terminal positions over the full period, two summaries are considered: mean PageRank conditional on activity and mean PageRank across all 60 months, assigning zero to inactive months. The first describes a terminal's average prominence when participating in the network; the second combines that prominence with its frequency of participation.

Table 4 reports the three highest full-period mean scores within each terminal role. Rasgas LNG Terminal 3 has the highest mean PageRank among exporters, at 0.0912, followed by Sabine Pass and Yamal, at 0.0449 and 0.0418, respectively. All three terminals are active in every month, so their mean scores reflect sustained network participation over the full observation period.

Among importers, Gate LNG Terminal has the highest full-period mean, at 0.0195, followed by Yung-An and Fos Tonkin. Fos Tonkin has a slightly higher mean during active months than Yung-An, but its lower number of active months reverses their ordering when inactivity is included. The distinction is more pronounced for terminals observed only occasionally: Sepetiba Bay FSRU has an active-month mean of 0.0191, close to Gate's, but is active in only one month. Consequently, its mean PageRank across the full panel is approximately 0.00032, highlighting how a high conditional mean alone should therefore not be interpreted as evidence of sustained prominence.

PageRank is relevant to the assessment of terminal prominence, but does not establish how easily a terminal could be replaced. A well-positioned terminal may trade with partners that have alternative sources or destinations, whereas a less prominent terminal may serve counterparties that rely heavily on its supply or handle a substantial share of its country's LNG receipts. Concentration and dependence are therefore

Table 4: Terminals with the highest mean role-specific PageRank across all 60 months, ranked separately by observed role.

Terminal	Country	Active months	Mean PageRank	
			Active months	All months
Exporters				
Rasgas LNG Terminal 3	Qatar	60	0.0912	0.0912
Sabine Pass LNG Terminal	United States	60	0.0449	0.0449
Yamal LNG Terminal	Russia	60	0.0418	0.0418
Importers				
Gate LNG Terminal	Netherlands	60	0.0195	0.0195
Yung-An LNG Terminal	Taiwan	57	0.0182	0.0172
Fos Tonkin LNG Terminal	France	55	0.0187	0.0171

Note: Arithmetic means of monthly PageRank scores. Active-month means exclude inactivity; all-month means assign zero to inactive months.

examined alongside PageRank to assess the trade relationships underlying a terminal's network position.

The counterparty concentration measures show substantial variation among active terminal–month observations. The median HHI is 0.556 when counterparties are individual terminals and 0.674 when they are aggregated by country. These median values correspond to approximately 1.8 and 1.5 equally weighted effective counterparties at the terminal and country levels, respectively. Thus, even when a terminal trades with several partners, its observed monthly flow may be concentrated among relatively few of them.

An HHI of one occurs in 43.5% of active terminal–month observations at the terminal level, indicating that the role-relevant flow involves a single terminal counterparty. At the country level, the corresponding proportion is 47.4%, indicating concentration in a single country, potentially across multiple terminals.

Terminal-level averages reveal different patterns behind these pooled results. As shown in Table 5, Rasgas and Sabine Pass have relatively low mean terminal-level HHIs, at 0.094 and 0.125, respectively, despite their prominent PageRank positions.

Their observed monthly exports are therefore distributed more broadly across counterparties than the pooled distribution might suggest. In contrast, Cameroon FLNG has a mean terminal-level HHI of 0.953 across 32 active months, indicating highly concentrated shipments during its observed activity.

Country-level concentration also distinguishes diversification across terminals from diversification across countries. For instance, the mean HHI of Gate increases from 0.362 at the terminal level to 0.531 at the country level, showing that some diversification across supplying terminals occurs within the same countries.

These findings suggest that an interruption involving a major partner could affect a substantial share of some terminals' existing trade. The measures nevertheless describe observed monthly relationships and do not establish whether alternative suppliers or destinations would be available. Moreover, the terminal-level averages summarise monthly concentration: they do not imply that a terminal trades with the same counterparties throughout the period.

Table 5: Mean counterparty concentration and role-specific dependence during active months for selected terminals, 2020–2024.

Terminal	Active months	Mean HHI		Mean dependence
		Terminal	Country	
Exporters				
Rasgas LNG Terminal 3	60	0.094	0.139	0.730
Sabine Pass LNG Terminal	60	0.125	0.179	0.508
Cameroon FLNG Terminal	32	0.953	0.969	0.350
Importers				
Gate LNG Terminal	60	0.362	0.531	0.910
Al Zour LNG Import Terminal	40	0.652	0.657	1.000
Zeebrugge LNG Terminal	59	0.558	0.599	1.000

Note: Terminals are selected to illustrate contrasting relationships, rather than to provide a ranking. All values are unweighted arithmetic means across active months.

Finally, role-specific dependence has a median of 0.406 across active terminal–month observations, but this pooled value combines different interpretations for the two terminal roles. For exporters, which have a median of 0.495, higher values indicate

that a larger share of their flow reaches importing terminals that rely heavily on its supply. An exporter can therefore have high dependence even when its shipments are distributed across several destinations, provided that it supplies a substantial share of those destinations' imports. Rasgas illustrates this combination: its low mean terminal-level HHI of 0.094 coexists with mean dependence of 0.730. Its exports are relatively diversified, yet its receiving terminals tend to rely substantially on its supply when weighted by their shares of its shipments. Cameroon FLNG illustrates the opposite pattern: its highly concentrated exports coexist with lower mean dependence of 0.350. Concentration in an exporter's own destinations therefore does not necessarily imply equally strong reliance by those destinations on its supply.

Among active importer months, the median dependence is 0.333. Here, higher values indicate that the terminal accounts for a larger share of its country's observed LNG receipts. A high value therefore identifies a terminal through which a substantial proportion of national LNG imports passes, making its continued operation potentially relevant to the country's ability to receive LNG.

The terminal-level summaries illustrate the extent of this concentration at individual terminals. Gate accounts for an average of 91.0% of the Netherlands' observed monthly LNG receipts across its 60 active months, while Al Zour and Zeebrugge have dependence equal to one in all their active months, numbering 40 and 59, respectively. In those months, all observed LNG receipts in their respective countries pass through these terminals.

The crucial distinction from the counterparty HHI measure described above lies in whose reliance is being described: HHI quantifies the concentration of a terminal's own trade across its partners, whereas role-specific dependence describes its counterparties' reliance on its supply or its country's reliance on that terminal for observed LNG receipts. An importer may receive all its LNG from one exporter, giving it an HHI of one, while handling only a small share of its country's total imports and therefore having low role-specific dependence. Conversely, a terminal may receive LNG from several suppliers while accounting for a large share of national LNG receipts. Gate

illustrates the latter pattern, combining a mean terminal-level HHI of 0.362 with mean dependence of 0.910.

Concentration and dependence are therefore retained alongside PageRank to capture distinct dimensions of terminal criticality: partner concentration, reliance on individual terminals, and prominence within the trade network. Together, these measures characterise terminal criticality without directly quantifying the consequences of a disruption.

4.4 Chokepoint network position

Unlike terminals, which serve as the origins or destinations of LNG voyages, chokepoints act as intermediate nodes along maritime routes. Their position can therefore be examined in terms of both exposure to LNG flows and location within the wider network, captured by PageRank and betweenness.

Table 6 provides summary statistics for the main flow-exposure and centrality measures considered in this section.

Table 6: Summary statistics for flow-exposure and centrality measures across active chokepoint–month observations, 2020–2024.

Measure	Mean	Median	95th percentile	Maximum
Throughput (MCM)	2.007	1.342	5.857	10.702
Voyages	11.08	8	31	55
Route exposure	8.30	7	21	31
Share of monthly network flow	0.0225	0.0148	0.0639	0.1132
PageRank	0.0172	0.0163	0.0299	0.0572
Betweenness	0.0126	0.0072	0.0408	0.0946

Note: Statistics are calculated across the 1,240 active chokepoint–month observations. Throughput is reported in million cubic metres.

4.4.1 Flow exposure

Flow exposure varies substantially across the 24 chokepoints and over months. As shown in Table 6, the distribution of monthly throughput is strongly right-skewed,

with the observed maximum of 10.70 million cubic metres recorded at the Strait of Hormuz in April 2022.

The same heterogeneity appears when flow is expressed relative to the total flow observed in the monthly network. Among active observations, the share of monthly network flow passing through a chokepoint has a median of approximately 1.48%, while the 95th percentile reaches 6.39%. The maximum is recorded at the Strait of Hormuz in June 2020, when approximately 11.32% of the month's observed LNG trade passed through the chokepoint.

The flow-share measure complements the absolute volume of LNG passing through a chokepoint. Because monthly LNG trade varies over the observation period, a given throughput may represent different levels of exposure depending on the total volume traded in that month. Expressing chokepoint throughput as a share of total monthly network flow therefore provides a relative measure of how much of the observed network flow is routed through each passage.

Beyond flow volume, voyage and route counts provide additional information on the frequency and variety of chokepoint use. Active chokepoints are traversed by a median of eight voyages per month and are exposed to a median of seven distinct routes, while the corresponding 95th percentiles are 31 voyages and 21 routes.

4.4.2 Centrality measures

Flow exposure alone does not describe the structural location of a chokepoint within the reconstructed network. Two passages carrying similar LNG volumes may connect different parts of the network and, consequently, differ in their centrality. PageRank and betweenness are therefore considered alongside flow exposure to capture these complementary aspects of chokepoint prominence.

As defined in Section 3.4.2, PageRank accounts for both the magnitude and direction of LNG flows and the importance of the nodes from which those flows originate.

It therefore complements flow exposure by capturing a chokepoint's relative position within each monthly network rather than simply the amount of LNG passing

through it. The distribution of raw PageRank values indicated relatively prominent positions for some chokepoints, with a median of 0.0163 among active observations and values reaching 0.0572. This motivated a closer examination of their magnitude relative to the size of each monthly network. Observed PageRank was therefore compared with a uniform benchmark of $1/N_t$, corresponding to an equal distribution of PageRank across the N_t active nodes.

Among active observations, the median PageRank was 1.54 times this monthly benchmark, and 85.6% of observations exceeded it. Active chokepoints therefore tended to receive a larger share of PageRank than would result from an equal distribution across active nodes.

While PageRank captures relative prominence within the flow network, betweenness focuses on a different aspect of centrality by indicating the extent to which a chokepoint lies on shortest paths connecting other nodes. A chokepoint may therefore receive a relatively high PageRank without occupying a particularly strong intermediary position, while another may have a lower PageRank but lie on a comparatively large number of shortest paths.

As shown in Table 6, its distribution is strongly right-skewed, with high levels of shortest-path intermediation concentrated in a comparatively small subset of active chokepoint–month observations.

Notably, positive activity does not necessarily imply positive betweenness: betweenness is equal to zero in 8 of the 1,240 active chokepoint–month observations. A chokepoint may therefore be traversed by observed LNG routes without lying on any of the shortest paths between other node pairs captured by this measure.

The distinction between the three measures is also visible when their average values are compared across individual chokepoints. Table 7 reports four passages selected to illustrate contrasting combinations of flow exposure, PageRank, and betweenness. All four are active throughout the 60-month observation period, so differences in their averages are not driven by different frequencies of activity.

The Strait of Hormuz has the largest mean flow share among the selected passages,

Table 7: Selected chokepoints illustrating different dimensions of network position, 2020–2024.

Chokepoint	Active months	Mean flow share	Mean PageRank	Mean betweenness
Strait of Hormuz	60	0.0766	0.0182	0.0181
Gibraltar Strait	60	0.0438	0.0263	0.0360
Taiwan Strait	60	0.0364	0.0309	0.0287
Suez Canal	60	0.0327	0.0211	0.0301

Note: Chokepoints are selected to illustrate contrasting relationships among the three measures rather than to provide a ranking. Means are calculated over active months.

at 7.66%, but substantially lower mean PageRank and betweenness than some passages carrying smaller shares of monthly network flow. The Taiwan Strait, by contrast, has a mean flow share of 3.64% but the highest mean PageRank of the four, at 0.0309. Gibraltar Strait combines a lower mean flow share than Hormuz, at 4.38%, with the highest mean betweenness, at 0.0360, while the Suez Canal also exhibits relatively high betweenness despite accounting for a mean 3.27% of monthly network flow.

These comparisons show that greater flow exposure does not necessarily correspond to higher PageRank or betweenness, highlighting the distinct dimensions of chokepoint criticality captured by these measures.

4.5 Structural position of nodes

The structural indicators defined in Section 3.4.3 complement the node-type-specific measures by describing connectivity in the monthly undirected projection of the reconstructed network. They distinguish nodes whose removal would disconnect parts of the network from nodes that are incident to structurally important edges or adjacent to articulation points. Rather than reflecting the volume of LNG transported through individual connections, these indicators depend only on their presence in the monthly network.

The distributions differ substantially between terminals and chokepoints. Articulation status occurs in 393 of the 4,552 active terminal–month observations (8.6%),

compared with 902 of the 1,240 active chokepoint–month observations (72.7%). Incidence to at least one global bridge is common in both groups, occurring in 64.7% of active terminal months and 72.5% of active chokepoint months. Local-only bridge incidence is more frequent among chokepoints, at 82.8%, than among terminals, at 26.1%.

These differences are consistent with the roles of terminals and chokepoints in the network: terminals are voyage origins or destinations, whereas chokepoints are intermediate nodes connecting successive route segments. Chokepoints consequently have more opportunities to connect branches of the observed network, although their classifications depend on the routes used in each month.

The analysis of articulation points further illustrates this distinction. Among terminals, Vysotsk repeatedly acts as an articulation point, being classified as an articulation point in 49 of its 59 active months. Among chokepoints, Gibraltar Strait, Taiwan Strait and the Strait of Hormuz are articulation points in all 60 months. Removing the node therefore disconnects the monthly undirected network in every observed month.

Table 8: Frequency of structural classifications for selected terminals and chokepoints, 2020–2024.

Node	Activemonths	Articulationpoint	Globalbridge	Local-onlybridge
Terminals				
Vysotsk LNG Terminal	59	49	58	0
Dunkirk LNG Terminal	56	0	56	0
Nigeria LNG Terminal	59	21	21	59
Sabine Pass LNG Terminal	60	24	25	58
Chokepoints				
Gibraltar Strait	60	60	60	58
Strait of Hormuz	60	60	58	36
Malacca Strait	60	54	54	60
Suez Canal	60	46	46	59

Note: Entries report the number of active months in which each classification applies. Global and local-only bridge columns indicate incidence to at least one edge of the corresponding type. Classifications can coexist at a node, although global bridges are excluded from the local-only edge set. Examples illustrate contrasting patterns rather than a ranking.

Since a global bridge is an edge whose removal separates its endpoints into different

connected components, incidence to such an edge can indicate a connection between groups of nodes, but it can also describe the attachment of a peripheral node to the rest of the network.

Dunkirk LNG Terminal provides a clear example of the latter case. In all its 56 active months, it has one neighbour in the undirected transport network and its sole incident edge is a global bridge. It is nevertheless never an articulation point: removing Dunkirk does not split the remaining network, whereas removing its incident edge isolates Dunkirk. Its classification therefore describes peripheral attachment rather than a role in connecting other nodes.

Conversely, a node can be an articulation point without being incident to a global bridge. The Strait of Hormuz is an articulation point in all 60 months but is incident to a global bridge in only 58. In the remaining two months, removing the node separates the network even though removing any individual incident edge does not. This distinction illustrates why node-removal and edge-removal indicators provide different information.

Local-only bridges identify another structural feature. Their endpoints have no common neighbour, but removing the edge does not disconnect the component because a longer alternative path remains. They therefore indicate the absence of an immediate two-edge alternative through a shared neighbour, rather than an edge that is indispensable to overall connectivity.

Nigeria LNG Terminal is incident to at least one local-only bridge in all its 59 active months, and Sabine Pass in 58 of 60. Among chokepoints, Malacca Strait has local-only bridge incidence in every month, while Suez Canal has it in 59 of 60 months (Table 8).

These classifications can coexist with articulation status or global bridge incidence because a node may have several incident edges with different structural properties. For instance, Malacca Strait is an articulation point and is incident to a global bridge in 54 months, while also being incident to a local-only bridge in all 60 months. This combination indicates that some connections have longer alternative paths, whereas

other connections, or the node as a whole, maintain connectivity between otherwise separated parts of the network. It is to be noted that the alternative paths identified in the undirected projection are not necessarily feasible substitutes for the original LNG movements: these indicators do not account for direction, navigability or transport capacity.

These results distinguish the frequent intermediate connectivity roles of chokepoints from the more varied structural positions of terminals.

4.6 Associations and modelling implications

Spearman rank correlations provide a preliminary assessment of the overlap among variables considered in the subsequent modelling analysis.

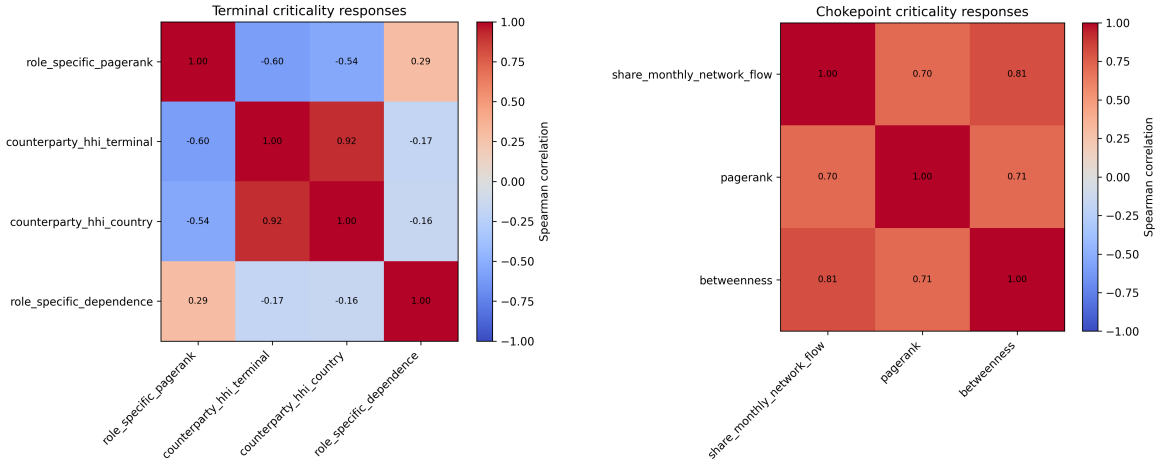
Among the terminal covariates, lagged log-transformed throughput is strongly associated with lagged log-transformed voyage count ($\rho_S = 0.989$), whereas its associations with other terminal characteristics are substantially weaker. Its correlation with lagged log-transformed mean voyage distance is 0.223 when the latter is defined, while the corresponding correlations with terminal capacity and terminal age are 0.423 and 0.199, respectively. Neighbour activity is also only weakly associated with these node-specific characteristics. These patterns suggest that, beyond the strong association between throughput and voyage count, the candidate covariates capture aspects of terminal activity that cannot be explained solely by operational scale.

A similar pattern is observed among the chokepoint covariates. Lagged log-transformed throughput is strongly associated with lagged log-transformed voyage count ($\rho_S = 0.992$), whereas its associations with the lagged share of active neighbours ($\rho_S = 0.118$) and lagged log-transformed mean voyage distance ($\rho_S = -0.057$) are substantially weaker. The seasonal terms also show little association with lagged throughput. Beyond the close correspondence between throughput and voyage count, the chokepoint covariates therefore capture distinct aspects of activity, route characteristics, and temporal variation.

The associations among the criticality measures differ between terminals and chokepoints. Among active terminal–month observations, terminal- and country-level counterparty HHIs are strongly correlated ($\rho_S = 0.920$), reflecting their related but distinct measures of trade concentration. Their associations with role-specific PageRank are negative and more moderate in magnitude (-0.596 and -0.545, respectively), while role-specific dependence is more weakly associated with the other terminal measures.

For active chokepoint–month observations, the three continuous measures show stronger positive associations. Flow share correlates with weighted betweenness at 0.810 and with PageRank at 0.704, while the correlation between PageRank and betweenness is 0.706. Thus, although the individual comparisons in Section 4.4 show that greater flow exposure does not necessarily correspond to higher centrality, the aggregate associations indicate that chokepoints carrying larger shares of observed LNG flows tend to have higher PageRank and betweenness values. The correlations remain below one, consistent with the measures capturing related but distinct dimensions of chokepoint criticality.

Figure 5 summarises the pairwise associations among the terminal and chokepoint criticality measures. Taken together, these patterns provide descriptive evidence of shared variation across multiple dimensions of node criticality while preserving meaningful differences between the individual measures. These findings provide the empirical motivation for investigating whether the measures' joint variation can be represented by a lower-dimensional structure in the modelling framework developed in Chapter 5.



(a) Terminal criticality measures

(b) Chokepoint criticality measures

Figure 5: Spearman rank correlations among the criticality measures for active node–month observations. Panel (a) reports correlations among terminal measures, while panel (b) reports correlations among chokepoint measures.

Chapter 5

Bayesian Reduced-Rank model for network measures

This chapter develops a Bayesian reduced-rank framework to investigate common patterns across multiple measures of node network position and their associations with node characteristics and lagged network conditions.

Separate rank-one latent factor models are estimated for terminals and chokepoints, reflecting their different functions and response measures within the LNG maritime network. Each model represents its own set of responses variables, through a shared latent factor, related to the corresponding set of covariates.

The analysis focuses on node-month observations with consecutive observed activity, using a static specification to examine associations between network position and the selected covariates. The resulting latent factors provide model-specific summaries of network position, whose interpretation and adequacy are evaluated through posterior inference, predictive checks and a prior sensitivity analysis.

5.1 General modelling framework

5.1.1 Estimation samples, responses and covariates

The terminal model-ready panel contains 9,540 observations for 159 terminals from January 2020 to December 2024.

For the current specification, an observation is retained when the terminal is active in month t, a lag from the preceding calendar month is available, and the terminal

was also active in that month. This restriction allows the analysis to focus on network position among consecutively active terminals, using information from the preceding month without modelling transitions between active and inactive states.

Excluding 4,988 inactive observations, 71 active observations without an available calendar lag, and 916 observations preceded by inactivity results in 3,565 observations active in consecutive months. A further 112 observations with missing terminal age are excluded, resulting in an estimation sample of size 3,453 data points for 137 terminals. Of these observations, 2,325 have lagged importer status and 1,128 have lagged exporter status.

The terminal model includes six responses: three bounded measures (role-specific PageRank, terminal counterparty HHI and role-specific dependence) and three binary indicators (articulation-point status, incident-global-bridge status and connection to an articulation point). These measures are expected to capture different aspects of network position and need not all increase with the same latent dimension.

Seven predictors are included: lagged historical-neighbour activity share, log-transformed lagged mean voyage distance, log-transformed lagged terminal capacity, monthly sine and cosine terms, terminal age, and an indicator for exporter status in the preceding month. Numeric covariates are standardised, while the exporter indicator remains on its zero–one scale. Accordingly, coefficients for numeric predictors correspond to one-standard-deviation differences, whereas the role coefficient compares lagged exporters with lagged importers, holding other covariates fixed.

The bounded responses are modelled on their original scales. PageRank lies strictly within $(0, 1)$, whereas HHI and dependence take values in $(0, 1]$. Neither HHI nor dependence has observed zeros, but HHI equals one in 34.09% of observations and dependence equals one in 20.76%. These features motivate a Beta likelihood for PageRank and one-inflated Beta likelihoods for HHI and dependence, as specified in Section 5.1.2.

The chokepoint panel contains 1,440 observations for 24 nodes. Applying the same calendar-lag and consecutive-activity criteria excludes 200 currently inactive observations, 21 active observations without an available calendar lag, and 56 observations

preceded by inactivity. The resulting estimation sample contains 1,163 observations for all 24 chokepoints.

The chokepoint model includes five responses: PageRank, betweenness, share of monthly network flow, articulation-point status, and connection to an articulation point. The incident-global-bridge indicator is excluded because it differs from articulation-point status in only three observations and their binary correlation is 0.993.

Four predictors are included: lagged historical-neighbour activity share, log-transformed lagged mean voyage distance, and monthly sine and cosine terms. All four are standardised within the chokepoint estimation sample.

PageRank and flow share lie strictly within $(0, 1)$, while betweenness has six exact zeros (0.516%) and otherwise takes values in $(0, 1)$. None of these responses equals one. Consequently, PageRank and flow share are modelled using Beta likelihoods, while betweenness uses a zero-inflated Beta likelihood.

5.1.2 Response likelihoods

The following response families are used for the baseline model and provide the likelihood framework for the proposed extensions. The expressions below use a scalar factor $f_{i,t}$ and its response-specific loading, as in Model 1; their latent linear predictors will require extension for the rank-two model. For bounded response j, the interior mean and precision are defined as

$$\operatorname{logit}(\mu_{i,t,j}) = \alpha_j + \lambda_j f_{i,t}, \tag{31}$$

$$A_{i,t,j} = \mu_{i,t,j}\phi_j, \qquad D_{i,t,j} = (1 - \mu_{i,t,j})\phi_j. \tag{32}$$

For terminals, $j = 1$ denotes PageRank, $j = 2$ HHI, and $j = 3$ dependence. PageRank follows

$$Y_{i,t,1} \mid f_{i,t} \sim \text{Beta}(A_{i,t,1}, D_{i,t,1}). \tag{33}$$

For HHI and dependence, the boundary probability has a separate loading:

$$\begin{aligned} \operatorname{logit}(\pi_{i,t,j}) &= c_j^T + d_j^T f_{i,t}, \qquad j = 2, 3, \\ Y_{i,t,j} \mid f_{i,t} &\sim \pi_{i,t,j}\delta_1 + (1 - \pi_{i,t,j}) \operatorname{Beta}(A_{i,t,j}, D_{i,t,j}). \end{aligned} \tag{34}$$

where δ_1 denotes a point mass at one. Interior observations contribute the Beta density multiplied by $1 - \pi_{i,t,j}$, while exact ones contribute the probability $\pi_{i,t,j}$. Observed values are not clipped or displaced from the boundary. The overall conditional mean is

$$\mathbb{E}(Y_{i,t,j} \mid f_{i,t}) = \pi_{i,t,j} + (1 - \pi_{i,t,j})\mu_{i,t,j}. \tag{35}$$

Thus the sign of λ_j alone does not determine how the overall mean changes with the factor.

For chokepoints, PageRank and flow share follow the interior Beta specification. For betweenness, let $\rho_{i,t,2}$ denote the probability of zero:

$$\begin{aligned} \operatorname{logit}(\rho_{i,t,2}) &= c_2^C + d_2^C f_{i,t}, \\ Y_{i,t,2} \mid f_{i,t} &\sim \rho_{i,t,2}\delta_0 + (1 - \rho_{i,t,2}) \operatorname{Beta}(A_{i,t,2}, D_{i,t,2}). \end{aligned} \tag{36}$$

Here δ_0 is a point mass at zero. The overall conditional mean is $(1 - \rho_{i,t,2})\mu_{i,t,2}$. The zero-probability loading is estimated separately from the interior-mean loading.

For binary response k,

$$B_{i,t,k} \mid f_{i,t} \sim \text{Bernoulli}(p_{i,t,k}), \tag{37}$$

$$\operatorname{logit}(p_{i,t,k}) = a_k + \ell_k f_{i,t}. \tag{38}$$

5.1.3 Prior specification

The following priors specify the shared response parameters and the rank-one baseline factor regression. Additional priors and identification constraints required by Models 2 and 3 will be stated with those extensions. The terminal and chokepoint models are estimated separately, with independent parameters.

The prior distributions were chosen to provide regularisation while allowing sufficient flexibility in the relationships between the latent factor and the observed responses. Let q denote the number of covariates, with $q = 7$ for terminals and $q = 4$ for chokepoints. The factor-regression coefficients were assigned independent zero-centred Normal priors, $\beta_h \sim \mathcal{N}(0, 0.5^2)$ for $h = 1, \ldots, q$, favouring moderate covariate associations on the scale of the latent factor.

The loading associated with PageRank was constrained to be positive to fix the orientation of the latent factor, while the remaining loadings were allowed to take either sign. For the three bounded responses, the loading priors were $\lambda_1 \sim \text{HalfNormal}(1)$ and $\lambda_j \sim \mathcal{N}(0, 1)$ for $j = 2, 3$. The binary-response loadings were assigned independent priors $\ell_k \sim \mathcal{N}(0, 1)$ for $k = 1, \ldots, K$, where $K = 3$ for terminals and $K = 2$ for chokepoints.

For the interior components of the terminal bounded responses, intercept priors were specified on the logit scale, with centres chosen to reflect differences in the empirical scales of the corresponding network measures. In particular, PageRank typically takes values close to zero, making a zero-centred intercept prior, corresponding to an inverse-logit value of 0.5, inappropriate. Its intercept prior was therefore centred at -4, corresponding to an inverse-logit value of approximately 0.018. The intercept priors for HHI and dependence were instead centred at zero, allowing their interior means to vary around a reference value of 0.5 when the factor is zero. These choices correspond to $\alpha_j \sim \mathcal{N}(m_j, 1.5^2)$ for $j = 1, 2, 3$, with $(m_1, m_2, m_3) = (-4, 0, 0)$.

Binary-response intercepts were assigned independent $a_k \sim \mathcal{N}(0, 2^2)$ priors, allowing a relatively broad range of probabilities at factor value zero. For terminal HHI and dependence, the intercepts governing the probability of an exact one were assigned $c_j^T \sim \mathcal{N}(0, 1.5^2)$ priors, while their separate factor loadings followed $d_j^T \sim \mathcal{N}(0, 1)$, for $j = 2, 3$.

The precision parameters of the bounded continuous components were assigned independent log-Normal priors, with $\log \phi_j \sim \mathcal{N}(\log 10, 0.7^2)$ for $j = 1, 2, 3$. This specification ensures positive precision values, with prior median 10 and an approximate

central 95% interval of $[2.54, 39.44]$, allowing different degrees of dispersion around the conditional interior means.

For chokepoints, the same prior families and scales were adopted, with parameters estimated independently from the terminal model. The interior intercept centres were set to $(m_1, m_2, m_3) = (-4, -5, -4)$ for PageRank, betweenness, and flow share, respectively, reflecting the relatively small values typically observed for these measures. In particular, the inverse-logit of -5 is approximately 0.0067, comparable to the empirical median of chokepoint betweenness.

The betweenness zero-component intercept was assigned $c_2^C \sim \mathcal{N}(-5, 1.5^2)$, expressing a prior preference for a low probability of exact zeros at factor value zero. Its separate factor loading followed $d_2^C \sim \mathcal{N}(0, 1)$. Given the small number of observed zeros, inference on this component may be particularly sensitive to the prior specification.

All response-parameter priors were specified independently of one another and of the factor-regression coefficients. These distributions were adopted as initial regularising priors rather than independently elicited prior information. Their prior predictive implications and the sensitivity of the posterior findings to alternative specifications are examined in Section 5.4.

5.2 Model specifications

5.2.1 Rank-one specification and posterior distribution

For each node type separately, let $\boldsymbol{x}_{i,t}$ be the standardised covariate vector and $f_{i,t}$ be a scalar latent factor. The factor regression is

$$f_{i,t} \mid \boldsymbol{x}_{i,t}, \boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{x}_{i,t}^{\mathsf{T}}\boldsymbol{\beta}, 1). \tag{39}$$

The residual factor variance is fixed at one to define its scale. The PageRank loading is constrained to be positive to orient the factor, while the other loadings are unrestricted. Conditional on the factors and parameters, the responses are assumed

independent, and factors for different node-months are independent given their covariates and regression coefficients.

Let $\boldsymbol{\gamma}$ collect the loadings of all response components:

$$\boldsymbol{\gamma} = \begin{cases} (\lambda_1, \lambda_2, \lambda_3, \ell_1, \ell_2, \ell_3, d_2^T, d_3^T)^\mathsf{T}, & \text{for terminals,} \\ (\lambda_1, \lambda_2, \lambda_3, \ell_1, \ell_2, d_2^C)^\mathsf{T}, & \text{for chokepoints.} \end{cases}$$

For component $m = 1, \ldots, M$, let $\eta_{i,t,m}$ denote the linear predictor on the logit scale: the logit of the interior Beta mean, the binary-response probability, or the boundary probability, according to the component. It can be written as $\eta_{i,t,m} = \xi_m + \gamma_m f_{i,t}$, where ξ_m and γ_m denote the corresponding intercept and loading, respectively, with $M = 8$ for terminals and $M = 6$ for chokepoints.

Substituting $f_{i,t} = \boldsymbol{x}_{i,t}^\mathsf{T}\boldsymbol{\beta} + z_{i,t}$, where $z_{i,t} \sim \mathcal{N}(0, 1)$ independently across node–month observations, gives

$$\eta_{i,t,m} = \xi_m + \boldsymbol{x}_{i,t}^\mathsf{T}(\gamma_m \boldsymbol{\beta}) + \gamma_m z_{i,t}. \tag{40}$$

Thus, the covariate coefficient vector for component m is $\gamma_m\boldsymbol{\beta}$. Stacking these vectors as rows gives

$$\begin{pmatrix} \gamma_1\beta_1 & \cdots & \gamma_1\beta_q \\ \vdots & \ddots & \vdots \\ \gamma_M\beta_1 & \cdots & \gamma_M\beta_q \end{pmatrix} = \boldsymbol{\gamma}\boldsymbol{\beta}^\mathsf{T},$$

where q denotes the number of covariates. Every row is a scalar multiple of $\boldsymbol{\beta}^\mathsf{T}$, so the matrix has rank at most one. This structure provides a parsimonious summary of how the covariates are associated with multiple network measures through a shared latent dimension, while allowing each response component to relate to that dimension with a different strength and sign.

Inference for this model is based on the joint posterior distribution of both the parameters and latent factors, which describes their uncertainty conditional on the observed responses and covariates. By Bayes' theorem, this distribution is proportional to the product of the response likelihood, the conditional distribution of the latent factors given the covariates and regression coefficients, and the parameter prior

densities (Gelman et al., 2013). These components are specified below, followed by the joint posterior distribution and the full conditional posterior kernels. Detailed derivations are provided in Appendix D.

Let $\mathcal{O}$ denote the set of node–month observations included in estimation. To simplify the subscripts, each observation $(i,t) \in \mathcal{O}$ is indexed by n, so that f_n, $\boldsymbol{x}_n$, Y_{nj}, and B_{nk} correspond to $f_{i,t}$, $\boldsymbol{x}_{i,t}$, $Y_{i,t,j}$, and $B_{i,t,k}$, respectively.

Let $\mathbf{f} = (f_n)_{n\in\mathcal{O}}$ collect the latent factors. The observed responses are Y_{nj} for the three bounded measures, $j = 1,2,3$, and B_{nk} for the binary measures, $k = 1,\ldots,K$, where $K = 3$ for terminals and $K = 2$ for chokepoints.

Let $\boldsymbol{\theta}$ collect the response-specific parameters: the bounded-response intercepts α_j, loadings λ_j, and precisions ϕ_j; the binary-response intercepts a_k and loadings ℓ_k; and the boundary-component parameters c_j and d_j. The latter denote c_j^T, d_j^T for terminal HHI and dependence, $j = 2,3$, or c_2^C, d_2^C for chokepoint betweenness.

Under the conditional-independence assumption stated above, the joint contribution of the responses for a single node–month is the product of the contributions of each bounded response, denoted here by $g_j(Y_{nj} \mid f_n, \boldsymbol{\theta})$, and the Bernoulli probabilities of the binary responses:

$$\begin{aligned} &p(\mathbf{Y}_n, \mathbf{B}_n \mid f_n, \boldsymbol{\theta}) \\ &\quad = \prod_{j=1}^{3} g_j(Y_{nj} \mid f_n, \boldsymbol{\theta}) \prod_{k=1}^{K} p_{nk}^{B_{nk}} (1-p_{nk})^{1-B_{nk}}. \end{aligned}$$

Combining the response contributions over all observations in $\mathcal{O}$ gives the likelihood

$$\begin{aligned} L(\boldsymbol{\theta}, \mathbf{f}) &= \prod_{n\in\mathcal{O}} p(\mathbf{Y}_n, \mathbf{B}_n \mid f_n, \boldsymbol{\theta}) \\ &= \prod_{n\in\mathcal{O}} \left[\prod_{j=1}^{3} g_j(Y_{nj} \mid f_n, \boldsymbol{\theta}) \prod_{k=1}^{K} p_{nk}^{B_{nk}} (1-p_{nk})^{1-B_{nk}} \right]. \end{aligned} \tag{41}$$

The factor regression in Equation 39 gives the joint conditional density of the latent factors:

$$p(\mathbf{f} \mid \boldsymbol{\beta}, \mathbf{X}) = \prod_{n\in\mathcal{O}} \mathcal{N}\big(f_n; \boldsymbol{x}_n^{\mathsf{T}}\boldsymbol{\beta}, 1\big), \tag{42}$$

where $\mathbf{X}$ is the covariate design matrix, with row $\boldsymbol{x}_n^\mathsf{T}$ for observation n.

Combining these components and denoting the observed responses and covariates by $\mathcal{D}$, the joint posterior density is therefore proportional to

$$p(\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{f} \mid \mathcal{D}) \propto L(\boldsymbol{\theta}, \mathbf{f})\, p(\mathbf{f} \mid \boldsymbol{\beta}, \mathbf{X})\, p(\boldsymbol{\theta}) p(\boldsymbol{\beta}). \tag{43}$$

where $p(\boldsymbol{\theta})$ and $p(\boldsymbol{\beta})$ are the products of the independent prior densities introduced in Section 5.1.3.

The full conditional posterior kernels follow by retaining the factors of Equation 43 that depend on the quantity under consideration, with all remaining unknown quantities held fixed. In the expressions below, *rest* denotes these remaining quantities together with the observed data and covariates.

Define $\mathcal{I}_j = \{n \in \mathcal{O} : 0 < Y_{nj} < 1\}$, so that for ordinary Beta responses, $\mathcal{I}_j = \mathcal{O}$, while for boundary-inflated responses it contains only the interior observations.

Binary-response loadings. For each binary-response loading, the $\mathcal{N}(0, 1)$ prior gives

$$\begin{aligned} p(\ell_k \mid \text{rest}) \propto & \left[\prod_{n \in \mathcal{O}} \frac{\exp\{B_{nk}(a_k + \ell_k f_n)\}}{1 + \exp(a_k + \ell_k f_n)}\right] \\ & \times \exp\left(-\frac{\ell_k^2}{2}\right), \qquad k = 1, \ldots, K. \end{aligned} \tag{44}$$

Bounded-response interior loadings. For the unrestricted interior-mean loadings λ_j, $j = 2, 3$, corresponding to terminal HHI and dependence or chokepoint betweenness and flow share, the full conditional kernels are

$$\begin{aligned} p(\lambda_j \mid \text{rest}) \propto & \left[\prod_{n \in \mathcal{I}_j} \frac{Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ & \times \exp\left(-\frac{\lambda_j^2}{2}\right), \qquad j = 2, 3, \end{aligned} \tag{45}$$

For PageRank, the HalfNormal prior imposes the positive-support restriction:

$$p(\lambda_1 \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{Y_{n1}^{\mu_{n1}\phi_1-1}(1-Y_{n1})^{(1-\mu_{n1})\phi_1-1}}{\Gamma(\mu_{n1}\phi_1)\Gamma((1-\mu_{n1})\phi_1)}\right] \times \exp\left(-\frac{\lambda_1^2}{2}\right) \mathbb{1}(\lambda_1 > 0). \tag{46}$$

Boundary-probability loadings. For terminal HHI and dependence, define $Z_{nj} = \mathbb{1}(Y_{nj} = 1)$, $j = 2, 3$. The full conditional kernels are

$$p(d_j^T \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1+\exp(c_j^T + d_j^T f_n)}\right] \times \exp\left[-\frac{(d_j^T)^2}{2}\right], \qquad j = 2, 3. \tag{47}$$

For chokepoint betweenness, let $Z_{n2} = \mathbb{1}(Y_{n2} = 0)$. The full conditional kernel is

$$p(d_2^C \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\exp\{Z_{n2}(c_2^C + d_2^C f_n)\}}{1+\exp(c_2^C + d_2^C f_n)}\right] \times \exp\left[-\frac{(d_2^C)^2}{2}\right]. \tag{48}$$

Factor-regression coefficients. The Gaussian factor regression and the independent $\mathcal{N}(0, 0.5^2)$ coefficient priors give the full conditional kernel

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto \exp\left\{-\frac{1}{2}\Big[\boldsymbol{\beta}^\top(\mathbf{X}^\top\mathbf{X} + 4\mathbf{I}_q)\boldsymbol{\beta} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{f}\Big]\right\}. \tag{49}$$

Recognising the kernel of a multivariate Normal distribution gives the conjugate full conditional

$$\boldsymbol{\beta} \mid \text{rest} \sim \mathcal{N}_q\Big((\mathbf{X}^\top\mathbf{X} + 4\mathbf{I}_q)^{-1}\mathbf{X}^\top\mathbf{f}, (\mathbf{X}^\top\mathbf{X} + 4\mathbf{I}_q)^{-1}\Big). \tag{50}$$

Matrix notation is used here for compactness, with $\mathbf{I}_q$ denoting the $q \times q$ identity matrix.

Latent factors. For each node–month observation, the full conditional of the factor combines its Gaussian regression density with all response contributions for that observation:

$$p(f_n \mid \text{rest}) \propto L_n(f_n) \exp\left[-\frac{1}{2}(f_n - \boldsymbol{x}_n^\top \boldsymbol{\beta})^2\right]. \tag{51}$$

For terminals, the observation-specific likelihood is

$$\begin{aligned} L_n^T(f_n) = p(Y_{n1} \mid f_n, \alpha_1, \lambda_1, \phi_1) \\ \times \prod_{j=2}^{3} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, c_j^T, d_j^T) \\ \times \prod_{k=1}^{3} p(B_{nk} \mid f_n, a_k, \ell_k). \end{aligned}$$

For chokepoints, it is

$$\begin{aligned} L_n^C(f_n) = p(Y_{n1} \mid f_n, \alpha_1, \lambda_1, \phi_1) \\ \times p(Y_{n2} \mid f_n, \alpha_2, \lambda_2, \phi_2, c_2^C, d_2^C) \\ \times p(Y_{n3} \mid f_n, \alpha_3, \lambda_3, \phi_3) \\ \times \prod_{k=1}^{2} p(B_{nk} \mid f_n, a_k, \ell_k). \end{aligned}$$

The inflated-response terms in these expressions denote the complete mixture contributions, including the boundary probabilities and, for interior observations, the Beta densities.

Binary-response intercepts. Under the prior $a_k \sim \mathcal{N}(0, 2^2)$,

$$\begin{aligned} p(a_k \mid \text{rest}) \propto \exp\Bigg\{ -\frac{a_k^2}{8} + a_k \sum_{n \in \mathcal{O}} B_{nk} \\ - \sum_{n \in \mathcal{O}} \log\left[1 + \exp(a_k + \ell_k f_n)\right] \Bigg\}. \end{aligned} \tag{52}$$

Bounded-response interior intercepts. For $\alpha_j \sim \mathcal{N}(m_j, 1.5^2)$, the full conditional kernels are

$$\begin{aligned} p(\alpha_j \mid \text{rest}) \propto & \left[\prod_{n\in\mathcal{I}_j} \frac{Y_{nj}^{\mu_{nj}\phi_j-1}(1-Y_{nj})^{(1-\mu_{nj})\phi_j-1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ & \times \exp\left[-\frac{(\alpha_j-m_j)^2}{2(1.5)^2}\right], \qquad j=1,2,3. \end{aligned} \tag{53}$$

Boundary-probability intercepts. For terminal HHI and dependence, $c_j^T \sim \mathcal{N}(0, 1.5^2)$ gives

$$\begin{aligned} p(c_j^T \mid \text{rest}) \propto & \left[\prod_{n\in\mathcal{O}} \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1+\exp(c_j^T + d_j^T f_n)}\right] \\ & \times \exp\left[-\frac{(c_j^T)^2}{2(1.5)^2}\right], \qquad j=2,3. \end{aligned} \tag{54}$$

For chokepoint betweenness, $c_2^C \sim \mathcal{N}(-5, 1.5^2)$ gives

$$\begin{aligned} p(c_2^C \mid \text{rest}) \propto \exp\Bigg\{ & -\frac{(c_2^C+5)^2}{2(1.5)^2} + c_2^C \sum_{n\in\mathcal{O}} Z_{n2} \\ & - \sum_{n\in\mathcal{O}} \log\left[1+\exp(c_2^C + d_2^C f_n)\right] \Bigg\}. \end{aligned} \tag{55}$$

Beta precision parameters. For each bounded-response component, under the prior $\log\phi_j \sim \mathcal{N}(\log 10, 0.7^2)$, the full conditional kernel is

$$\begin{aligned} p(\phi_j \mid \text{rest}) \propto & \left[\prod_{n\in\mathcal{I}_j} \frac{\Gamma(\phi_j) Y_{nj}^{\mu_{nj}\phi_j-1}(1-Y_{nj})^{(1-\mu_{nj})\phi_j-1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ & \times \frac{1}{\phi_j} \exp\left[-\frac{(\log\phi_j - \log 10)^2}{2(0.7)^2}\right], \\ & \qquad \phi_j > 0, \qquad j=1,2,3. \end{aligned} \tag{56}$$

Apart from the Normal coefficient full conditional in Equation 50, the full conditionals above do not generally have standard conjugate forms. The joint posterior density is known up to a normalising constant, but the integrals required for normalisation and marginalisation are analytically intractable. Inference is therefore based on numerical sampling using the No-U-Turn Sampler (NUTS), whose implementation

and associated diagnostic criteria are described in Section 5.3.

5.2.2 Node-specific intercepts

5.2.3 Rank-two specification

5.3 Posterior computation

5.3.1 NUTS sampling algorithm and implementation

The models were implemented in Python 3.12.15 using PyMC 5.20.1 (Abril-Pla et al., 2023), with PyTensor 2.27.1 as the computational backend. Posterior inference was performed using the No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014), an adaptive variant of Hamiltonian Monte Carlo (HMC) (Neal, 2011). Unlike traditional random-walk MCMC methods, HMC exploits gradient information to construct proposals that can explore high-dimensional posterior distributions more efficiently. NUTS extends HMC by automatically determining the trajectory length, avoiding the need to manually specify the number of integration steps. The principles underlying HMC and its NUTS extension are outlined below.

Let $\boldsymbol{\omega} = (\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{f})$ collect all unknown response parameters, regression coefficients, and latent factors. The sampler operates on their unconstrained representation $\mathbf{q} = T(\boldsymbol{\omega})$. By the change-of-variables formula, the transformed log-posterior density is

$$\mathcal{L}(\mathbf{q}) = \log p_{\boldsymbol{\omega}}\big(T^{-1}(\mathbf{q}) \mid \mathcal{D}\big) + \log \left| \det \frac{\partial T^{-1}(\mathbf{q})}{\partial \mathbf{q}} \right|. \tag{57}$$

Following Neal (2011), an auxiliary momentum vector $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$ is introduced, where $\mathbf{M}$ is a positive-definite mass matrix. Intuitively, HMC treats the current parameter values $\mathbf{q}$ as the position of a moving particle. The negative transformed log-posterior defines a potential-energy landscape, with lower values in regions of higher transformed posterior density. The momentum allows the particle to move through this landscape, while the gradient of $\mathcal{L}(\mathbf{q})$ guides its trajectory.

The negative log-density of the Gaussian momentum, up to an additive constant, defines the kinetic energy $\frac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{M}^{-1}\mathbf{v}$. Together with the potential energy $-\mathcal{L}(\mathbf{q})$, this

defines the Hamiltonian:

$$H(\mathbf{q}, \mathbf{v}) = -\mathcal{L}(\mathbf{q}) + \frac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{M}^{-1}\mathbf{v}. \tag{58}$$

The corresponding equations of motion are

$$\frac{d\mathbf{q}}{d\tau} = \nabla_{\mathbf{v}} H(\mathbf{q}, \mathbf{v}) = \mathbf{M}^{-1}\mathbf{v}, \qquad \frac{d\mathbf{v}}{d\tau} = -\nabla_{\mathbf{q}} H(\mathbf{q}, \mathbf{v}) = \nabla_{\mathbf{q}}\mathcal{L}(\mathbf{q}),$$

with τ denoting an artificial integration time.

These equations are approximated numerically using leapfrog integration: for a step size ϵ, each integration step comprises the updates

$$\begin{aligned} \mathbf{v}_{1/2} &= \mathbf{v} + \frac{\epsilon}{2}\nabla_{\mathbf{q}}\mathcal{L}(\mathbf{q}), \\ \mathbf{q}' &= \mathbf{q} + \epsilon\mathbf{M}^{-1}\mathbf{v}_{1/2}, \\ \mathbf{v}' &= \mathbf{v}_{1/2} + \frac{\epsilon}{2}\nabla_{\mathbf{q}}\mathcal{L}(\mathbf{q}'). \end{aligned} \tag{59}$$

These updates act on the complete state vector, so regression coefficients, response parameters, and latent factors are explored jointly rather than through sequential parameter-specific full conditional updates.

NUTS constructs trajectories by recursively expanding a tree of leapfrog states in forward or backward directions until the no-U-turn criterion detects that the trajectory is doubling back, a numerical stopping criterion is met, or the maximum tree depth is reached. A state from the constructed trajectory is then selected according to a rule that preserves the target distribution (Hoffman and Gelman, 2014).

Algorithm 1 summarises the sampling procedure for the rank-one baseline model.

Algorithm 1 NUTS sampling for one-rank baseline model

Require: Data $\mathcal{D}$; chains C; warm-up iterations W; retained draws S; target acceptance δ.

for each chain $c = 1, \ldots, C$ **do**
  Initialise the unconstrained parameter vector $\mathbf{q}$.
  Initialise step size ϵ and diagonal mass matrix $\mathbf{M}$.
  for $m = 1, \ldots, W + S$ **do**
    Sample momentum $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{M})$.
    Initialise the trajectory at $(\mathbf{q}, \mathbf{v})$.
    repeat
      Sample a direction $d \sim \text{Uniform}(\{-1, +1\})$.
      Extend the tree in direction d by doubling the number of leapfrog steps.
      Update the candidate state according to the NUTS selection rule.
    until a stopping criterion is met
    Set $\mathbf{q}$ to the selected candidate position.
    if $m \leq W$ **then**
      Adapt ϵ to approach target acceptance δ.
      Adapt $\mathbf{M}$ to account for differences in parameter scales.
    else
      Store the joint posterior draw $\boldsymbol{\omega}^{(c,m-W)} = T^{-1}(\mathbf{q})$.
      Compute factor means $\mathbf{x}_n^\top \boldsymbol{\beta}$ and residuals $f_n - \mathbf{x}_n^\top \boldsymbol{\beta}$ for all $n \in \mathcal{O}$.
    end if
  end for
end for
return S posterior draws per chain.

Each node-type model was estimated using four parallel chains, with 5,000 warm-up iterations and 10,000 retained draws per chain, yielding 40,000 posterior draws per node-type model. Earlier runs with 1,500 warm-up iterations and 2,000 retained draws per chain were used for initial estimation and comparison. The use of multiple chains allows convergence to be assessed by comparing within-chain and between-chain variability (Gelman et al., 2013), while the number of iterations was increased to allow more extensive sampler adaptation during warm-up and improve the Monte Carlo precision of posterior summaries through a larger number of retained draws.

The target acceptance probability was set to 0.80, corresponding to the target acceptance rate used during step-size adaptation in the warm-up phase (Hoffman and Gelman, 2014). The random seed was 20261008.

5.3.2 Sampling diagnostics

Posterior summaries and convergence diagnostics were computed using ArviZ 0.21.0, with NumPy 1.26.4, pandas 2.3.3, and Matplotlib 3.9.2 used for data processing and visualisation.

The assessment covered regression coefficients, response parameters, latent factors, factor means, and factor residuals. Several complementary diagnostics were considered together to examine agreement across chains, sampling efficiency, within-chain stationarity, and the Monte Carlo precision of posterior estimates.

Agreement across chains was assessed using rank-normalised $\widehat{R}$, with values close to one indicating consistency between within-chain and between-chain variability. Bulk and tail effective sample sizes (ESS) were used to assess how much information the correlated draws provided about the centre and tails of the posterior distribution, respectively (Vehtari et al., 2021).

Post-warm-up divergences were also monitored, as they signal numerical difficulties in following Hamiltonian trajectories and may indicate that some posterior regions have not been adequately explored (Betancourt, 2017).

The Geweke and Heidelberger–Welch diagnostics provided additional checks on individual chains (Geweke, 1992; Heidelberger and Welch, 1983).

The Geweke diagnostic compares the means of an early and a late segment of a chain, standardising their difference using a standard error that accounts for autocorrelation. Under suitable stationarity conditions, the statistic has an asymptotic standard Normal distribution, with large absolute values suggesting inconsistency between the two segments.

The Heidelberger–Welch procedure first assesses stationarity using a test based on the Cramér–von Mises statistic. If necessary, the test is repeated after removing progressively larger initial portions of the chain, up to half of the available draws. Once stationarity is accepted, a half-width test checks whether the Monte Carlo confidence interval for the posterior mean, computed from the remaining draws, is sufficiently

narrow relative to the magnitude of the estimated mean.

Trace plots and autocorrelation functions (ACFs) for selected parameters complemented these numerical checks. Trace plots were inspected for persistent trends, differences between chains, and whether the chains explored similar posterior regions. ACFs showed how dependence between draws changed with increasing lag, with faster decay generally indicating more efficient sampling (Vehtari et al., 2021).

5.4 Prior sensitivity analysis

To assess the implications of the prior assumptions and the sensitivity of posterior findings, two alternative prior profiles were investigated on the same estimation samples. The analysis considers both the prior predictive distributions and the sensitivity of posterior inference to these modifications.

5.4.1 Alternative prior specifications

Three prior specifications were considered: the baseline described above, an alternative with more concentrated priors for selected parameters, and a third specification with response-specific priors for the Beta precision parameters. The alternative specifications were informed by the observed response distributions and prior predictive simulations, rather than independently elicited prior information.

Under the baseline priors, prior predictive simulations generated relatively broad distributions for several network measures, particularly those characterised by small observed values, such as PageRank, betweenness and flow share.

In response to these findings, the second specification introduced more concentrated prior distributions for selected intercepts and factor loadings. Specifically, the standard deviations of the logit-intercept priors for the small-valued continuous responses were reduced from 1.5 to 0.7, while the PageRank loading scale and the loading standard deviations for betweenness and flow share were reduced from 1 to 0.5. For the boundary components, intercept prior standard deviations were reduced from 1.5 to 1, and boundary-loading standard deviations from 1 to 0.5. The prior centres were

retained, as were the original priors for the terminal HHI and dependence interior components, regression coefficients, and binary-response parameters.

The third specification retained the modifications introduced in the second specification and additionally introduced response-specific priors for the Beta precision parameters.

The corresponding prior medians were set to $(\widetilde{\phi}_1, \widetilde{\phi}_2, \widetilde{\phi}_3) = (300, 10, 10)$ for terminal PageRank, HHI, and dependence, respectively, and to $(\widetilde{\phi}_1, \widetilde{\phi}_2, \widetilde{\phi}_3) = (500, 100, 100)$ for chokepoint PageRank, betweenness, and flow share, respectively. For each precision prior, the log-scale standard deviation was fixed at 0.7.

Chapter 6

Results and discussion

6.1 Sampling diagnostics: baseline results

The baseline models were estimated using four chains, each with 5,000 warm-up iterations and 10,000 retained draws. Neither run presented post-warm-up divergences. For the terminal model, the maximum reported rank-normalised $\widehat{R}$ was 1.00 rounded to two decimal points, with minimum bulk and tail effective sample sizes (ESS) of 2,931 and 6,726, respectively. For the chokepoint model, the maximum reported $\widehat{R}$ was also 1.00, with minimum bulk and tail ESS of 1,693 and 3,698, respectively. The lowest ESS values in the chokepoint model were associated with the betweenness precision parameter.

Both models therefore satisfied the prespecified screening criteria of no divergences, maximum $\widehat{R} \leq 1.01$, and minimum bulk and tail ESS of at least 400. The selected trace plots and within-chain autocorrelation functions in Appendix E.1 complement these numerical results. The betweenness loading and precision show relatively persistent autocorrelation in the chokepoint model. Overall, the diagnostics support satisfactory sampling behaviour, although they cannot establish model adequacy or guarantee complete exploration of the posterior distribution.

6.2 Baseline model results

6.2.1 Terminal posterior analysis

This section presents the posterior results for the terminal model, focusing on covariate associations, response relationships and the interpretation of the latent factor. Posterior predictive checks are then used to assess how well the model reproduces the observed response distributions and associations.

Terminal posterior associations and factor interpretation

The posterior estimates of the factor-regression coefficients are presented in Table 9, together with their 95% highest-density intervals (HDIs).

The most pronounced positive associations concern lagged exporter status and terminal capacity. Conditional on the other predictors, having been classified as an exporter in the preceding month corresponds to an increase of approximately 1.4 units in the expected factor value. This is substantial relative to the residual standard deviation of the factor, which is fixed at one.

Neighbour activity and voyage distance in the previous month are also positively related to the factor, but their estimated associations are considerably weaker. By contrast, the posterior intervals for terminal age and the seasonal terms include zero, leaving their individual contributions uncertain.

Table 9: Factor-regression coefficients for the terminal model. Numeric predictors are standardised; exporter status is a zero–one indicator.

Predictor	Posterior mean	95% HDI
Lagged neighbour activity share	0.097	$[0.056, 0.136]$
$\log(1 + \text{lagged voyage distance})$	0.062	$[0.021, 0.106]$
Month sine	-0.032	$[-0.069, 0.005]$
Month cosine	-0.003	$[-0.041, 0.033]$
$\log(1 + \text{capacity})$	0.409	$[0.365, 0.454]$
Terminal age	-0.019	$[-0.059, 0.020]$
Lagged exporter indicator	1.395	$[1.296, 1.490]$

The response loadings in Table 10 provide further insight into the interpretation

of the terminal factor, by describing associations with the interior means of bounded responses, the probabilities of binary outcomes, and the boundary probabilities.

The clearest pattern combines increasing PageRank with decreasing counterparty concentration. In particular, higher factor values correspond to both a lower interior mean for HHI and a reduced probability of HHI being exactly one.

The structural indicators show contrasting relationships: the probability of being an articulation point increases with the factor, while that of having an incident global bridge decreases. For connection to an articulation point, the posterior interval includes zero, providing no clear evidence of an association.

Table 10: Loadings on the latent factor. Bounded interior means and all probabilities use logit links.

Response component	Posterior mean	95% HDI
PageRank interior mean	0.359	$[0.348, 0.370]$
HHI interior mean	-0.535	$[-0.569, -0.502]$
Dependence interior mean	0.420	$[0.393, 0.446]$
Articulation-point probability	0.771	$[0.680, 0.860]$
Incident-global-bridge probability	-0.638	$[-0.719, -0.561]$
Connected-to-articulation probability	0.109	$[-0.017, 0.239]$
Probability of HHI equal to one	-3.986	$[-4.727, -3.258]$
Probability of dependence equal to one	-0.545	$[-0.642, -0.451]$

The magnitude of the HHI relationship becomes clearer when expressed in the original response units. Moving from a factor value of 0 to 1, the posterior median expected HHI falls from approximately 0.66 to 0.38, with a median difference of -0.29 (95% equal-tailed credible interval: $[-0.31, -0.27]$), incorporating both the interior Beta component and the point mass at one.

Dependence exhibits a more complex pattern. Its interior mean increases with the factor, whereas the probability of an exact value of one declines. Combining these components, as defined in Equation (35), yields a mildly U-shaped conditional expectation over the displayed factor range. The fitted response functions are displayed in Figure 6 to allow the different response components and their combined implications to be examined across the estimated factor range.

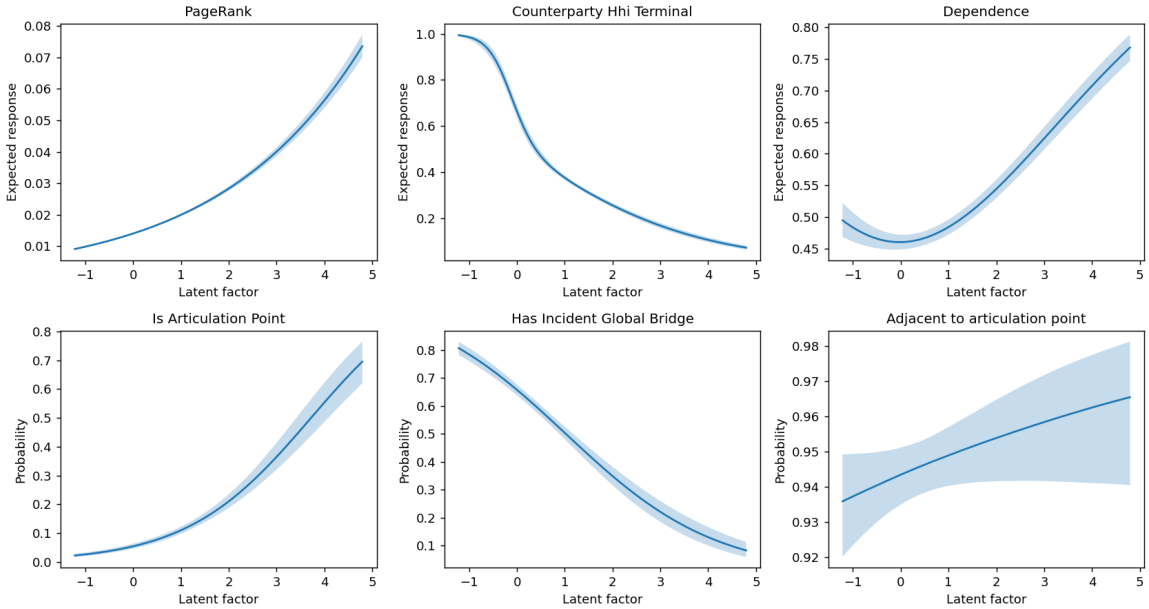


Figure 6: Fitted response functions against the terminal latent factor. Lines are posterior medians and bands are equal-tailed 95% credible intervals for conditional expected responses or probabilities. HHI and dependence means include the point mass at one. The range spans the first to the 99th percentile of posterior mean factor scores.

To relate these posterior patterns to observed network positions, two terminal-month observations with posterior mean factor values closest to the 5th and 95th percentiles of the estimation sample are considered: Zhuhai LNG Terminal in September 2022 has an estimated factor value of approximately -1.0, compared with 2.7 for Sabine Pass LNG Terminal in June 2024. Their observed PageRank values are 0.011 and 0.039, respectively, while their HHI values are 1.00 and 0.17. Zhuhai is classified as an importer and has an incident global bridge, whereas Sabine Pass is an exporter without one. These differences are consistent with the estimated relationships between the latent factor and the terminal responses, although the two observations also differ in commercial role.

While this example provides an illustration of the factor's interpretation, the extent to which the model reproduces the observed response patterns more generally is examined through the posterior predictive checks in the following subsection.

Terminal posterior predictive assessment

Posterior predictive assessment evaluates how well the fitted model reproduces the observed response distributions and associations by comparing statistics computed from the terminal-month observations with those obtained from replicated responses conditional on the posterior factors. These comparisons are conducted both overall and within lagged-role groups, with 95% equal-tailed intervals summarising the variability of the statistics across replications.

The model closely reproduces the overall response means and the overall frequencies of HHI and dependence equal to one. The observed overall binary proportions also lie within their replicated intervals. Nevertheless, discrepancies appear in dispersion, quantiles and role-specific boundary frequencies. For instance, PageRank dispersion is overestimated for lagged importers and underestimated for lagged exporters. For exporters, the observed 95th percentile of PageRank is 0.0714, compared with a replicated mean of 0.0620 and a 95% replicated interval of $[0.0587, 0.0654]$.

Table 11: Selected checks of the proportion of bounded responses equal to one, by lagged role. All entries are percentages.

Response	Lagged role	Observed	Replicated mean	95% interval
HHI	All	34.09	34.10	[32.73, 35.48]
HHI	Importer	41.89	43.41	[41.59, 45.25]
HHI	Exporter	18.00	14.90	[13.39, 16.40]
Dependence	All	20.76	20.78	[18.97, 22.68]
Dependence	Importer	26.45	23.96	[21.72, 26.28]
Dependence	Exporter	9.04	14.24	[11.97, 16.58]

As shown in Table 11, among exporters, the model also underestimates the frequency of HHI equal to one and overestimates the frequency of dependence equal to one. The importer dependence boundary frequency is also slightly above its replicated interval. These discrepancies suggest that a shared factor with common response functions does not fully capture role-specific response behaviour.

The assessment then turns to associations between responses, comparing observed pairwise Spearman correlations with those obtained from 200 replicated datasets selected evenly across the four chains.

Table 12 reports selected results. The most pronounced discrepancy concerns articulation-point and incident-global-bridge status, which are positively associated in the observed data but negatively associated in the replications. This pattern is particularly evident among exporters.

Table 12: Selected joint posterior predictive checks. Entries are Spearman correlations. Replicated intervals use 200 conditional in-sample replicates.

Response pair	Lagged role	Observed	Rep. median	95% interval
Articulation–bridge	All	0.290	-0.123	[−0.162, −0.081]
Articulation–bridge	Importer	0.137	-0.034	[−0.076, 0.003]
Articulation–bridge	Exporter	0.546	-0.155	[−0.211, −0.091]
HHI–bridge	All	0.533	0.288	[0.249, 0.340]
PageRank–dependence	All	0.314	0.158	[0.116, 0.195]

This mismatch is consistent with restrictions of the fitted specification. The single factor has opposite loadings on articulation-point and incident-bridge probabilities, while conditional independence allows no additional residual association between

these responses. The model therefore struggles to reproduce their positive observed co-occurrence; these checks alone do not establish which restriction is primarily responsible.

6.2.2 Chokepoint posterior analysis

The current section examines how the latent factor relates to network characteristics and measures of centrality, flow exposure and structural importance for the chokepoint specification. The adequacy of the fitted model is subsequently evaluated through posterior predictive checks.

Chokepoint posterior associations and factor interpretation

For chokepoints, the factor-regression results in Table 13 indicate relatively modest associations with the available predictors.

Greater neighbour activity and longer average voyage distances in the preceding month are associated with higher factor values. However, the posterior interval for voyage distance has a lower endpoint close to zero, suggesting that this association is less clearly established.

The seasonal coefficients provide limited evidence of seasonal variation in the conditional factor mean: the 95% HDI for the negative cosine coefficient narrowly excludes zero, whereas the corresponding interval for the sine coefficient includes zero.

Table 13: Factor-regression coefficients for the chokepoint model. Numeric predictors are standardised.

Predictor	Posterior mean	95% HDI
Lagged neighbour activity share	0.103	$[0.042, 0.170]$
$\log(1 + \text{lagged voyage distance})$	0.065	$[0.005, 0.128]$
Month sine	-0.036	$[-0.098, 0.023]$
Month cosine	-0.064	$[-0.128, -0.004]$

Turning to the responses, Table 14 shows a consistent pattern of positive loadings across the interior means and binary structural probabilities, with all 95% HDIs excluding zero. These relationships are also illustrated by the fitted response curves in

Figure 7.

The factor is positively related to PageRank, betweenness and flow share, indicating that higher scores correspond to chokepoints occupying more central positions and carrying larger proportions of the network's LNG flows. Moreover, both articulation-point status and connection to an articulation point become more probable as the factor increases, a finding that differs from the terminal specification, where the latter association is not clearly established. The negative loading on the probability of an exact zero in betweenness is consistent with this pattern, as higher factor values are associated with a lower probability of zero betweenness, further reinforcing the positive relationship between the factor and its overall conditional expectation.

Table 14: Loadings on the latent factor for the chokepoint model. Bounded interior means and all probabilities use logit links.

Response component	Posterior mean	95% HDI
PageRank interior mean	0.263	$[0.243, 0.282]$
Betweenness interior mean	1.025	$[0.969, 1.081]$
Flow-share interior mean	0.619	$[0.578, 0.660]$
Articulation-point probability	0.719	$[0.561, 0.884]$
Connected-to-articulation probability	1.206	$[0.939, 1.480]$
Probability of betweenness equal to zero	-1.656	$[-2.556, -0.767]$

To illustrate how these relationships are reflected in individual observations, consider Cape of Good Hope in May 2021 and Taiwan Strait in October 2021. Their posterior mean factor values, located near the 5th and 95th percentiles of the estimation sample, are approximately -1.6 and 1.6, respectively. Taiwan Strait has substantially higher observed PageRank (0.035 versus 0.007), betweenness (0.040 versus 0.0006) and flow share (3.6% versus 0.3%) Nevertheless, both chokepoint-month observations are classified as articulation points and are connected to articulation points. Their different factor values therefore reflect variation in the continuous measures even though these binary structural indicators coincide.

This comparison illustrates how the estimated factor can summarise different aspects of chokepoint centrality, flow exposure and structural position.

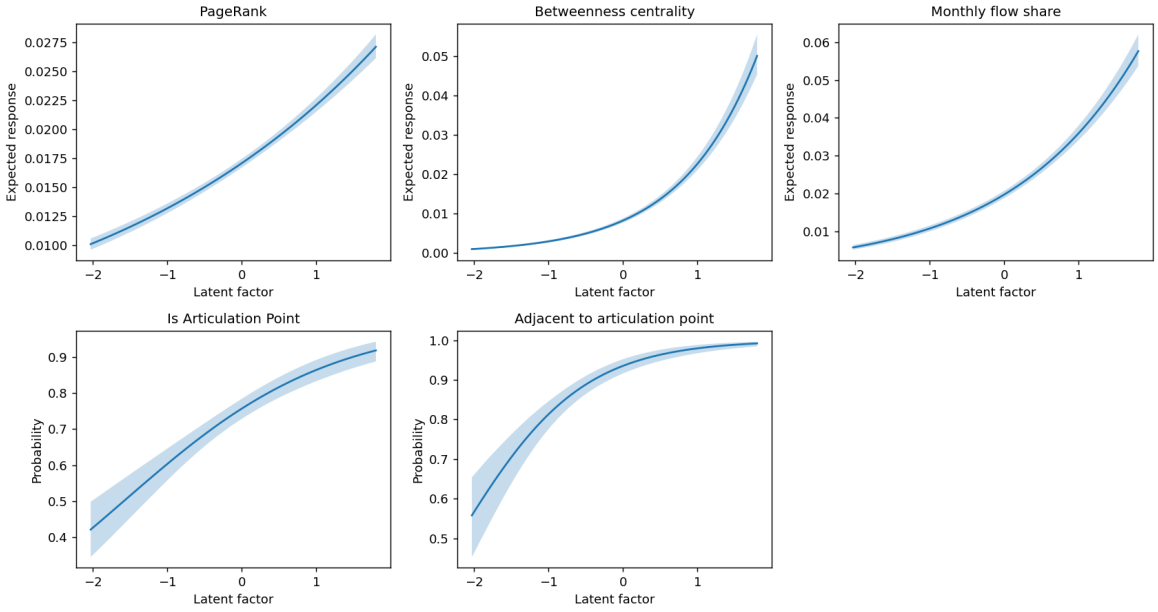


Figure 7: Fitted response functions against the chokepoint latent factor. Lines are posterior medians and bands are equal-tailed 95% credible intervals for conditional expected responses or binary probabilities. The betweenness mean includes the mass at zero. The range spans the first to the 99th percentile of posterior mean factor scores.

Chokepoint posterior predictive assessment

Following the same conditional, in-sample approach used for terminals, the chokepoint posterior predictive checks indicate that the model reproduces the overall response means and binary proportions reasonably closely.

Most reported PageRank statistics fall within their replicated intervals and the model also adequately reproduces the low frequency of zero betweenness (0.516% in the observed data), even though the observed fifth percentile of betweenness lies slightly below its corresponding replicated interval.

The largest marginal discrepancies concern flow share. The observed standard deviation (0.0201) exceeds the 95% replicated interval of $[0.0171, 0.0194]$, indicating that the model underestimates its dispersion. The observed median (0.0159) falls below its replicated interval of $[0.0178, 0.0201]$, whereas the observed 95th percentile (0.0643) exceeds the corresponding interval of $[0.0558, 0.0634]$. These discrepancies indicate that, despite reproducing the overall mean, the model does not fully capture the observed distribution of the measure.

Beyond the marginal distributions, posterior predictive checks also examine whether the model reproduces the observed associations between responses. Pairwise Spearman correlations are compared with those obtained from 200 replicated datasets selected evenly across the four chains. As shown in Table 15, the model reproduces the PageRank–betweenness association reasonably closely, but underestimates the associations involving flow share and the PageRank–articulation association.

Table 15: Selected joint predictive checks for chokepoints. Entries are Spearman correlations; intervals use 200 conditional in-sample replicates.

Response pair	Observed	Rep. median	95% interval
PageRank–betweenness	0.685	0.656	$[0.619, 0.692]$
PageRank–flow share	0.672	0.535	$[0.494, 0.575]$
Betweenness–flow share	0.800	0.722	$[0.686, 0.752]$
PageRank–articulation	0.287	0.201	$[0.139, 0.268]$

6.3 Prior sensitivity results

6.3.1 Prior predictive assessment

Prior predictive simulations were used to examine the response distributions implied by the three prior specifications before conditioning on the observed responses. Table 16 compares the observed standard deviations with the mean standard deviations across prior predictive replications under each specification.

Table 16: Observed standard deviations and mean prior predictive standard deviations under the three prior specifications.

Node type	Response	Observed	Baseline	Second	Third
Terminal	PageRank	0.0133	0.1065	0.0584	0.0289
Chokepoint	PageRank	0.0069	0.0959	0.0542	0.0228
	Betweenness	0.0135	0.0599	0.0329	0.0154
	Flow share	0.0201	0.0870	0.0546	0.0278

Under the baseline specification, several continuous responses exhibited substantially greater prior predictive dispersion than observed in the data.

The second specification reduced the simulated dispersion of PageRank, betweenness and flow share, with further reductions under the third specification. The latter produced standard deviations closer to the observed values, particularly for chokepoint betweenness, although substantial differences remained for PageRank.

The modifications had a more limited effect on the terminal HHI and dependence distributions, whose prior predictive boundary frequencies remained relatively high.

These results indicate that the alternative specifications imply less dispersed prior predictive distributions for several responses, without eliminating all differences from the empirical distributions.

6.3.2 Posterior sensitivity and predictive robustness

To assess the sensitivity of posterior inference to more concentrated priors, the models were re-estimated under the third specification and compared with the baseline results.

The alternative terminal fit produced no divergent transitions, a maximum reported $\widehat{R}$ of 1.01, a minimum bulk effective sample size (ESS) of 782, and a minimum tail ESS of 2,037. For chokepoints, no divergences were observed and the maximum reported $\widehat{R}$ was 1.01, with a minimum tail ESS of 780. However, the minimum bulk ESS was 359 for the betweenness precision parameter, below the screening threshold of 400. Estimates involving this parameter should therefore be interpreted with some caution due to reduced Monte Carlo precision.

Table 17 compares the posterior means and 95% HDIs of all regression coefficients under the two specifications.

Table 17: Posterior means and 95% highest-density intervals (HDIs) for regression coefficients under the baseline and alternative prior specifications.

Covariate	Baseline	Alternative
Terminals		
Lagged neighbour activity	0.096 [0.057, 0.137]	0.093 [0.055, 0.132]
Log lagged voyage distance	0.062 [0.020, 0.105]	0.058 [0.015, 0.099]
Month sine	-0.032 [-0.070, 0.004]	-0.036 [-0.071, 0.003]
Month cosine	-0.004 [-0.042, 0.032]	-0.005 [-0.042, 0.032]
Log capacity	0.409 [0.363, 0.453]	0.408 [0.363, 0.453]
Terminal age	-0.019 [-0.057, 0.021]	-0.022 [-0.061, 0.019]
Lagged exporter indicator	1.392 [1.296, 1.488]	1.407 [1.311, 1.504]
Chokepoints		
Lagged neighbour activity	0.103 [0.042, 0.170]	0.103 [0.040, 0.168]
Log lagged voyage distance	0.065 [0.005, 0.128]	0.066 [0.007, 0.131]
Month sine	-0.036 [-0.098, 0.023]	-0.035 [-0.098, 0.028]
Month cosine	-0.064 [-0.128, -0.004]	-0.065 [-0.125, -0.004]

The regression coefficients remained relatively stable across the two specifications, with similar posterior means and credible intervals.

For terminals, the positive associations with lagged neighbour activity, voyage distance, capacity and lagged exporter status were preserved, while the intervals for terminal age and the seasonal terms continued to include zero. For chokepoints, the positive associations with lagged neighbour activity and voyage distance were also preserved, and the cosine coefficient remained negative, with its interval just excluding zero.

Greater sensitivity emerged for selected response-specific parameters. In particular, the terminal loading on the probability of HHI equal to one changed from -3.982 to -3.116, while the chokepoint loading on the probability of zero betweenness changed from -1.659 to -0.980. Both retained their negative directions, although their magnitudes changed.

Posterior predictive checks were also compared to examine whether the alternative specification improved the reproduction of selected observed statistics. These checks were conditional and in-sample, as in the preceding analyses. Table 18 reports the corresponding replicated statistics under both specifications.

Table 18: Selected posterior predictive statistics under the baseline and alternative prior specifications. Replicated values are posterior predictive means, except for correlations, for which medians are reported. Checks were based on 200 replicated datasets.

Model	Statistic	Observed	Baseline	Alternative
Terminal	Articulation–bridge correlation	0.290	-0.122	-0.121
Terminal	Exporter HHI = 1 (%)	18.00	14.89	14.92
Terminal	Exporter dependence = 1 (%)	9.04	14.23	14.31
Chokepoint	Flow-share standard deviation	0.0201	0.0182	0.0182
Chokepoint	Flow-share median	0.0159	0.0190	0.0190

Findings show that the alternative specification produced little change in the reported posterior predictive statistics, leaving the main discrepancies largely unresolved: for terminals, the replicated association between articulation-point and incident-global-bridge status remained negative, despite its positive observed correlation, and the boundary frequencies of HHI and dependence among exporters remained inaccurately reproduced; for chokepoints, both specifications underestimated the observed dispersion of flow share and overestimated its median.

Given the limited changes in posterior inference and predictive performance, together with the lower bulk ESS in the alternative chokepoint fit, the baseline specification was retained for the main analysis, while the alternative provides a sensitivity assessment of the results under more concentrated priors.

6.4 Model 2: Node-specific intercept results

6.5 Model 3: Rank-two results

6.6 Comparative discussion and limitations

6.6.1 Model limitations and implications

The bounded rank-one models provide an initial Bayesian framework for examining common patterns across multiple measures of node importance in the LNG maritime network. Both models accommodate the bounded nature of the responses and their observed boundary values, while the posterior estimates satisfy the implemented MCMC diagnostic screening criteria.

The terminal latent factor is associated with network centrality and counterparty diversification, whereas the chokepoint factor reflects centrality, flow exposure and structural position. These interpretations suggest that a shared latent dimension can summarise several aspects of network position, although its meaning differs between the two node types.

Nevertheless, satisfactory sampling diagnostics do not guarantee an adequate representation of the observed response distributions. The posterior predictive checks identified limitations in the ability of the terminal model to reproduce role-specific distributions and associations between structural indicators. For chokepoints, the main discrepancies concern the distribution of flow share and its relationships with other centrality measures.

The prior sensitivity analysis further showed that, although the main regression coefficients were relatively stable, some response-specific parameters were more sensitive to prior choices. Moreover, the alternative specification did not substantially improve the posterior predictive results, highlighting that the limitations identified remain relevant under the considered prior modifications.

These findings motivate several possible extensions. For terminals, allowing response-specific intercepts or factor loadings to vary by terminal role could help capture differences between importers and exporters that are not fully represented by the current model. This distinction may be particularly relevant for counterparty concentration: for importing terminals, a high HHI indicates dependence on a limited number of LNG suppliers, potentially increasing vulnerability to supply disruptions; for exporting terminals, by contrast, a low HHI indicates a more diversified set of destinations, potentially reflecting broader commercial reach within international LNG trade. The current specification imposes a common relationship between HHI and the latent factor across terminal roles, potentially overlooking such differences.

Another possible extension would be to relax the assumption of conditional independence between responses given the latent factor. This could allow the model to capture residual associations that are not adequately represented by the shared factor alone.

More flexible likelihood specifications could also improve the representation of bounded responses, especially where the observed dispersion or distributional shape is not adequately reproduced, as in the case of chokepoint flow share.

Finally, node-specific random effects could account for persistent heterogeneity across terminals or chokepoints and introduce dependence among repeated observations of the same node. A further extension would be to develop a joint model for terminals and chokepoints, which are currently analysed separately despite belonging to the same maritime network. Such a framework could allow for dependencies between the two node types while retaining response specifications appropriate to their different characteristics.

These extensions are not mutually exclusive, and the posterior predictive discrepancies do not establish which modelling assumptions are primarily responsible for the observed shortcomings. Their adequacy would therefore need to be evaluated through further model comparisons and predictive assessments.

An additional limitation concerns the estimation sample. The models are fitted

only to node-month observations corresponding to consecutive active months. Consequently, the latent factors describe network position conditional on sustained observed activity, without modelling the process determining whether a node is active. Incorporating an activity component would extend the analysis to both network participation and node importance conditional on participation.

Taken together, these limitations suggest that the estimated latent factors should be interpreted as preliminary summaries of network position rather than comprehensive measures of node criticality. The present models therefore provide a baseline for subsequent methodological developments.

Chapter 7

Conclusions and future work

Appendix A

Chokepoint geometry registry

Table A.1 documents the geometries used to assess route crossings for all 28 initial PortWatch chokepoints. Published spatial geometries were used directly where available. Geometries based on published routeing coordinates were reconstructed as polygons. For the remaining passages, operational corridors or crossing gates were constructed using the geographic and navigational references indicated below. These constructed geometries serve to classify route crossings and do not represent official maritime boundaries. Four of the 28 passages were not traversed under the final route-assignment specification and were excluded from the monthly network.

Table A.1: Geometry registry for the 28 initial chokepoints

ID	Chokepoint	Geometry source	Geometry used for route classification
CP001	Suez Canal	World Bank	Published polygon.
CP002	Panama Canal	Marine Regions reference	Operational canal centreline with an 8 km buffer.
CP003	Bosporus Strait*	Marine Regions reference	Operational strait centreline with a 5 km buffer.
CP004	Bab el-Mandeb Strait	World Bank	Published polygon.
CP005	Malacca Strait	Marine Regions, IHO	Detailed source MultiPolygon.
CP006	Strait of Hormuz	Marine Regions, SeaVoX	Detailed source MultiPolygon.
CP007	Cape of Good Hope	World Bank	Published polygon.
CP008	Gibraltar Strait	Marine Regions, IHO	Detailed source MultiPolygon.
CP009	Dover Strait	Marine Regions, SeaVoX	Detailed source MultiPolygon.

ID	Chokepoint	Geometry source	Geometry used for route classification
CP010	Oresund Strait	Marine Regions reference	Crossing gate, 45 × 8 km, oriented east–west.
CP011	Taiwan Strait	Marine Regions reference	Crossing gate, 220 × 12 km, oriented northwest–southeast.
CP012	Korea Strait	Marine Regions reference	Crossing gate, 130 × 10 km, oriented northwest–southeast.
CP013	Tsugaru Strait	Marine Regions reference	Crossing gate, 55 × 8 km, oriented north–south and centred on the Marine Regions location.
CP014	Luzon Strait	Marine Regions reference	Crossing gate, 220 × 12 km, oriented north–south.
CP015	Lombok Strait	IMO routeing coordinates	Polygon reconstructed from three traffic-separation-scheme sections.
CP016	Ombai Strait	IMO sea-lane reference	Crossing gate, 115 × 12 km, oriented north–south.
CP017	Bohai Strait	NGA GeoNames reference	Crossing gate, 120 × 15 km, oriented north–south.
CP018	Torres Strait	IMO routeing coordinates	Polygon reconstructed from the Prince of Wales Channel two-way route.
CP019	Sunda Strait	IMO routeing coordinates	Polygon reconstructed from two traffic-separation-scheme sections.
CP020	Makassar Strait	Marine Regions, IHO	Detailed source MultiPolygon.
CP021	Magellan Strait	Marine Regions reference	Crossing gate, 55 × 8 km, oriented north–south.
CP022	Yucatan Channel	NOAA/NCEI reference	Crossing gate, 230 × 15 km, oriented east–west.
CP023	Windward Passage	Marine Regions reference	Crossing gate, 105 × 12 km, oriented northwest–southeast.
CP024	Mona Passage	Marine Regions reference	Crossing gate, 145 × 12 km, oriented east–west.
CP025	Balabac Strait*	Marine Regions reference	Crossing gate, 100 × 10 km, oriented north–south.
CP026	Bering Strait*	IMO routeing coordinates	Polygon reconstructed from the two-way route through the strait.
CP027	Mindoro Strait	Marine Regions reference	Crossing gate, 95 × 10 km, oriented northeast–southwest.

CP028	Kerch Strait*	Marine Regions reference	Crossing gate, 26 × 5 km, oriented east–west.

Notes: IDs correspond to the study registry; the original PortWatch identifiers and reference coordinates are retained in the accompanying digital file. Gate dimensions are span × thickness. Crossing gates are centred on the PortWatch reference point. The asterisk identifies passages excluded from the final monthly network because no reconstructed route crossed them under the selected 25 km tolerance. The complete geometry registry and the GeoJSON file containing the final geometries are available in the project's GitHub repository.

Appendix B

Sensitivity analysis of chokepoint assignments

B.1 Purpose and method

The identification of the chokepoints traversed by each reconstructed maritime route may be affected by small spatial discrepancies between the `SeaRoute` paths and the operational chokepoint geometries. Such discrepancies can arise from the spatial resolution and simplification of the routing network, the geographic representation adopted for each passage, and small differences between a reconstructed route centreline and the corresponding transit geometry.

A sensitivity analysis was therefore conducted to quantify how the route–chokepoint assignments changed under alternative spatial specifications. Four classifications were compared: exact geometric intersection and intersections obtained after applying uniform buffers of 5, 10, and 25 km around each of the 28 candidate chokepoint geometries. The 25-km specification corresponds to the classification retained for the final network.

For each chokepoint, distances and buffers were calculated in a local azimuthal-equidistant projection centred on the geometry, allowing the tolerance distance to be measured in metres rather than in geographic degrees. Each of the 1,037 reconstructed origin–destination routes was then compared with the original and buffered chokepoint geometries. For every route–chokepoint pair, the analysis recorded the minimum geometric distance, exact-intersection status, and whether the route fell within each alternative tolerance. The same tolerance was applied to every candidate chokepoint, thereby avoiding passage-specific adjustments based on the observed routes.

B.2 Sensitivity results

Table B.1 reports, for each candidate chokepoint, the number of reconstructed routes identified through exact intersection and under the three alternative tolerance buffers. The final column gives the number of assignments added by the 25-km specification relative to exact intersection.

Table B.1: Sensitivity of route–chokepoint assignments to alternative spatial tolerances

Chokepoint	**Exact**	**5 km**	**10 km**	**25 km**	**Additional at 25 km**
Suez Canal	151	151	151	151	0
Panama Canal	148	148	148	148	0
Bosporus Strait	0	0	0	0	0
Bab el-Mandeb Strait	147	147	147	147	0
Malacca Strait	133	133	133	133	0
Strait of Hormuz	122	122	122	122	0
Cape of Good Hope	65	65	65	65	0
Gibraltar Strait	216	222	222	222	6
Dover Strait	91	91	91	91	0
Oresund Strait	47	47	47	47	0
Taiwan Strait	152	152	152	152	0
Korea Strait	3	3	3	68	65
Tsugaru Strait	50	50	52	52	2
Luzon Strait	43	56	56	56	13
Lombok Strait	69	69	69	74	5
Ombai Strait	57	57	57	58	1
Bohai Strait	35	35	35	35	0
Torres Strait	5	5	5	5	0
Sunda Strait	51	51	51	51	0
Makassar Strait	58	58	58	58	0
Magellan Strait	2	2	2	2	0
Yucatan Channel	137	137	137	137	0
Windward Passage	10	10	10	10	0
Mona Passage	17	17	17	17	0
Balabac Strait	0	0	0	0	0
Bering Strait	0	0	0	0	0
Mindoro Strait	0	0	8	63	63
Kerch Strait	0	0	0	0	0
Total assignments	**1,809**	**1,828**	**1,838**	**1,964**	**155**
Observed chokepoints	**23**	**23**	**24**	**24**	—

Exact intersection identified 1,809 route–chokepoint assignments involving 23 of the 28 candidate passages. Under the final 25-km specification, the number of assignments increased to 1,964 and the number of identified chokepoints increased to 24, resulting in 155 additional assignments, or an 8.6% increase relative to exact intersection.

The assignments were stable across all four specifications for 17 of the 24 chokepoints retained in the final network. Most of the additional assignments under the final specification were concentrated in Korea Strait and Mindoro Strait, indicating that the results for these two passages were particularly sensitive to the selected spatial tolerance.

Four candidate chokepoints—Bosporus Strait, Balabac Strait, Bering Strait, and Kerch Strait—were not traversed under any of the evaluated specifications: their minimum distances from the closest reconstructed routes were approximately 71.3, 162.0, 2,617.3, and 829.7 km, respectively. They were therefore excluded from the final monthly network. The exclusion indicates that none of the reconstructed LNG routes in the study period passed within 25 km of their operational geometries, but it does not imply that the passages have no relevance to maritime traffic outside the analysed LNG network.

The 25-km tolerance was retained because it accommodates spatial misalignments between the simplified `SeaRoute` paths and the operational transit geometries while applying a uniform and transparent classification rule to all candidate passages. The exact-intersection assignments were preserved separately for reproducibility, and the sensitivity procedure did not alter the reconstructed route geometries.

Appendix C

Model-ready panels variables

This appendix documents the variables contained in the three model-ready panels described in Section 3.5. Variables are grouped according to their analytical role rather than their physical order in the CSV files. Flow variables are measured monthly, binary indicators take the value one when the stated condition is satisfied, and variables carrying the suffix `lag1` refer to month $t-1$. Log-transformed variables were calculated using $\log(1+x)$.

The three panels cover January 2020–December 2024. Consequently, all one-month lagged variables are unavailable in January 2020. Measures based on historical neighbours are additionally unavailable until a node has acquired at least one neighbour through a connection observed before the current month. Commercial criticality measures that require realised trade are defined only for active terminal–months.

C.1 Node activity panel

Table C.1: Variables in the node activity panel

Variable	Definition	Role
`node_id`	Unique node identifier.	Identifier
`period_month`	Calendar month of observation.	Identifier
`node_name`	Node name.	Identifier
`node_type`	Node category: terminal or chokepoint.	Covariate
`country`	Country in which a terminal is located; not applicable to chokepoints.	Covariate
`active`	Equals one when the node participates in the monthly network and zero otherwise.	Outcome
`incoming_flow`	Total LNG volume entering the node in month (t).	Descriptive
`outgoing_flow`	Total LNG volume leaving the node in month (t).	Descriptive

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Table C.1 continued

Variable	Definition	Role
incident_flow	Sum of incoming and outgoing LNG volume incident to the node.	Descriptive
node_throughput	Node throughput, adjusted for double counting at chokepoints.	Descriptive
in_degree	Number of distinct incoming neighbours in the monthly directed network.	Descriptive
out_degree	Number of distinct outgoing neighbours in the monthly directed network.	Descriptive
total_degree	Sum of monthly in-degree and out-degree.	Descriptive
in_voyages	Number of incoming voyage movements associated with the node.	Descriptive
out_voyages	Number of outgoing voyage movements associated with the node.	Descriptive
incident_voyage_traversals	Sum of incoming and outgoing voyage traversals incident to the node.	Descriptive
total_voyages	Number of voyages associated with the node, adjusted for double counting at chokepoints.	Descriptive
in_route_exposure	Number of reconstructed routes entering the node.	Descriptive
out_route_exposure	Number of reconstructed routes leaving the node.	Descriptive
total_route_exposure	Total reconstructed-route exposure of the node.	Descriptive
betweenness	Normalised weighted betweenness centrality in the monthly segment-level network.	Descriptive
pagerank	Weighted PageRank in the monthly directed segment-level network.	Descriptive
share_monthly_network_flow	Share of total monthly node throughput associated with the node.	Descriptive
months_since_last_active	Number of months elapsed since the node was last active; unavailable before its first observed active month.	Supplementary
infrastructure_type	Terminal infrastructure role, classified as import or export; not applicable to chokepoints.	Covariate
capacity_mtpa	Annual terminal processing capacity in million tonnes per annum; not applicable to chokepoints.	Covariate
log1p_capacity_mtpa	Log-transformed annual terminal processing capacity.	Covariate
start_year	Year in which the terminal began operating; not applicable to chokepoints.	Covariate
activity_lag1	Activity status in the preceding month.	Covariate

Continued on next page

Table C.1 continued

Variable	Definition	Role
throughput_lag1	Node throughput in the preceding month.	Covariate
log1p_throughput_lag1	Log-transformed node throughput in the preceding month.	Covariate
voyage_count_lag1	Number of voyages associated with the node in the preceding month.	Covariate
log1p_voyage_count_lag1	Log-transformed voyage count in the preceding month.	Covariate
historical_neighbor_count_lag1	Number of neighbours connected to the node by at least one edge observed up to (t-1).	Covariate
active_neighbor_count_lag1	Number of historical neighbours active in (t-1).	Covariate
neighbor_activity_lag1	Share of historical neighbours active in (t-1).	Covariate
lag_available	Equals one when preceding-month values can be calculated and zero in January 2020.	Availability
year	Calendar year.	Time index
month	Calendar month, coded from 1 to 12.	Time index
month_sin	Sine transformation of the calendar month.	Covariate
month_cos	Cosine transformation of the calendar month.	Covariate

C.2 Terminal criticality panel

Table C.2: Variables in the terminal criticality panel

Variable	Definition	Role
terminal_id	Unique terminal identifier.	Identifier
node_name	Terminal name.	Identifier
period_month	Calendar month of observation.	Identifier
year	Calendar year.	Time index
month	Calendar month, coded from 1 to 12.	Time index
node_type	Node category, equal to terminal throughout this panel.	Descriptor
country	Country in which the terminal is located.	Covariate
infrastructure_type	Terminal infrastructure role, classified as import or export.	Covariate
capacity_mtpa	Annual processing capacity in million tonnes per annum.	Covariate
log1p_capacity_mtpa	Log-transformed annual processing capacity.	Covariate

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Table C.2 continued

Variable	Definition	Role
unit_count	Number of infrastructure units associated with the terminal.	Covariate
start_year	Year in which the terminal began operating.	Covariate
terminal_age	Difference between the observation year and the terminal start year; unavailable before the start year.	Covariate
terminal_role	Observed monthly role: exporter, importer, or inactive.	Descriptor
active	Equals one when the terminal participates in LNG trade in month (t).	Conditioning indicator
export_share	Share of terminal throughput represented by outgoing flow; defined for active terminal–months.	Descriptor
outgoing_flow	Total LNG volume exported by the terminal.	Descriptive
incoming_flow	Total LNG volume imported by the terminal.	Descriptive
throughput	Sum of outgoing and incoming LNG volume.	Descriptive
outgoing_voyages	Number of voyages departing from the terminal.	Descriptive
incoming_voyages	Number of voyages arriving at the terminal.	Descriptive
voyage_count	Total number of voyages associated with the terminal.	Descriptive
counterparty_count	Number of distinct terminal counterparties.	Descriptive
counterparty_country_count	Number of distinct counterparty countries.	Descriptive
counterparty_hhi_terminal	Role-specific HHI of flows across terminal counterparties.	Outcome
counterparty_hhi_country	Role-specific HHI of flows across counterparty countries.	Outcome
effective_counterparties_terminal	Effective number of terminal counterparties, calculated as the inverse of the terminal-level HHI.	Supplementary
effective_counterparties_country	Effective number of counterparty countries, calculated as the inverse of the country-level HHI.	Supplementary
max_dependence_generated	Maximum recipient dependence generated by an exporting terminal.	Supplementary
weighted_dependence_generated	Exporter-generated dependence weighted by the distribution of its outgoing flow.	Supplementary
country_terminal_dependence	Importing terminal's share of total LNG imports received by its country.	Supplementary
role_specific_dependence	Dependence measure selected according to the terminal's observed monthly role.	Outcome

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Table C.2 continued

Variable	Definition	Role
import_pagerank	Weighted PageRank calculated in the observed direction of LNG flows.	Supplementary
export_pagerank	Weighted PageRank calculated after reversing the direction of LNG flows.	Supplementary
role_specific_pagerank	Directional PageRank selected according to the terminal's observed monthly role.	Outcome
degree_undirected	Degree in the undirected projection of the active monthly network.	Structural
is_articulation_point	Equals one when removal of the terminal increases the number of connected components.	Outcome
global_bridge_count	Number of global bridges incident to the terminal.	Structural
global_bridge_share	Share of incident undirected edges classified as global bridges.	Structural
local_bridge_only_count	Number of incident local bridges after excluding global bridges.	Structural
local_bridge_only_share	Share of incident undirected edges classified as local-only bridges.	Structural
has_incident_global_bridge	Equals one when the terminal is incident to at least one global bridge.	Outcome
connected_to_articulation_point	Equals one when the terminal is directly connected to an articulation point.	Outcome
terminal_role_lag1	Observed terminal role in the preceding month: exporter, importer, or inactive.	Covariate
activity_lag1	Terminal activity status in the preceding month.	Covariate
throughput_lag1	Terminal throughput in the preceding month.	Covariate
log1p_throughput_lag1	Log-transformed terminal throughput in the preceding month.	Covariate
voyage_count_lag1	Terminal voyage count in the preceding month.	Covariate
log1p_voyage_count_lag1	Log-transformed terminal voyage count in the preceding month.	Covariate
mean_voyage_distance_lag1	Role-relevant mean voyage distance in the preceding month; undefined when no applicable voyage was observed.	Covariate
log1p_mean_voyage_distance_lag1	Log-transformed preceding-month role-relevant mean voyage distance; undefined when no applicable voyage was observed.	Covariate
historical_neighbor_count_lag1	Number of historical neighbours observed up to (t-1).	Covariate
active_neighbor_count_lag1	Number of historical neighbours active in (t-1).	Covariate

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Table C.2 continued

Variable	Definition	Role
lag_available	Equals one when the preceding calendar month is available and zero for January 2020.	Availability
neighbor_activity_lag1	Share of historical neighbours active in (t-1).	Covariate
month_sin	Sine transformation of the calendar month.	Covariate
month_cos	Cosine transformation of the calendar month.	Covariate

C.3 Chokepoint criticality panel

Table C.3: Variables in the chokepoint criticality panel

Variable	Definition	Role
node_id	Unique chokepoint identifier.	Identifier
period_month	Calendar month of observation.	Identifier
node_name	Chokepoint name.	Identifier
node_type	Node category, equal to chokepoint throughout this panel.	Descriptor
active	Equals one when the chokepoint is traversed in month (t).	Conditioning indicator
incoming_flow	Total LNG volume entering the chokepoint.	Descriptive
outgoing_flow	Total LNG volume leaving the chokepoint.	Descriptive
incident_flow	Sum of incoming and outgoing LNG volume incident to the chokepoint.	Descriptive
node_throughput	Chokepoint throughput, equal to one half of incident flow.	Descriptive
in_degree	Number of distinct incoming neighbours.	Descriptive
out_degree	Number of distinct outgoing neighbours.	Descriptive
total_degree	Sum of directed in-degree and out-degree.	Descriptive
in_voyages	Number of incoming voyage traversals.	Descriptive
out_voyages	Number of outgoing voyage traversals.	Descriptive
incident_voyage_traversals	Sum of incoming and outgoing voyage traversals.	Descriptive
total_voyages	Chokepoint voyage count adjusted for double counting across incoming and outgoing segments.	Descriptive
in_route_exposure	Number of reconstructed routes entering the chokepoint.	Descriptive
out_route_exposure	Number of reconstructed routes leaving the chokepoint.	Descriptive

Continued on next page

Table C.3 continued

Variable	Definition	Role
total_route_exposure	Total reconstructed-route exposure of the chokepoint.	Descriptive
betweenness	Normalised weighted betweenness centrality.	Outcome
pagerank	Weighted PageRank in the monthly directed segment-level network.	Outcome
share_monthly_network_flow	Share of total monthly node throughput associated with the chokepoint.	Outcome
months_since_last_active	Number of months since the chokepoint was last active; unavailable before its first active month.	Supplementary
degree_undirected	Degree in the undirected projection of the active monthly network.	Structural
is_articulation_point	Equals one when removal of the chokepoint increases the number of connected components.	Outcome
global_bridge_count	Number of global bridges incident to the chokepoint.	Structural
global_bridge_share	Share of incident undirected edges classified as global bridges.	Structural
local_bridge_only_count	Number of incident local bridges after excluding global bridges.	Structural
local_bridge_only_share	Share of incident undirected edges classified as local-only bridges.	Structural
has_incident_global_bridge	Equals one when the chokepoint is incident to at least one global bridge.	Outcome
connected_to_articulation_point	Equals one when the chokepoint is directly connected to an articulation point.	Outcome
activity_lag1	Chokepoint activity status in the preceding month.	Covariate
throughput_lag1	Chokepoint throughput in the preceding month.	Covariate
log1p_throughput_lag1	Log-transformed chokepoint throughput in the preceding month.	Covariate
voyage_count_lag1	Chokepoint voyage count in the preceding month.	Covariate
log1p_voyage_count_lag1	Log-transformed chokepoint voyage count in the preceding month.	Covariate
mean_voyage_distance_lag1	Mean voyage distance for voyages traversing the chokepoint in the preceding month; undefined when no applicable voyage was observed.	Covariate
log1p_mean_voyage_distance_lag1	Log-transformed preceding-month mean voyage distance.	Covariate

Continued on next page

Table C.3 continued

Variable	Definition	Role
historical_neighbor_count_lag1	Number of historical neighbours observed up to (t-1).	Covariate
active_neighbor_count_lag1	Number of historical neighbours active in (t-1).	Covariate
neighbor_activity_lag1	Share of historical neighbours active in (t-1).	Covariate
lag_available	Equals one when preceding-month information can be calculated.	Availability
year	Calendar year.	Time index
month	Calendar month, coded from 1 to 12.	Time index
month_sin	Sine transformation of the calendar month.	Covariate
month_cos	Cosine transformation of the calendar month.	Covariate

Appendix D

Full conditional posterior distributions

This appendix derives the full conditional posterior distributions for the rank-one baseline model. The derivations apply separately to terminals and chokepoints, using the response likelihoods and original prior specification introduced in Sections 5.1.2 and 5.1.3. Each full conditional is obtained from the joint posterior in Equation 43 by retaining the factors that depend on the quantity under consideration. Multiplicative terms that are constant with respect to that quantity are absorbed into the normalising constant.

The notation follows the main text: $n \in \mathcal{O}$ indexes the node–month observations, $N = |\mathcal{O}|$, and $\mathcal{I}_j = \{n \in \mathcal{O} : 0 < Y_{nj} < 1\}$ contains the interior observations for bounded response j. Throughout, *rest* denotes all remaining unknown quantities together with the observed data and covariates.

D.1 Factor-regression coefficients

Let $\mathbf{X}$ be the $N \times q$ covariate design matrix, with row $\mathbf{x}_n^\top$, and let $\mathbf{f}$ collect the corresponding latent factors.

Conditional on $\mathbf{f}$, the response likelihood does not depend on $\boldsymbol{\beta}$. Retaining the Gaussian factor-regression density and the independent $\mathcal{N}(0, 0.5^2)$ coefficient priors therefore gives

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto p(\mathbf{f} \mid \boldsymbol{\beta}, \mathbf{X}) p(\boldsymbol{\beta}).$$

Writing the relevant densities explicitly,

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \exp\left\{ -\frac{1}{2}(f_n - \mathbf{x}_n^\top \boldsymbol{\beta})^2 \right\} \right] \times \left[\prod_{r=1}^{q} \exp\left\{ -\frac{\beta_r^2}{2(0.5)^2} \right\} \right].$$

Combining the exponential terms and using matrix notation yields

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto \exp\left\{ -\frac{1}{2}\Big[(\mathbf{f}-\mathbf{X}\boldsymbol{\beta})^{\mathsf{T}}(\mathbf{f}-\mathbf{X}\boldsymbol{\beta}) + 4\boldsymbol{\beta}^{\mathsf{T}}\boldsymbol{\beta}\Big]\right\}.$$

The first quadratic term expands as

$$(\mathbf{f}-\mathbf{X}\boldsymbol{\beta})^{\mathsf{T}}(\mathbf{f}-\mathbf{X}\boldsymbol{\beta}) = \mathbf{f}^{\mathsf{T}}\mathbf{f} - 2\boldsymbol{\beta}^{\mathsf{T}}\mathbf{X}^{\mathsf{T}}\mathbf{f} + \boldsymbol{\beta}^{\mathsf{T}}\mathbf{X}^{\mathsf{T}}\mathbf{X}\boldsymbol{\beta}.$$

Removing $\mathbf{f}^{\mathsf{T}}\mathbf{f}$, which does not depend on $\boldsymbol{\beta}$, gives

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto \exp\left\{ -\frac{1}{2}\Big[\boldsymbol{\beta}^{\mathsf{T}}(\mathbf{X}^{\mathsf{T}}\mathbf{X}+4\mathbf{I}_q)\boldsymbol{\beta} - 2\boldsymbol{\beta}^{\mathsf{T}}\mathbf{X}^{\mathsf{T}}\mathbf{f}\Big]\right\},$$

where $\mathbf{I}_q$ is the $q \times q$ identity matrix.

Define

$$\mathbf{V}_\beta = (\mathbf{X}^{\mathsf{T}}\mathbf{X}+4\mathbf{I}_q)^{-1}, \qquad \mathbf{m}_\beta = \mathbf{V}_\beta\mathbf{X}^{\mathsf{T}}\mathbf{f}.$$

Since $\mathbf{V}_\beta^{-1}\mathbf{m}_\beta = \mathbf{X}^{\mathsf{T}}\mathbf{f}$, completing the square gives

$$\begin{aligned}&\boldsymbol{\beta}^{\mathsf{T}}\mathbf{V}_\beta^{-1}\boldsymbol{\beta} - 2\boldsymbol{\beta}^{\mathsf{T}}\mathbf{X}^{\mathsf{T}}\mathbf{f} \\ &\quad = (\boldsymbol{\beta}-\mathbf{m}_\beta)^{\mathsf{T}}\mathbf{V}_\beta^{-1}(\boldsymbol{\beta}-\mathbf{m}_\beta) - \mathbf{m}_\beta^{\mathsf{T}}\mathbf{V}_\beta^{-1}\mathbf{m}_\beta.\end{aligned}$$

The final term is constant with respect to $\boldsymbol{\beta}$. Thus,

$$p(\boldsymbol{\beta} \mid \text{rest}) \propto \exp\left\{-\frac{1}{2}(\boldsymbol{\beta}-\mathbf{m}_\beta)^{\mathsf{T}}\mathbf{V}_\beta^{-1}(\boldsymbol{\beta}-\mathbf{m}_\beta)\right\},$$

which identifies the conjugate full conditional

$$\boldsymbol{\beta} \mid \text{rest} \sim \mathcal{N}_q\big((\mathbf{X}^{\mathsf{T}}\mathbf{X}+4\mathbf{I}_q)^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{f},\ (\mathbf{X}^{\mathsf{T}}\mathbf{X}+4\mathbf{I}_q)^{-1}\big).$$

D.2 Response loadings

D.2.1 Binary-response loadings

For binary response k,

$$B_{nk} \mid f_n, a_k, \ell_k \sim \text{Bernoulli}(p_{nk}), \qquad \text{logit}(p_{nk}) = a_k + \ell_k f_n.$$

Holding the factors and other response parameters fixed, only the likelihood of binary response k and the prior of ℓ_k depend on this loading. Therefore,

$$p(\ell_k \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} p(B_{nk} \mid f_n, a_k, \ell_k)\right] p(\ell_k).$$

The Bernoulli probability is

$$p(B_{nk} \mid f_n, a_k, \ell_k) = p_{nk}^{B_{nk}} (1 - p_{nk})^{1 - B_{nk}}.$$

Using

$$p_{nk} = \frac{\exp(a_k + \ell_k f_n)}{1 + \exp(a_k + \ell_k f_n)}, \qquad 1 - p_{nk} = \frac{1}{1 + \exp(a_k + \ell_k f_n)},$$

this becomes

$$p(B_{nk} \mid f_n, a_k, \ell_k) = \frac{\exp\{B_{nk}(a_k + \ell_k f_n)\}}{1 + \exp(a_k + \ell_k f_n)}.$$

The prior $\ell_k \sim \mathcal{N}(0, 1)$ has density

$$p(\ell_k) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\ell_k^2}{2}\right).$$

Combining likelihood and prior, and absorbing the Normal normalising constant, gives

$$p(\ell_k \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \frac{\exp\{B_{nk}(a_k + \ell_k f_n)\}}{1 + \exp(a_k + \ell_k f_n)}\right] \exp\left(-\frac{\ell_k^2}{2}\right).$$

This expression applies to $k = 1, 2, 3$ for terminals and $k = 1, 2$ for chokepoints.

D.2.2 Ordinary Beta-response loadings

For an ordinary Beta response,

$$Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j \sim \text{Beta}\big(\mu_{nj}\phi_j, (1 - \mu_{nj})\phi_j\big), \qquad \text{logit}(\mu_{nj}) = \alpha_j + \lambda_j f_n.$$

Its density is

$$p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j) = \frac{\Gamma(\phi_j)}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)} Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}.$$

The loading enters this density through μ_{nj}. Retaining the relevant likelihood and prior gives

$$p(\lambda_j \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j)\right] p(\lambda_j).$$

Unrestricted loading of chokepoint flow share

For chokepoint flow share, $j = 3$ and $\lambda_3 \sim \mathcal{N}(0, 1)$, with density

$$p(\lambda_3) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\lambda_3^2}{2}\right).$$

Substituting the Beta likelihood and Normal prior gives

$$p(\lambda_3 \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \frac{\Gamma(\phi_3) Y_{n3}^{\mu_{n3}\phi_3 - 1}(1 - Y_{n3})^{(1-\mu_{n3})\phi_3 - 1}}{\Gamma(\mu_{n3}\phi_3)\Gamma((1-\mu_{n3})\phi_3)}\right] \times \exp\left(-\frac{\lambda_3^2}{2}\right).$$

Since ϕ_3 is held fixed, $\Gamma(\phi_3)$ does not depend on λ_3 and can be absorbed into the normalising constant. The resulting kernel is

$$p(\lambda_3 \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \frac{Y_{n3}^{\mu_{n3}\phi_3 - 1}(1 - Y_{n3})^{(1-\mu_{n3})\phi_3 - 1}}{\Gamma(\mu_{n3}\phi_3)\Gamma((1-\mu_{n3})\phi_3)}\right] \times \exp\left(-\frac{\lambda_3^2}{2}\right).$$

Positive PageRank loading

For PageRank in both node types, $j = 1$ and $\lambda_1 \sim \text{HalfNormal}(1)$. Its prior density is

$$p(\lambda_1) = \sqrt{\frac{2}{\pi}} \exp\left(-\frac{\lambda_1^2}{2}\right) \mathbb{1}(\lambda_1 > 0).$$

Combining this prior with the PageRank likelihood gives

$$p(\lambda_1 \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\Gamma(\phi_1)Y_{n1}^{\mu_{n1}\phi_1-1}(1-Y_{n1})^{(1-\mu_{n1})\phi_1-1}}{\Gamma(\mu_{n1}\phi_1)\Gamma((1-\mu_{n1})\phi_1)}\right] \times \exp\left(-\frac{\lambda_1^2}{2}\right)\mathbb{1}(\lambda_1 > 0).$$

Removing the factors $\Gamma(\phi_1)$, which are constant with respect to λ_1, yields

$$p(\lambda_1 \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{Y_{n1}^{\mu_{n1}\phi_1-1}(1-Y_{n1})^{(1-\mu_{n1})\phi_1-1}}{\Gamma(\mu_{n1}\phi_1)\Gamma((1-\mu_{n1})\phi_1)}\right] \times \exp\left(-\frac{\lambda_1^2}{2}\right)\mathbb{1}(\lambda_1 > 0).$$

D.2.3 Interior-mean loadings of boundary-inflated Beta responses

Terminal HHI and dependence

For terminal HHI and dependence, $j = 2, 3$, define $Z_{nj} = \mathbb{1}(Y_{nj} = 1)$. The boundary probability satisfies

$$\text{logit}(\pi_{nj}) = c_j^T + d_j^T f_n.$$

Separating the boundary and interior observations, the likelihood for response j is

$$\prod_{n\in\mathcal{O}} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, c_j^T, d_j^T)$$
$$= \left[\prod_{\substack{n\in\mathcal{O}:\\ Y_{nj}=1}} \pi_{nj}\right] \left[\prod_{n\in\mathcal{I}_j} (1-\pi_{nj}) \frac{\Gamma(\phi_j)Y_{nj}^{\mu_{nj}\phi_j-1}(1-Y_{nj})^{(1-\mu_{nj})\phi_j-1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right].$$

The probabilities π_{nj} depend on c_j^T, d_j^T, and f_n, but not on λ_j. Their contributions are therefore constant in the full conditional of λ_j. Consequently,

$$p(\lambda_j \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{I}_j} \frac{\Gamma(\phi_j)Y_{nj}^{\mu_{nj}\phi_j-1}(1-Y_{nj})^{(1-\mu_{nj})\phi_j-1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] p(\lambda_j).$$

Substituting the $\mathcal{N}(0,1)$ prior and absorbing the factors $\Gamma(\phi_j)$ gives

$$p(\lambda_j \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{I}_j} \frac{Y_{nj}^{\mu_{nj}\phi_j-1}(1-Y_{nj})^{(1-\mu_{nj})\phi_j-1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \times \exp\left(-\frac{\lambda_j^2}{2}\right), \qquad j=2,3.$$

Thus, exact ones do not contribute to the full conditional of the interior-mean loading when the factors and boundary parameters are held fixed.

Chokepoint betweenness

For chokepoint betweenness, define $Z_{n2} = \mathbb{1}(Y_{n2}=0)$, with

$$\operatorname{logit}(\rho_{n2}) = c_2^C + d_2^C f_n.$$

The response likelihood is

$$\prod_{n\in\mathcal{O}} p(Y_{n2} \mid f_n, \alpha_2, \lambda_2, \phi_2, c_2^C, d_2^C) = \left[\prod_{\substack{n\in\mathcal{O}:\\ Y_{n2}=0}} \rho_{n2}\right]\left[\prod_{n\in\mathcal{I}_2}(1-\rho_{n2})\frac{\Gamma(\phi_2)Y_{n2}^{\mu_{n2}\phi_2-1}(1-Y_{n2})^{(1-\mu_{n2})\phi_2-1}}{\Gamma(\mu_{n2}\phi_2)\Gamma((1-\mu_{n2})\phi_2)}\right].$$

Since ρ_{n2} does not depend on λ_2, only the interior Beta densities and the loading prior remain:

$$p(\lambda_2 \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{I}_2} \frac{\Gamma(\phi_2)Y_{n2}^{\mu_{n2}\phi_2-1}(1-Y_{n2})^{(1-\mu_{n2})\phi_2-1}}{\Gamma(\mu_{n2}\phi_2)\Gamma((1-\mu_{n2})\phi_2)}\right] p(\lambda_2).$$

Using $\lambda_2 \sim \mathcal{N}(0,1)$ and removing terms constant with respect to λ_2 yields

$$p(\lambda_2 \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{I}_2} \frac{Y_{n2}^{\mu_{n2}\phi_2-1}(1-Y_{n2})^{(1-\mu_{n2})\phi_2-1}}{\Gamma(\mu_{n2}\phi_2)\Gamma((1-\mu_{n2})\phi_2)}\right] \times \exp\left(-\frac{\lambda_2^2}{2}\right).$$

As in the one-inflated case, only interior observations contribute to this loading conditional.

D.2.4 Loadings of the probability of mass at one

For terminal HHI and dependence, the indicators $Z_{nj} = \mathbb{1}(Y_{nj} = 1)$ satisfy

$$Z_{nj} \mid f_n, c_j^T, d_j^T \sim \text{Bernoulli}(\pi_{nj}), \qquad \text{logit}(\pi_{nj}) = c_j^T + d_j^T f_n, \qquad j = 2, 3.$$

When deriving the full conditional of d_j^T, the interior Beta densities are constant because their parameters do not depend on d_j^T. The remaining likelihood consists of the boundary probabilities for all observations:

$$p(d_j^T \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \pi_{nj}^{Z_{nj}} (1 - \pi_{nj})^{1 - Z_{nj}} \right] p(d_j^T).$$

From the logistic specification,

$$\pi_{nj} = \frac{\exp(c_j^T + d_j^T f_n)}{1 + \exp(c_j^T + d_j^T f_n)}, \qquad 1 - \pi_{nj} = \frac{1}{1 + \exp(c_j^T + d_j^T f_n)}.$$

Hence,

$$\pi_{nj}^{Z_{nj}} (1 - \pi_{nj})^{1 - Z_{nj}} = \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1 + \exp(c_j^T + d_j^T f_n)}.$$

The prior $d_j^T \sim \mathcal{N}(0, 1)$ has density

$$p(d_j^T) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(d_j^T)^2}{2} \right).$$

Combining these terms gives

$$\begin{aligned} p(d_j^T \mid \text{rest}) \propto & \left[\prod_{n \in \mathcal{O}} \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1 + \exp(c_j^T + d_j^T f_n)} \right] \\ & \times \exp\left(-\frac{(d_j^T)^2}{2} \right), \qquad j = 2, 3. \end{aligned}$$

Both boundary and interior observations contribute: the former have $Z_{nj} = 1$, while the latter have $Z_{nj} = 0$.

D.2.5 Loading of the probability of mass at zero

For chokepoint betweenness, $Z_{n2} = \mathbb{1}(Y_{n2} = 0)$ satisfies

$$Z_{n2} \mid f_n, c_2^C, d_2^C \sim \text{Bernoulli}(\rho_{n2}), \qquad \text{logit}(\rho_{n2}) = c_2^C + d_2^C f_n.$$

The interior Beta densities do not depend on d_2^C. Retaining the boundary-indicator

likelihood and the loading prior therefore gives

$$p(d_2^C \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \rho_{n2}^{Z_{n2}}(1-\rho_{n2})^{1-Z_{n2}}\right] p(d_2^C).$$

Using

$$\rho_{n2} = \frac{\exp(c_2^C + d_2^C f_n)}{1+\exp(c_2^C + d_2^C f_n)}, \qquad 1-\rho_{n2} = \frac{1}{1+\exp(c_2^C + d_2^C f_n)},$$

the likelihood contribution becomes

$$\rho_{n2}^{Z_{n2}}(1-\rho_{n2})^{1-Z_{n2}} = \frac{\exp\{Z_{n2}(c_2^C + d_2^C f_n)\}}{1+\exp(c_2^C + d_2^C f_n)}.$$

The prior $d_2^C \sim \mathcal{N}(0,1)$ has density

$$p(d_2^C) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(d_2^C)^2}{2}\right).$$

The resulting full conditional kernel is

$$p(d_2^C \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\exp\{Z_{n2}(c_2^C + d_2^C f_n)\}}{1+\exp(c_2^C + d_2^C f_n)}\right] \times \exp\left(-\frac{(d_2^C)^2}{2}\right).$$

Here exact zeros have $Z_{n2} = 1$ and interior observations have $Z_{n2} = 0$, so both contribute to inference on the zero-probability loading.

D.3 Response intercepts

D.3.1 Bounded-response interior intercepts

For each bounded response, the interior intercept enters the Beta mean through

$$\text{logit}(\mu_{nj}) = \alpha_j + \lambda_j f_n.$$

Its prior is $\alpha_j \sim \mathcal{N}(m_j, 1.5^2)$, where $(m_1, m_2, m_3) = (-4, 0, 0)$ for terminals and $(-4, -5, -4)$ for chokepoints.

For ordinary Beta responses, all observations contribute to the likelihood of α_j. For boundary-inflated responses, the boundary probabilities do not depend on α_j, so

only the interior Beta densities remain. Both cases can therefore be written as

$$p(\alpha_j \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{I}_j} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, 0 < Y_{nj} < 1)\right] p(\alpha_j).$$

For ordinary Beta responses, $\mathcal{I}_j = \mathcal{O}$.

The interior density is

$$\begin{aligned} &p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, 0 < Y_{nj} < 1) \\ &\quad = \frac{\Gamma(\phi_j) Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}. \end{aligned}$$

The prior density is

$$p(\alpha_j) = \frac{1}{1.5\sqrt{2\pi}} \exp\left\{-\frac{(\alpha_j - m_j)^2}{2(1.5)^2}\right\}.$$

Substituting these densities gives

$$\begin{aligned} p(\alpha_j \mid \text{rest}) \propto &\left[\prod_{n\in\mathcal{I}_j} \frac{\Gamma(\phi_j) Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ &\times \exp\left\{-\frac{(\alpha_j - m_j)^2}{2(1.5)^2}\right\}. \end{aligned}$$

Since ϕ_j is held fixed, the factors $\Gamma(\phi_j)$ do not depend on α_j. Absorbing them into the normalising constant yields

$$\begin{aligned} p(\alpha_j \mid \text{rest}) \propto &\left[\prod_{n\in\mathcal{I}_j} \frac{Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ &\times \exp\left\{-\frac{(\alpha_j - m_j)^2}{2(1.5)^2}\right\}, \qquad j = 1, 2, 3. \end{aligned}$$

The likelihood dependence on α_j enters through μ_{nj}.

D.3.2 Binary-response intercepts

For binary response k,

$$B_{nk} \mid f_n, a_k, \ell_k \sim \text{Bernoulli}(p_{nk}), \qquad \text{logit}(p_{nk}) = a_k + \ell_k f_n.$$

Only the likelihood of response k and the prior of a_k depend on this intercept.

Therefore,

$$p(a_k \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} p(B_{nk} \mid f_n, a_k, \ell_k)\right] p(a_k).$$

Using the logistic specification,

$$\begin{aligned} p(B_{nk} \mid f_n, a_k, \ell_k) &= p_{nk}^{B_{nk}}(1-p_{nk})^{1-B_{nk}} \\ &= \frac{\exp\{B_{nk}(a_k+\ell_k f_n)\}}{1+\exp(a_k+\ell_k f_n)}. \end{aligned}$$

The prior $a_k \sim \mathcal{N}(0, 2^2)$ has density

$$p(a_k) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{a_k^2}{8}\right).$$

Combining likelihood and prior gives

$$p(a_k \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\exp\{B_{nk}(a_k+\ell_k f_n)\}}{1+\exp(a_k+\ell_k f_n)}\right] \exp\left(-\frac{a_k^2}{8}\right).$$

Equivalently, collecting the exponential terms and removing those independent of a_k gives

$$\begin{aligned} p(a_k \mid \text{rest}) \propto \exp\Bigg\{ &-\frac{a_k^2}{8} + a_k \sum_{n\in\mathcal{O}} B_{nk} \\ &- \sum_{n\in\mathcal{O}} \log[1+\exp(a_k+\ell_k f_n)] \Bigg\}. \end{aligned}$$

This expression applies to $k = 1, 2, 3$ for terminals and $k = 1, 2$ for chokepoints.

D.3.3 Intercepts of the probability of mass at one

For terminal HHI and dependence, define $Z_{nj} = \mathbb{1}(Y_{nj} = 1)$, with

$$Z_{nj} \mid f_n, c_j^T, d_j^T \sim \text{Bernoulli}(\pi_{nj}), \qquad \text{logit}(\pi_{nj}) = c_j^T + d_j^T f_n, \qquad j = 2, 3.$$

The interior Beta densities do not depend on c_j^T. Retaining the boundary-indicator likelihood and the intercept prior gives

$$p(c_j^T \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \pi_{nj}^{Z_{nj}}(1-\pi_{nj})^{1-Z_{nj}}\right] p(c_j^T).$$

The logistic link implies

$$\pi_{nj}^{Z_{nj}}(1-\pi_{nj})^{1-Z_{nj}} = \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1+\exp(c_j^T + d_j^T f_n)}.$$

The prior $c_j^T \sim \mathcal{N}(0, 1.5^2)$ has density

$$p(c_j^T) = \frac{1}{1.5\sqrt{2\pi}} \exp\left\{ -\frac{(c_j^T)^2}{2(1.5)^2} \right\}.$$

Substituting gives

$$p(c_j^T \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \frac{\exp\{Z_{nj}(c_j^T + d_j^T f_n)\}}{1+\exp(c_j^T + d_j^T f_n)} \right] \times \exp\left\{ -\frac{(c_j^T)^2}{2(1.5)^2} \right\}, \qquad j = 2, 3.$$

Equivalently, removing terms independent of c_j^T yields

$$p(c_j^T \mid \text{rest}) \propto \exp\left\{ -\frac{(c_j^T)^2}{2(1.5)^2} + c_j^T \sum_{n\in\mathcal{O}} Z_{nj} - \sum_{n\in\mathcal{O}} \log\left[1+\exp(c_j^T + d_j^T f_n)\right] \right\}, \qquad j = 2, 3.$$

D.3.4 Intercept of the probability of mass at zero

For chokepoint betweenness, define $Z_{n2} = \mathbb{1}(Y_{n2} = 0)$, with

$$Z_{n2} \mid f_n, c_2^C, d_2^C \sim \text{Bernoulli}(\rho_{n2}), \qquad \text{logit}(\rho_{n2}) = c_2^C + d_2^C f_n.$$

The interior Beta densities do not depend on c_2^C. Its full conditional therefore satisfies

$$p(c_2^C \mid \text{rest}) \propto \left[\prod_{n\in\mathcal{O}} \rho_{n2}^{Z_{n2}}(1-\rho_{n2})^{1-Z_{n2}} \right] p(c_2^C).$$

Using the logistic link,

$$\rho_{n2}^{Z_{n2}}(1-\rho_{n2})^{1-Z_{n2}} = \frac{\exp\{Z_{n2}(c_2^C + d_2^C f_n)\}}{1+\exp(c_2^C + d_2^C f_n)}.$$

The prior $c_2^C \sim \mathcal{N}(-5, 1.5^2)$ has density

$$p(c_2^C) = \frac{1}{1.5\sqrt{2\pi}} \exp\left\{ -\frac{(c_2^C + 5)^2}{2(1.5)^2} \right\}.$$

Combining these expressions gives

$$p(c_2^C \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{O}} \frac{\exp\{Z_{n2}(c_2^C + d_2^C f_n)\}}{1 + \exp(c_2^C + d_2^C f_n)} \right] \times \exp\left\{ -\frac{(c_2^C + 5)^2}{2(1.5)^2} \right\}.$$

Equivalently, removing terms independent of c_2^C yields

$$p(c_2^C \mid \text{rest}) \propto \exp\left\{ -\frac{(c_2^C + 5)^2}{2(1.5)^2} + c_2^C \sum_{n \in \mathcal{O}} Z_{n2} - \sum_{n \in \mathcal{O}} \log\left[1 + \exp(c_2^C + d_2^C f_n)\right] \right\}.$$

D.4 Beta precision parameters

For each bounded response, the precision ϕ_j enters both Beta shape parameters:

$$A_{nj} = \mu_{nj}\phi_j, \qquad D_{nj} = (1 - \mu_{nj})\phi_j.$$

When deriving its full conditional, the interior means μ_{nj} are held fixed because they depend on α_j, λ_j, and f_n, rather than on ϕ_j.

For ordinary Beta responses, all observations contribute to the precision likelihood. For boundary-inflated responses, the boundary probabilities do not depend on ϕ_j, so only the interior observations contribute. Retaining the relevant likelihood and prior gives

$$p(\phi_j \mid \text{rest}) \propto \left[\prod_{n \in \mathcal{I}_j} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, 0 < Y_{nj} < 1) \right] p(\phi_j).$$

The interior Beta density is

$$p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, 0 < Y_{nj} < 1) = \frac{\Gamma(\phi_j) Y_{nj}^{\mu_{nj}\phi_j - 1} (1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1 - \mu_{nj})\phi_j)}.$$

The prior is specified on the logarithmic scale:

$$\log \phi_j \sim \mathcal{N}(\log 10, 0.7^2).$$

Expressed as a density for ϕ_j, this becomes

$$p(\phi_j) = \frac{1}{0.7\,\phi_j\sqrt{2\pi}} \exp\left\{-\frac{(\log \phi_j - \log 10)^2}{2(0.7)^2}\right\}, \qquad \phi_j > 0.$$

The factor $1/\phi_j$ follows from the change-of-variables derivative $|d \log \phi_j / d\phi_j| = 1/\phi_j$.

Substituting the likelihood and prior densities, and absorbing constants independent of ϕ_j, gives the full conditional kernel

$$\begin{aligned} p(\phi_j \mid \text{rest}) \propto & \left[\prod_{n \in \mathcal{I}_j} \frac{\Gamma(\phi_j) Y_{nj}^{\mu_{nj}\phi_j - 1}(1 - Y_{nj})^{(1-\mu_{nj})\phi_j - 1}}{\Gamma(\mu_{nj}\phi_j)\Gamma((1-\mu_{nj})\phi_j)}\right] \\ & \times \frac{1}{\phi_j} \exp\left\{-\frac{(\log \phi_j - \log 10)^2}{2(0.7)^2}\right\}, \qquad \phi_j > 0, \quad j = 1, 2, 3. \end{aligned}$$

D.5 Latent factors

To derive the full conditional of a particular latent factor f_n, the regression coefficients, response parameters, and all other latent factors are held fixed. The parameter priors and the likelihood contributions from other observations are therefore constant with respect to f_n. Retaining the factors of the joint posterior that depend on f_n gives

$$p(f_n \mid \text{rest}) \propto p(\mathbf{Y}_n, \mathbf{B}_n \mid f_n, \boldsymbol{\theta}) p(f_n \mid \boldsymbol{\beta}, \mathbf{x}_n).$$

The Gaussian factor-regression density is

$$p(f_n \mid \boldsymbol{\beta}, \mathbf{x}_n) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(f_n - \mathbf{x}_n^{\mathsf{T}}\boldsymbol{\beta}\right)^2\right\}.$$

Conditional independence of the responses gives

$$\begin{aligned} & p(\mathbf{Y}_n, \mathbf{B}_n \mid f_n, \boldsymbol{\theta}) \\ & \quad = \prod_{j=1}^{3} g_j(Y_{nj} \mid f_n, \boldsymbol{\theta}) \prod_{k=1}^{K} p(B_{nk} \mid f_n, a_k, \ell_k), \end{aligned}$$

where g_j is the bounded-response contribution introduced in the main text.

Denote this observation-specific likelihood by $L_n(f_n)$. For terminals, it is

$$\begin{aligned} L_n^T(f_n) = p(Y_{n1} \mid f_n, \alpha_1, \lambda_1, \phi_1) \\ \times \prod_{j=2}^{3} p(Y_{nj} \mid f_n, \alpha_j, \lambda_j, \phi_j, c_j^T, d_j^T) \\ \times \prod_{k=1}^{3} p(B_{nk} \mid f_n, a_k, \ell_k). \end{aligned}$$

The bounded-response terms are the ordinary Beta density for PageRank and the one-inflated Beta contributions for HHI and dependence.

For chokepoints,

$$\begin{aligned} L_n^C(f_n) = p(Y_{n1} \mid f_n, \alpha_1, \lambda_1, \phi_1) \\ \times p(Y_{n2} \mid f_n, \alpha_2, \lambda_2, \phi_2, c_2^C, d_2^C) \\ \times p(Y_{n3} \mid f_n, \alpha_3, \lambda_3, \phi_3) \\ \times \prod_{k=1}^{2} p(B_{nk} \mid f_n, a_k, \ell_k). \end{aligned}$$

Here PageRank and flow share contribute ordinary Beta densities, while betweenness contributes the zero-inflated Beta likelihood.

Combining the observation-specific likelihood with the Gaussian factor-regression density and removing its constant prefactor yields

$$p(f_n \mid \text{rest}) \propto L_n(f_n) \exp\left\{ -\frac{1}{2} \left(f_n - \mathbf{x}_n^\top \boldsymbol{\beta} \right)^2 \right\},$$

where $L_n = L_n^T$ for terminals and $L_n = L_n^C$ for chokepoints.

The factor enters the interior means, binary probabilities, and boundary probabilities. Its full conditional therefore combines information from all responses observed for the same node–month. Although the factor-regression density is Gaussian, the response likelihood is a non-Gaussian function of f_n, so the resulting full conditional does not have a standard conjugate form.

Appendix E

Model diagnostics

E.1 MCMC convergence diagnostics

E.1.1 Trace plots

The following figures present posterior densities and trace plots for selected global parameters of the baseline terminal and chokepoint models, estimated using four chains with 5,000 warm-up iterations and 10,000 retained draws per chain. The displayed pages include parameters selected for the autocorrelation diagnostics, together with additional global parameters appearing on the same plotting pages. Complete plots are retained in the accompanying model outputs.

For terminals, the selection covers the lagged exporter coefficient, the PageRank loading, and the loading for the probability of counterparty HHI equal to one. For chokepoints, it covers the PageRank and betweenness loadings and the betweenness precision.

The displayed chains occupy broadly overlapping ranges, and their estimated posterior densities are similar. These visual checks complement the numerical diagnostics reported in Chapter 5; they do not by themselves establish convergence.

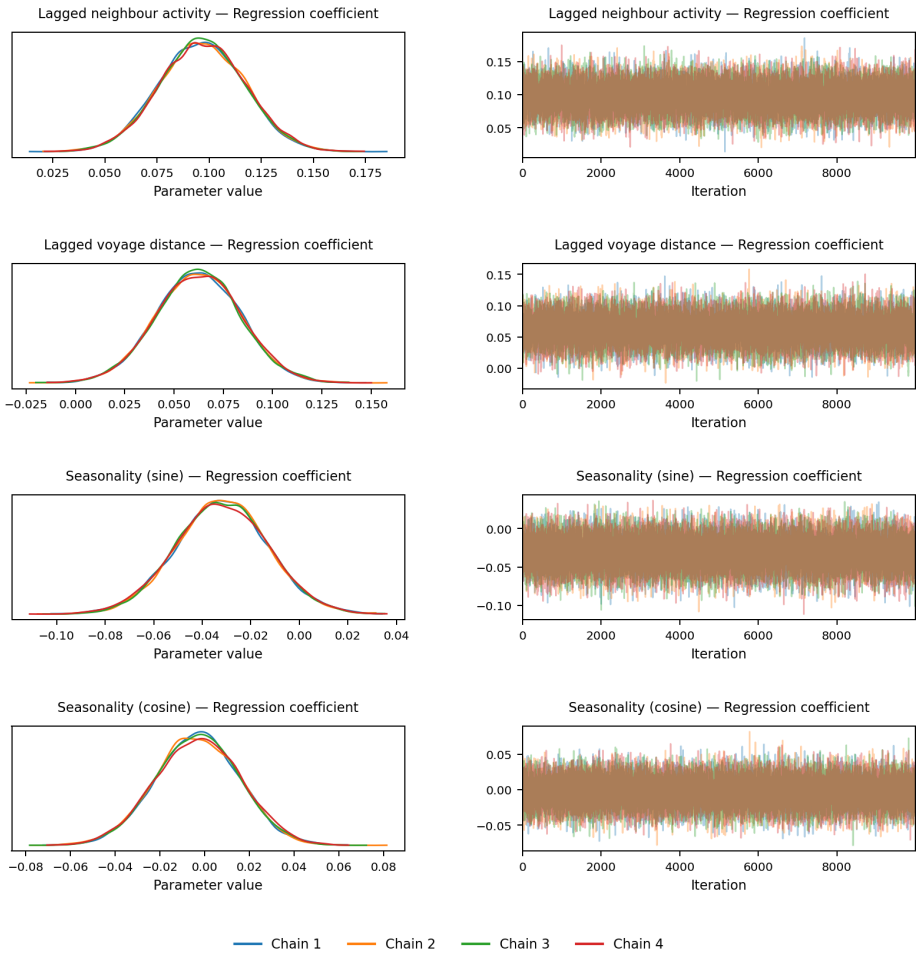

Figure 8: Posterior densities and trace plots for selected regression coefficients in the baseline terminal model.

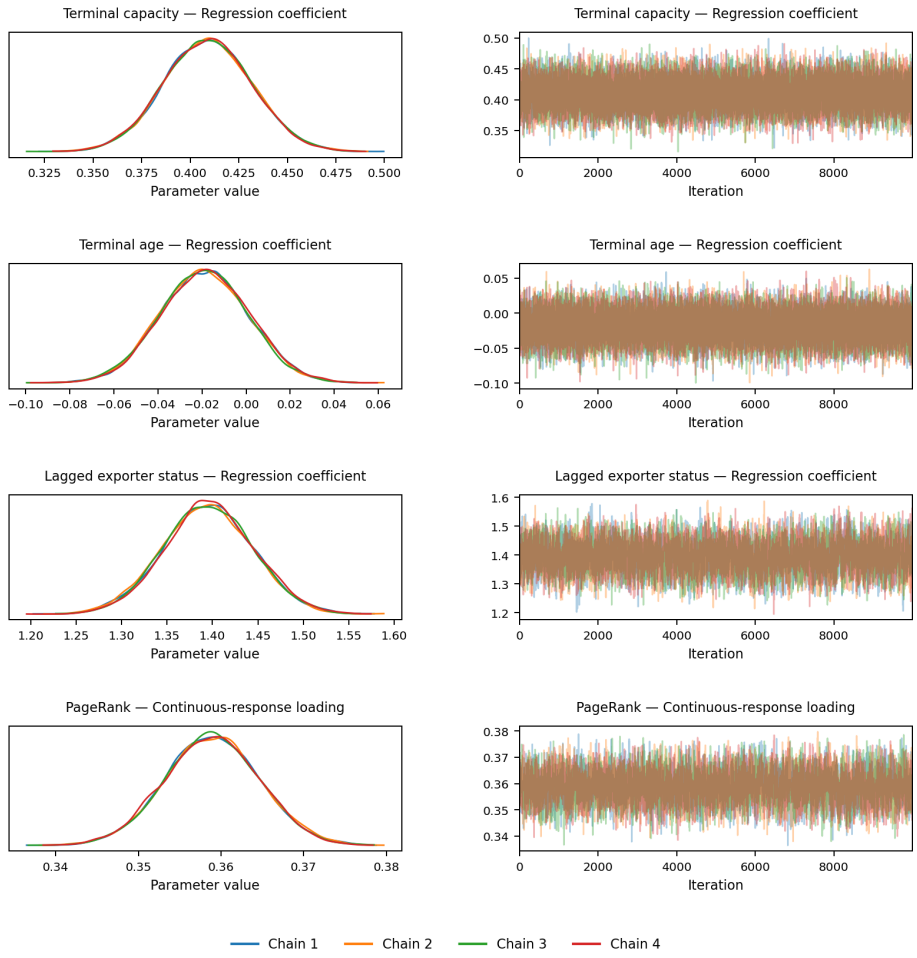

Figure 9: Posterior densities and trace plots for additional terminal regression coefficients and the PageRank loading in the baseline terminal model.

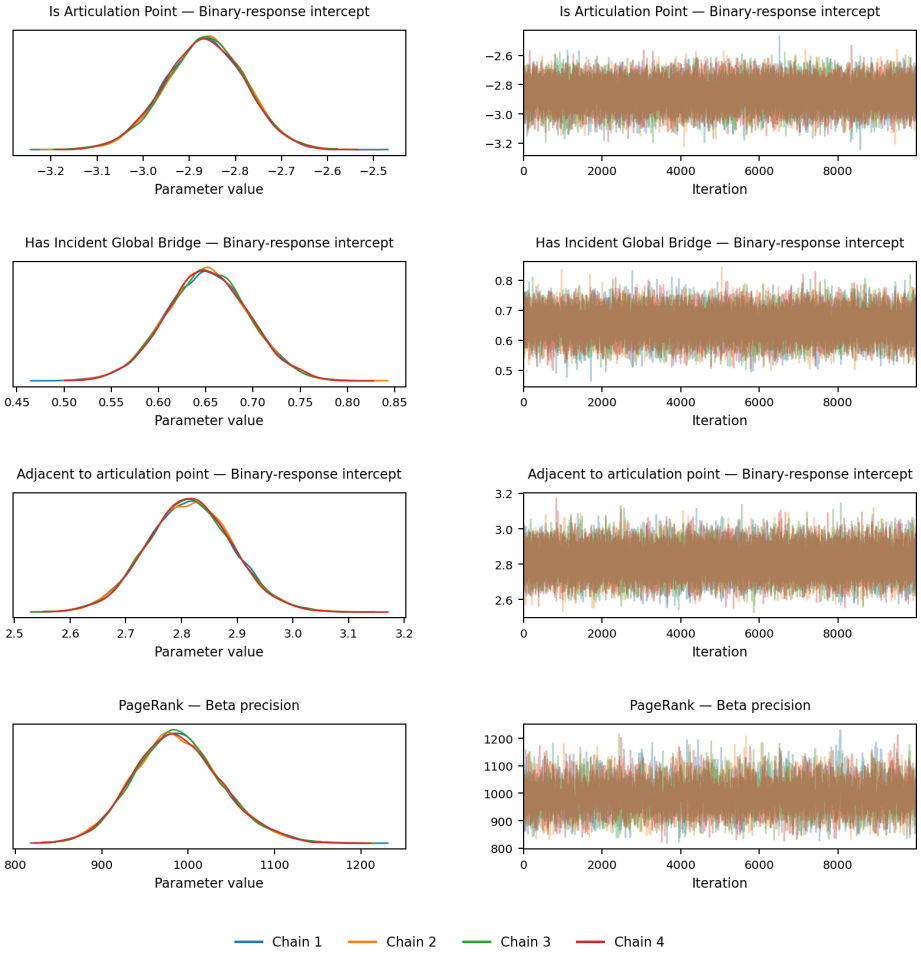

Figure 10: Posterior densities and trace plots for selected response parameters in the baseline terminal model, including parameters associated with the boundary-inflated components.

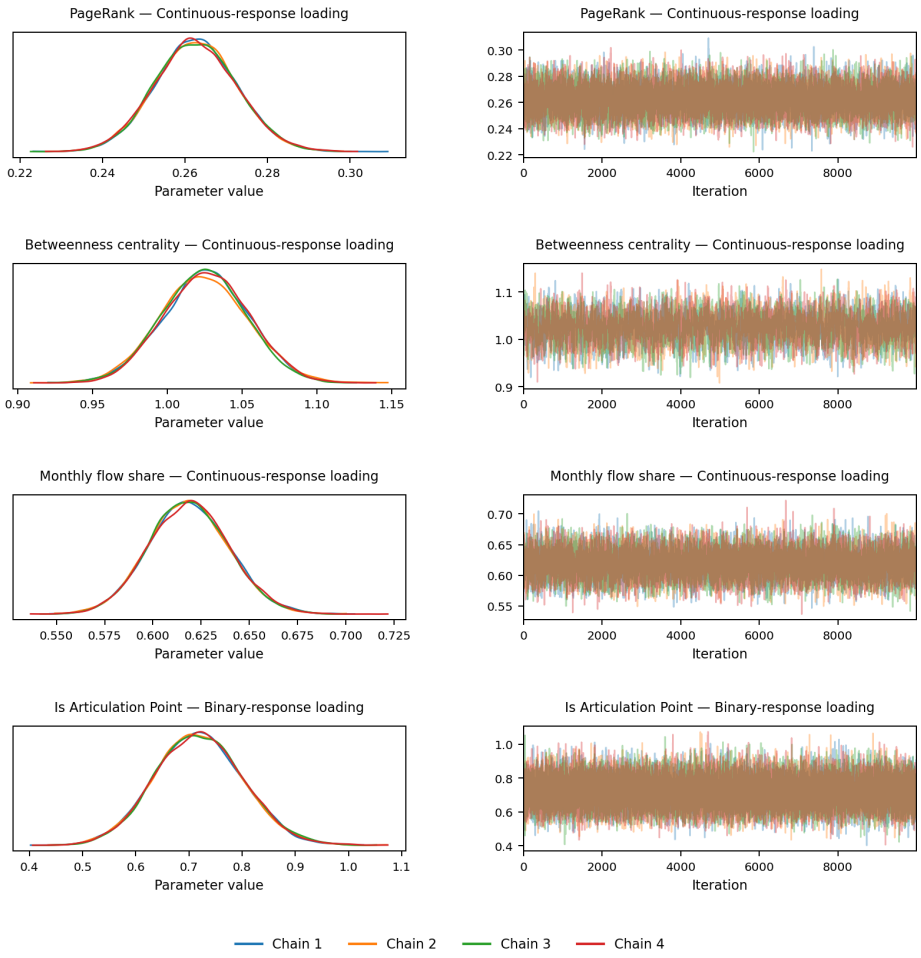

Figure 11: Posterior densities and trace plots for the PageRank, betweenness and flow-share loadings, and the articulation-point loading in the baseline chokepoint model.

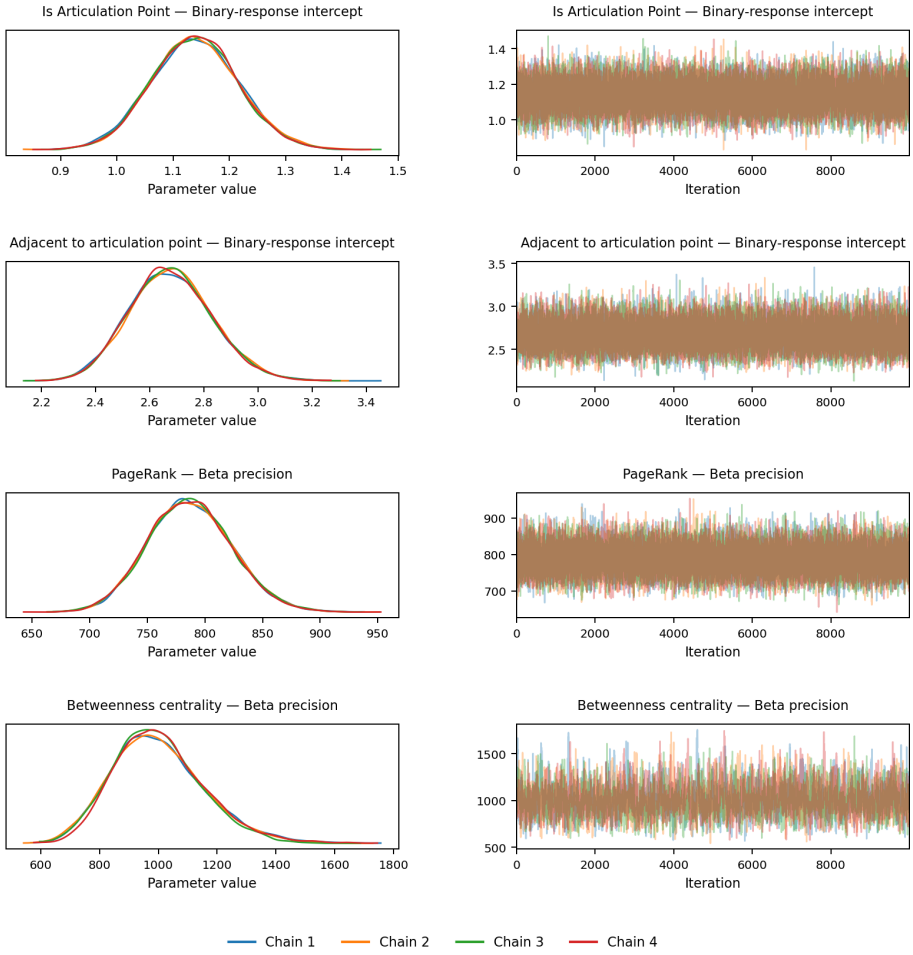

Figure 12: Posterior densities and trace plots for selected response parameters, including the PageRank and betweenness precision parameters, in the baseline chokepoint model.

E.1.2 Autocorrelation diagnostics

The following figures show within-chain autocorrelation functions for selected global parameters. Parameters were selected using the lowest bulk ESS among regression coefficients and response loadings, the lowest bulk and tail ESS among all global parameters, and the PageRank anchor loading. Duplicate selections were removed.

For terminals, the selected parameters are the lagged exporter coefficient, the PageRank loading, and the loading for the probability of HHI equal to one. Their autocorrelation decreases over the initial lags, with small fluctuations at longer lags.

For chokepoints, the displayed parameters are the betweenness loading and precision, and the PageRank loading. The betweenness parameters show a more gradual decay, particularly in some chains, while the PageRank loading exhibits relatively less persistent autocorrelation. The ACF for the lagged neighbour-activity coefficient, which was also generated by the selection procedure, is omitted from the appendix because of its relatively high effective sample size and rapid autocorrelation decay.

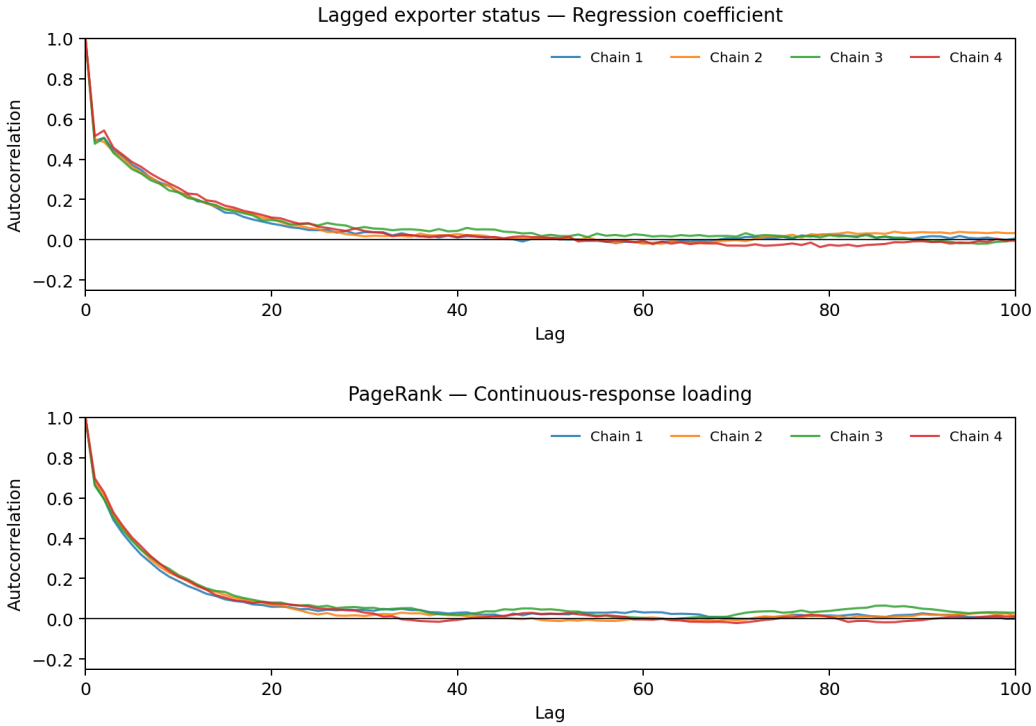

Figure 13: Within-chain autocorrelation functions for the lagged exporter coefficient and PageRank loading in the baseline terminal model.

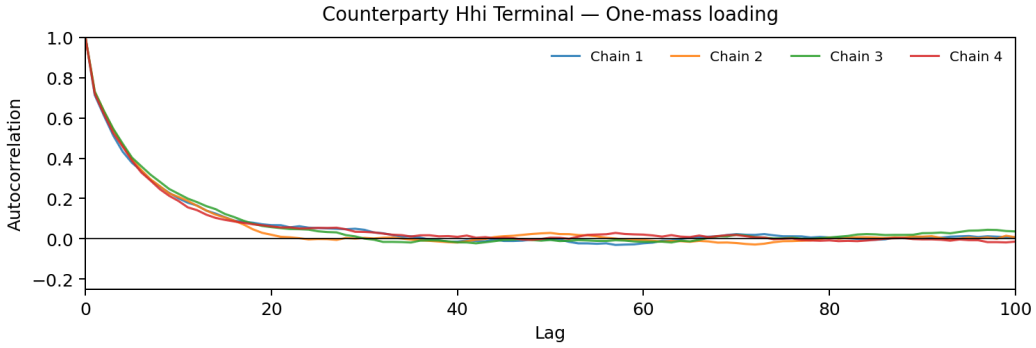

Figure 14: Within-chain autocorrelation functions for the loading governing the probability of counterparty HHI equal to one in the baseline terminal model.

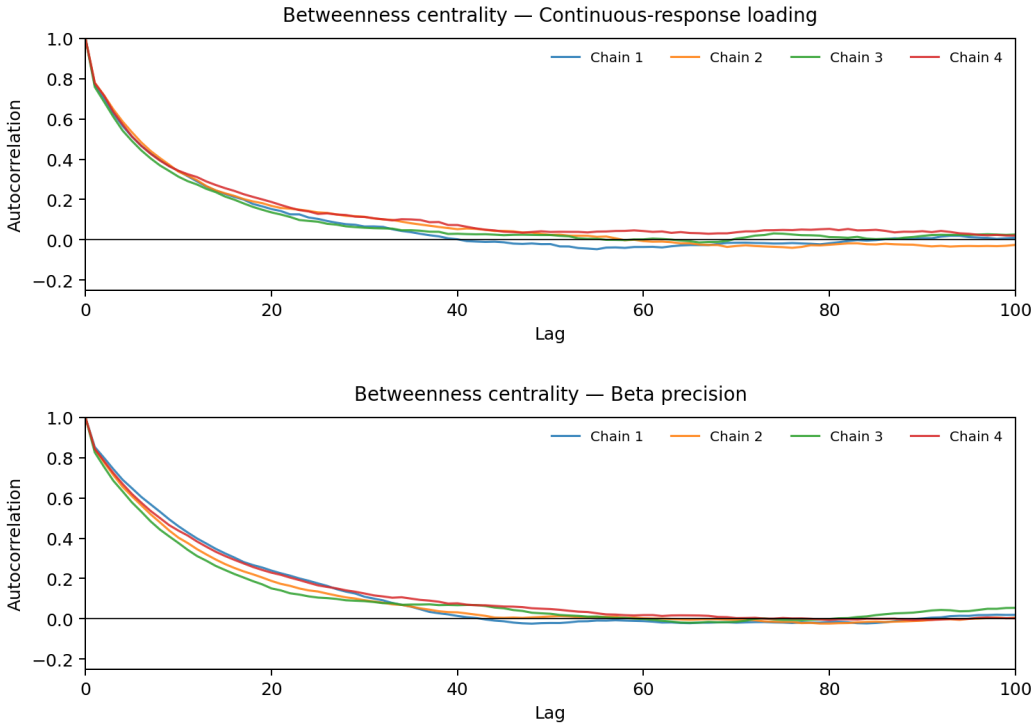

Figure 15: Within-chain autocorrelation functions for the betweenness loading and Beta precision in the baseline chokepoint model.

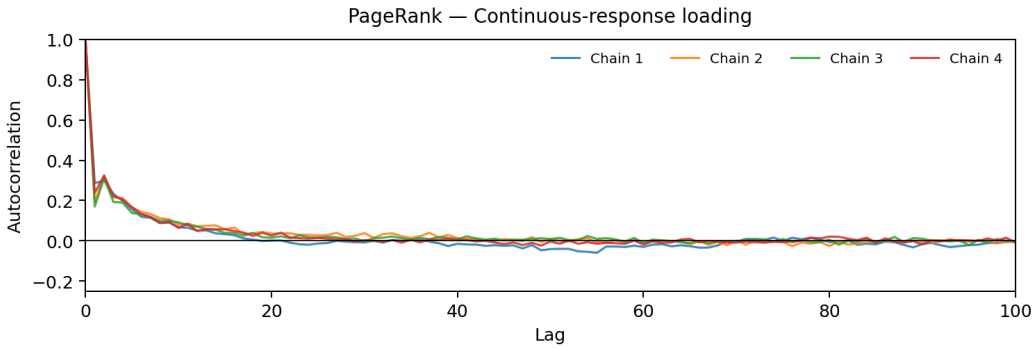

Figure 16: Within-chain autocorrelation functions for the PageRank loading in the baseline chokepoint model.

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